

Irradiation-assisted stress corrosion cracking

Gary S. Was^{1,*}, Yugo Ashida¹ and Peter L. Andresen²

¹University of Michigan, Ann Arbor, Michigan, USA,
e-mail: gsw@umich.edu

²General Electric Global Research, Schenectady, New York, USA

*Corresponding author

Abstract

Irradiation-assisted stress corrosion cracking (IASCC) refers to the process(es) by which irradiation enhances the inherent susceptibility of alloys to stress corrosion cracking (SCC). This article begins with an introduction to the IASCC problem experienced in light water reactors, followed by a summary of existing plant and laboratory data that reveal its dependence on irradiation, environment and alloy parameters. The specific effects of irradiation on IASCC are classified into two categories: the effects on water chemistry and the effects on microstructure. Water chemistry effects include radiolysis and its effects on corrosion potential, and effects of corrosion potential on IASCC. Microstructure effects include radiation-induced segregation (RIS), irradiated microstructure, swelling and creep, and hydrogen and helium generation. Leading mechanisms proposed to explain the role of radiation in the SCC process are: radiolysis and crack tip strain rate, grain boundary chromium depletion, irradiation hardening, localized deformation and RIS of minor elements. As nuclear power plants operate for longer, an increased incidence of SCC can be expected unless active mitigation steps are taken. Drawing on the accumulated understanding over the past four decades of the key processes believed to control IASCC, a set of attributes of an ideal, IASCC-resistant alloy are identified as a first step.

Keywords: boiling water reactors; corrosion potential; crack growth; crack initiation; high temperature water; localized deformation; nickel-base alloys; nuclear materials; pressurized water reactors; radiation creep relaxation; radiation effects; radiation hardening; radiation-induced segregation (RIS), radiolysis; stainless steels; stress corrosion cracking; water chemistry.

1. Introduction

Nuclear power is rapidly becoming established as a key source of clean, low-cost energy throughout the world. Today, some 441 nuclear power plants are in operation worldwide, supplying about 17% of the world's electricity needs. More than 60 nuclear reactors are under construction, and another ~400 are planned or proposed. The success of this technology critically

depends on the safe, reliable and economical operation of these plants. Stress corrosion cracking (SCC) of core internal components must be mitigated or managed to ensure the long-term success of nuclear energy. SCC has occurred in all water cooled reactors, including boiling water reactors (BWR) and pressurized water reactors (PWR), with a greater incidence in unirradiated, out-of-core components, especially between 1970 and 1990. As these materials, component designs and water chemistries have improved, and operating time and fluence has increased, an increasing percentage of cracking problems has occurred in irradiated components (IASCC).

The core of a nuclear reactor is an extreme environment consisting of high temperature water, imposed stresses and strains, intense radiation fields, and a relatively corrosive medium (Figure 1). Most irradiated core components consist of austenitic stainless steels and nickel-base alloys exposed in environments that span oxygenated to hydrogenated water at ~270–340°C. Initially, the affected components were primarily small components (bolts, springs, etc.) or components designed for replacement (fuel rods, control blades, or instrumentation tubes). However, in the last ~20 years, IASCC has been observed in structural components such as PWR baffle former bolts and BWR core shrouds and top guides.

Extensive literature exists for SCC under unirradiated conditions, and the basic factors and dependencies are well defined and reasonably well modeled for austenitic stainless steel and nickel alloys (e.g., Alloys 600 and its weld metals) (Andresen & Was, 2008; Bruemmer et al., 1992; Cowan & Gordon, 1977; Ford & Andresen, 1994; Jones, 2001; Staehle, Forty, & VanRooyen, 1967; Staehle, Hochmann, McCright, & Slatern, 1977). A complete consensus on the underlying mechanism of cracking has not emerged, although the well-behaved continuum in crack growth rate response vs. material/composition (including from stainless steels to nickel alloys), water chemistry, temperature and radiation suggests that a common crack growth mechanism is operative (Andresen, 1996, 2001; Andresen & Ford, 1988; Andresen & Was, 2008; Ford & Andresen, 1988, 1994).

Our understanding has evolved from the view that SCC occurs under very specific and unique conditions, to the view that a continuum in response exists (Andresen, 1996, 2001; Andresen & Ford, 1988; Andresen & Was, 2008; Ford & Andresen, 1988, 1994). With steady improvement in laboratory and plant detection of SCC, it is clear that SCC occurs under a wide range of conditions, but also at a wide range of growth rates. Figures 2 and 3 show examples of the effect of environment (corrosion potential and water purity), material condition (sensitized vs. cold worked) and stress (stress intensity factor) on SCC growth rates; the solid curves are the predicted response (Andresen & Ford, 1988; Andresen & Was, 2008; Ford & Andresen, 1988, 1994). SCC occurs even at low corrosion potential (Figure 2), and thus the

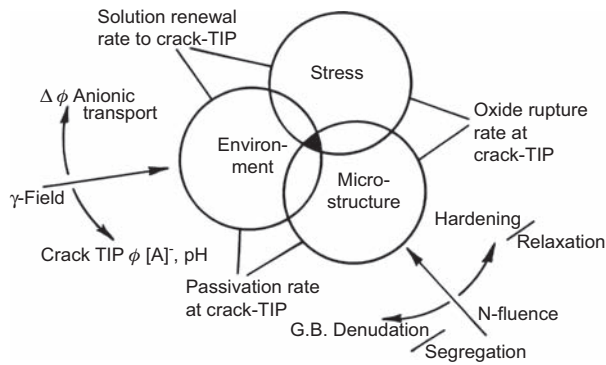


Figure 1 Schematic of the primary engineering parameters that effect SCC – stress, microstructure and environment – and the underlying crack tip processes that control SCC. The primary ways in which radiation affects SCC is also shown: segregation, hardening, relaxation and radiolysis. Radiolysis can increase the corrosion potential, which in turn increases the potential gradient ($\Delta\phi$) and the crack tip potential ϕ , anion concentration [A] and pH.

behavior in BWRs and PWRs is linked, with the primary differences being dissolved H_2 , temperature and the dissolved ion chemistry (boron and lithium are added to PWR primary water) (Andresen, 1996; Andresen & Was, 2008; Andresen, Hickling, Ahluwalia, & Wilsone, 2007; Ford & Andresen, 1994). Of these, temperature has a universal effect, variations in dissolved H_2 are particularly important in nickel alloys, and boron/lithium has little or no effect in deaerated water (Andresen & Was, 2008; Andresen et al., 2007).

Early plant (Figure 4) and laboratory (Figure 5) observations showed that the same basic dependencies existed for unirradiated and irradiated stainless steels, and that increasing fluence produces a well-behaved increase in SCC susceptibility

(Figure 6). Figure 4 shows a strong effect of water purity for both unirradiated and irradiated BWR components, and Figure 5 shows a very similar response to corrosion potential to that in Figure 2. Thus, it was proposed that radiation enhances SCC primarily in four ways: segregation, hardening, relaxation and radiolysis (Figure 1). At high fluence, differential swelling can produce re-loading, and helium generation and development, or dissolution of phases in the microstructure can play a role. The neutron fluence where these processes have an effect is shown in Figure 7, along with the current end-of-life fluence for various BWR and PWR components. The primary radiation effects on materials operate in a similar range of fluence, and thus their individual contributions can be difficult to distinguish. An example of their interaction in altering SCC growth rate is shown in the prediction of cracking of a weld in a BWR core shroud (Figure 8), in which the individual effects are plotted, along with the resulting crack length vs. time. While many of the enhancements in SCC susceptibility from irradiation dose (neutron fluence) have been well established, it remains possible that additional factors will emerge at high fluence (e.g., >30 dpa, or displacements per atom).

Intergranular (IG) SCC is promoted in austenitic stainless steels above a “threshold” fluence (Figures 5 and 6). This occurs in oxygenated (e.g., BWR) water above 2 to 5×10^{20} n/cm² ($E > 1$ MeV), which corresponds to about 0.3–0.7 dpa, and the fluence depends on the stress, water chemistry (esp. sulfate and chloride), operating time and other factors. Attempts to reproduce the same level of IG cracking in inert environments have been unsuccessful, confirming that it is an environmental cracking phenomenon, not simply a change in the microstructure, mechanical properties and the overall response of the irradiated material in an inert environment.

Cracking in hydrogenated water [i.e., BWR hydrogen water chemistry (HWC) or PWR water] is typically observed

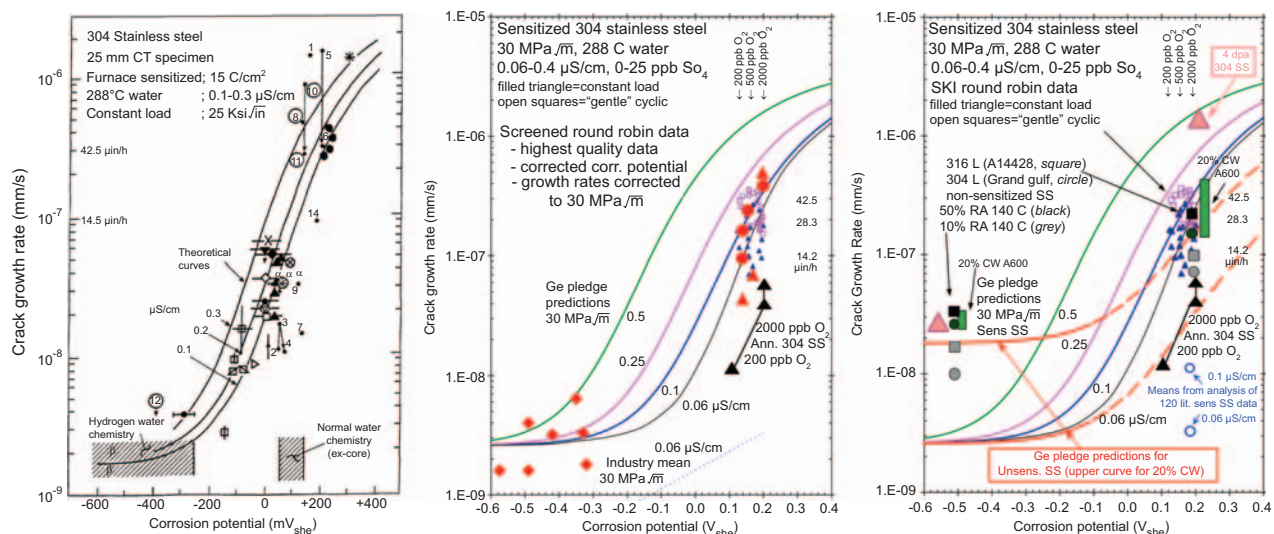


Figure 2 SCC growth rate vs. corrosion potential for stainless steels tested in 288°C high purity water containing 2000 ppb O_2 and 95–3000 ppb H_2 . Dissolved O_2 strongly influences corrosion potential, which in turn affects crack chemistry and growth rate of sensitized stainless steels (two graphs at left), as well as cold worked stainless steels and Alloy 600 (large rectangular symbols on right graph) and irradiated stainless steel (large triangular symbols). Cold worked or irradiated materials have an elevated yield strength, which causes an increase in growth rate at both low and high potential.

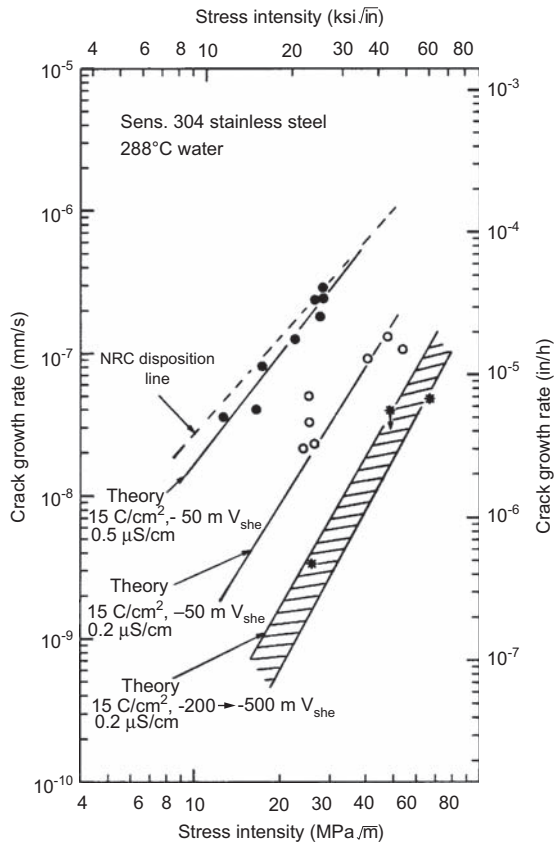


Figure 3 Effect of stress intensity factor on SCC growth rate for sensitized stainless steel exposed in various water chemistries at 288°C.

at roughly a 4X higher fluence than in oxidizing water. The *threshold* fluence again depends on stress, temperature (Figure 9), operating time and other factors. For both BWR and PWR conditions, the same basic dependencies exist for unirradiated and irradiated materials.

It is important to distinguish the results of different kinds of SCC testing. Crack growth testing typically uses fracture mechanics specimens, commonly a compact tension (CT) specimen. It has the significant advantage of continuous, on-line monitoring of crack length vs. time, usually based on an electric current, potential drop technique. It can provide resolution of about 1 μm , and can accurately characterize the inherent resistance to crack advance (i.e., the stage beyond the ill-defined *initiation* stage), as well as the dependencies on corrosion potential, stress intensity factor, etc. Smooth specimen tensile tests, whether using constant load, constant displacement, or slow strain rate (SSRT) techniques, are simpler to perform, but represent some combination of initiation and growth. SSRT tests impose failure, and can over-state or under-state the SCC susceptibility. Constant load or displacement tests generally require periodic interruption for examination, and “initiation” can represent microscopic cracks or complete failure.

Work over the last 25 years has enabled many aspects of IASCC phenomenology to be explained and predicted based

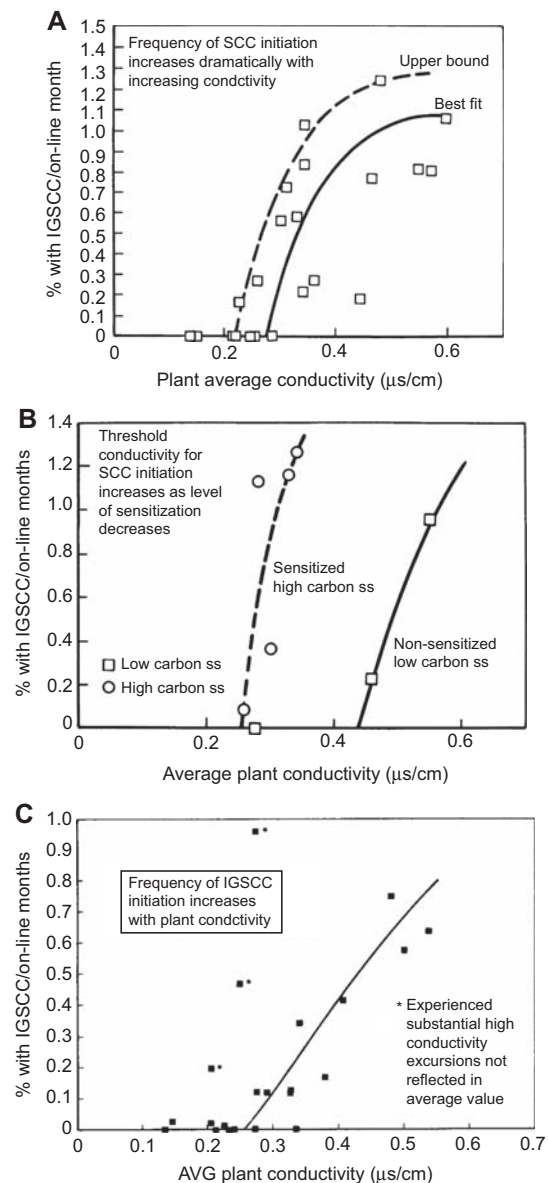


Figure 4 The effects of average plant water purity are shown in field correlations of the core component cracking behavior for (A) stainless steel intermediate (IRM) range monitor dry tubes, (B) creviced stainless steel safe ends, and (C) creviced Inconel 600 shroud head bolts, which also shows the predicted response vs. conductivity.

on the experience with IGSCC of non-irradiated stainless steel in high temperature water environments. This continuum approach has successfully accounted for radiation effects on water chemistry and its influence on electrochemical corrosion potential. However, all radiation-induced microstructural and microchemical changes that promote IASCC are neither fully known nor fully reproducible in similar materials. Well-controlled data from well characterized irradiated materials remain insufficient as a result of the inherent experimental difficulties and financial limitations. Many of the important metallurgical, mechanical and environmental aspects that are believed to play a role in the cracking process are illustrated

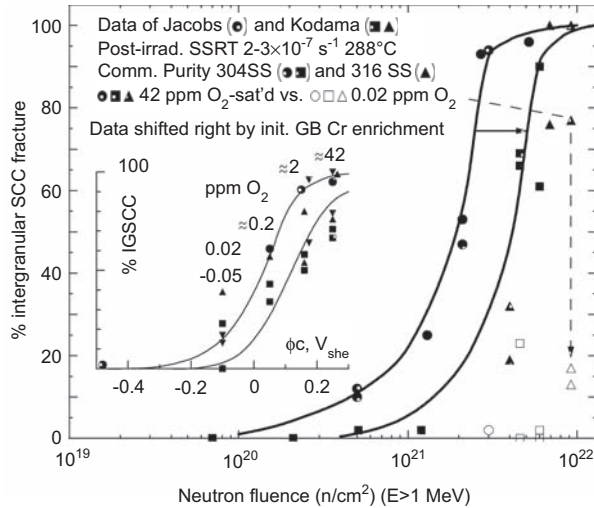


Figure 5 Dependence of IASCC on fast neutron fluence as measured in SSRT tests at $3.7 \times 10^{-7} \text{ s}^{-1}$ on pre-irradiated type 304 stainless steel in 288°C water. The effect of corrosion potential via changes in dissolved oxygen is shown at a fluence of $\approx 2 \times 10^{21} \text{ n/cm}^2$.

in Figure 1. Only persistent material changes are required for IASCC to occur, but in-core processes such as radiation creep and radiolysis also have an important effect on IASCC.

2. Plant data on IASCC

Even early observations exhibited characteristics and dependencies consistent with unirradiated materials, and appropriately led to the description of irradiation *assisted* stress corrosion cracking. Subsequent observation made this clearer e.g., by showing a similar effect of the water purity (as measured solution conductivity) on unirradiated and irradiated stainless steel (Figure 4). The initial reports of IASCC occurred in the early 1960s (Andresen & Ford, 1995; Andresen, Ford, Murphy, & Perks, 1990a; Armijo, Low, & Wolff, 1965; Brown,

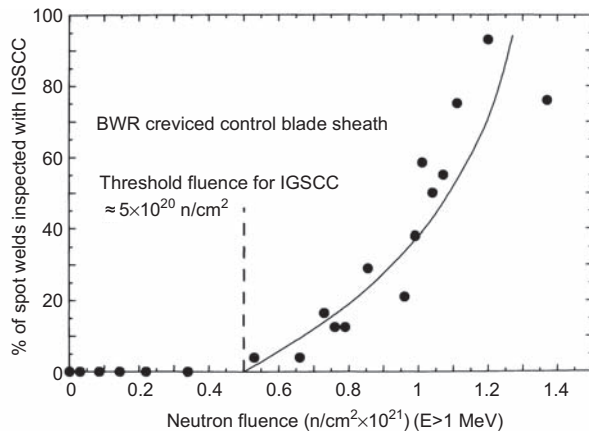


Figure 6 Dependence of IASCC on fast neutron fluence for creviced control blade sheath in high conductivity BWRs.

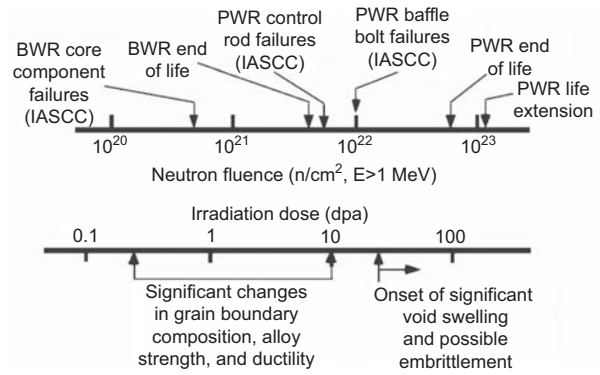


Figure 7 Neutron fluence effects on irradiation-assisted stress corrosion cracking susceptibility of 300 series stainless steels in LWR environments.

Storhok, & Gates, 1967; Bruemmer et al., 1999; Cheng, 1964, 1970, 1975; Cowan & Gordon, 1977; Duncan et al., 1965; Ford & Andresen, 1994; Hanninen & Aho-Mantila, 1988; Jacobs & Wozadlo, 1985; Multer, 1975; Pashos et al., 1964; Pasupathi & Klingensmith, 1981; Schaffer, 1962; Scott, 1994; Storrer & Locke, 1970; Was & Andresen, 1992a; 2007) and involved intergranular cracking of stainless steel fuel cladding initiating from the water side. Post-irradiation mechanical tests, by comparison, were performed in inert environments at various strain rates and temperatures and exhibited mostly ductile, transgranular cracking. Grain boundary carbide precipitation was rarely observed, indicating that thermal sensitization was not an adequate explanation, although it was later found that radiation-induced segregation (RIS) did occur, and resulted in chromium depletion at grain boundaries.

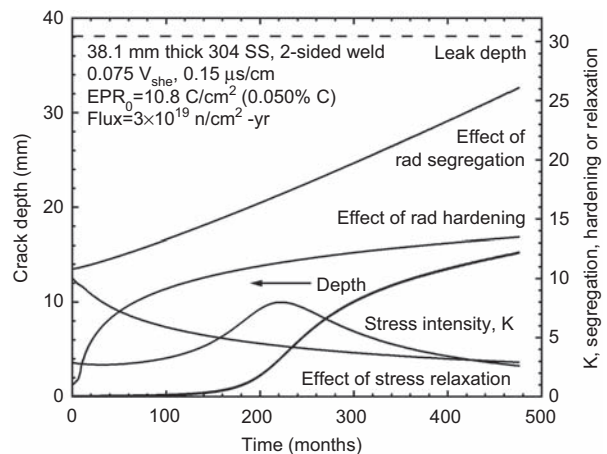


Figure 8 Predicted effect of radiation segregation, radiation hardening and radiation creep relaxation on a BWR core shroud, where the through-wall weld residual stress profile is the primary source of stress. Less aggressive water chemistry (corrosion potential and water purity) would result in less crack advance early in life, which would give a greater opportunity for radiation creep relaxation. The leak depth is the wall thickness of the shroud. While radiation hardening continues to increase the yield strength, its effect on crack growth tends to saturate (see Figure 28).

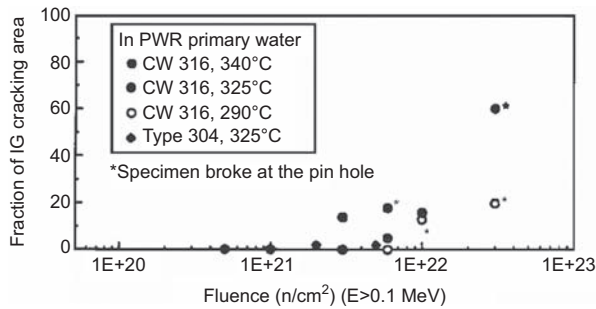


Figure 9 Percentage IGSCC vs. fluence for cold worked type 316 stainless steel tested at various temperatures. Despite the low potential environment of PWR primary water, at high fluence (especially at high temperature) there is significant susceptibility to SCC.

Time-to-failure and stress were found to be correlated, with the earliest failures occurring in thin-walled rods with small fuel-to-cladding gaps, where fuel swelling strains were largest. Areas of high heat flux – where the highest fluence developed and the greatest fuel-cladding interaction (highest stresses and strains) existed – were the regions of highest cracking incidence. The stainless steel cladding in PWR service exhibited a lower cracking incidence of intergranular failure, and some believed that cracking occurred because of off-chemistry conditions or stress rupture. But in the last two decades, it has become clear that SCC is also *irradiation assisted* in PWR water.

The incidence of cracking in stainless steel and nickel alloys core components steadily increased. In 1976, neutron source holders exhibited IASCC, and in 1978 cracking occurred in control rod absorber tubes. Low stress components also cracked, such as instrument dry tubes and control blade handles and sheaths (Figure 5), generally at higher fluences and in creviced locations (Andresen, Vasatis, & Ford, 1990b; Brown & Gordon, 1988; Gordon & Brown, 1990; Was & Andresen, 1992a, 2007). Incidents of IASCC, for example in BWR core shrouds (Andresen & Ford, 1995; Andresen et al., 1990; Ford & Andresen, 1994; Horn, Gordon, Ford, & Cowan, 1997) and PWR baffle bolts (Heuze, Goltrant, & Cauvin, 1998; Scott, 1994; Scott, Meunier, Deydier, Silvestre, & Trenty, 2000) continued to emerge.

Table 1 summarizes the failures of reactor internal components. IASCC is clearly not confined to a particular reactor design, material, component or water chemistry. For example, stainless steel fuel cladding failures were reported years ago in commercial PWRs and in PWR test reactors (Andresen et al., 1990; Brown Jr et al., 1967; Cheng, 1964, 1970, 1975; Garzarolli, Rubel, & Steinberg, 1984; Hanninen & Aho-Mantila, 1988; Multer, 1975; Pasupathi & Klingensmith, 1981; Schaffer, 1962; Storrer & Locke, 1970). At the West Milton PWR test loop, intergranular failure of vacuum annealed type 304 stainless steel fuel cladding was observed (Cheng, 1970) in 316°C ammoniated water (pH 10) when the cladding was stressed above yield. Similarly, IASCC was observed in creviced stainless steel fuel element ferrules in the

Table 1 IASCC service experience.

Component	Material	Reactor type	Possible sources of stress
Fuel cladding	304 SS	BWR	Fuel swelling
Fuel cladding	304 SS	PWR	Fuel swelling
Fuel cladding*	20% Cr/25% Ni/Nb	AGR	Fuel swelling
Fuel cladding ferrules	20% Cr/25% Ni/Nb	SGHWR	Fabrication
Neutron source holders	304 SS	BWR	Welding & Be swelling
Instrument dry tubes	304 SS	BWR	Fabrication
Control rod absorber tubes	304/304L/316L SS	BWR	B ₄ C swelling
Fuel bundle cap screws	304 SS	BWR	Fabrication
Control rod follower rivets	304 SS	BWR	Fabrication
Control blade handle	304 SS	BWR	Low stress
Control blade sheath	304 SS	BWR	Low stress
Control blades	304 SS	PWR	Low stress
Plate type control blade	304 SS	BWR	Low stress
Various bolts**	A-286	PWR & BWR	Service
Steam separator dryer bolts **	A-286	BWR	Service
Shroud head bolts **	600	BWR	Service
Various bolts	X-750	BWR & PWR	Service
Guide tube support pins	X-750	PWR	Service
Jet pump beams	X-750	BWR	Service
Various springs	X-750	BWR & PWR	Service
Various springs	718	PWR	Service
Baffle former bolts	316 SS Cold Work	PWR	Torque, differential swelling
Core shroud	304/316/347 /L SS	BWR	Weld residual stress
Top guide	304 SS	BWR	Low stress (bending)

*Cracking in AGR fuel occurred during storage in spent fuel pond. **Cracking of core internal occurs away from high neutron and gamma fluxes.

Winfrith SGHWR (Garzarolli, Alter, Dewes, & Nelson, 1988; Norris, Baker, Taylor, & Titchmarsh, 1987a), a 100 MWe plant in which light water is boiled in pressure tubes, where the coolant chemistry is similar to other boiling water reactor designs. The 20% Cr/25% Ni/Nb stainless steel differs from type 304 primarily in nickel and niobium content, and compared to most heats of 304 of that time, it had lower sulfur ($\approx 0.006\%$) and phosphorus ($\approx 0.005\%$) contents. The ferrules were designed for a five-year exposure during which the peak fast neutron fluence is $3\text{--}5 \times 10^{21} \text{ n/cm}^2$.

Swelling tube experiments were performed using a varying mix of Al_2O_3 and B_4C within the tubes to produce different swelling rates; B_4C swells as neutrons transmute boron to helium. Nominally identical strings of specimens were inserted into the core of BWRs and PWRs in place of fuel rods (Garzarolli, Alter, & Dewes, 1986; Garzarolli et al., 1988). No difference between BWR and PWR exposure was observed in the strain to SCC initiation of commercial and high purity heats of types 304, 316 and 348 stainless steel and Alloys X-750, 718 and 625.

For many decades, the oxidizing potential in the BWR core was higher than in PWRs. However, in the last ~ 15 years, most BWRs have deployed an electrocatalytic technology called NobleChem™ (Andresen, 1995; Andresen, Diaz, & Hettiarachchi, 2004; Hettiarachchi, Wozadlo, Diaz, Andresen, & Cowan, 1995) to create a low corrosion potential. While the actual corrosion potentials remain several hundred millivolts lower in PWRs because of the higher H_2 fugacity and the presence of the boron/lithium chemistry, what is important in SCC is that there is no differential aeration cell present in either case to produce a shifted crack chemistry.

PWR cracking was often initially attributed to low ductility stress rupture, a mechanism that is also relevant to BWRs. Small amounts of intergranular cracking of irradiated stainless steels was reported in a few laboratory in SSRT tests in $\approx 300^\circ\text{C}$ inert environments (Manahan, Kohli, Santucci, & Sipush, 1989), although in many related experiments (Jacobs, Hale, & Siegler, 1986; Onchi, Dohi, Soneda, Navas, & Castano, 2003) no intergranular failure was found. Small amounts of intergranular cracking in inert tensile tests is not surprising and, since the early 1990s, the plant and laboratory IASCC data show that cracking is environmentally assisted and follows a well-behaved continuum in response over ranges in fluence, corrosion potential, temperature, stress, etc. (Andresen & Ford, 1995; Andresen et al., 1990; Bruemmer et al., 1999; Ford & Andresen, 1994).

Compared to BWRs, PWRs operate at a higher temperature, at $\approx 10\times$ higher peak neutron flux and fluence in core structural components, at $\approx 100\times$ higher hydrogen fugacity, and with borated-lithiated water chemistry. The higher flux produces higher gamma heating and the possibility of localized boiling and thermal concentration cells in crevices, which could lead to aggressive local chemistries. To address this concern, some baffle plate and former bolt designs have been modified to allow the bulk water to communicate more readily in the bolt shank area.

In both PWRs and BWRs, high fluence brings the possibility of high silicon segregation to grain boundaries, and

laboratory data on irradiated alloys and unirradiated alloys with elevated silicon show that the SCC response can be unaffected by high vs. low corrosion potential (Andresen & Morra, 2005b; Andresen, Morra, & Catlin, 2004).

SCC of Alloy 600 shroud head bolts was first observed in 1986, and in 1984 stainless steel safe ends and in-core instrumentation tubes first exhibited SCC, with creviced components cracking somewhat sooner (Brown & Gordon, 1988; Cowan & Gordon, 1977; Gordon & Brown, 1990), as summarized in Figure 4. The highest radiation exposure occurred for the intermediate range and source range monitor (IRM/SRM) dry tubes, which contain flux monitors housed in thin-walled, annealed stainless steel tubes. Cracking initiated in the crevice between the spring housing tube and the guide plug at fluences between 0.5 and $1.0 \times 10^{22} \text{ n/cm}^2$ ($E > 1 \text{ MeV}$). Wedging stresses from the thick oxide observed in the crevice were implicated, as other (applied and residual) stresses were negligible.

Average coolant conductivity is the primary distinguishing factor among plants, and it correlates strongly with cracking incidence (Figure 4A–C). Each point in Figure 4 represents inspection results for one BWR plant, and data are normalized using reactor operating time (i.e., “percentage of components with intergranular cracks divided by the on-line exposure time”). The scatter in Figure 4A was attributed to variations in fluence and specific ion chemistry, as well as limitations in the resolution of underwater visual inspection. Plants with high conductivity early in operation fall on a different distribution, as this effect is not adequately reflected in the average (Figure 4C) because cracks initiate and grow fast at high conductivity, creating a stress intensity factor that can continue to drive the crack as the water purity improves. IASCC and water purity were also correlated in SCC of shroud head bolts (Figure 4C) and creviced safe ends (Figure 4B). The strong influence of conductivity on cracking of stainless steel has also been shown in laboratory tests and plant recirculation piping, where predictive modeling (Andresen & Ford, 1988; Andresen et al., 1990; Ford & Andresen, 1988, 1994; Ford, Taylor, Andresen, & Ballinger, 1987; Ford et al., 1990) has been compared to field data on the operational time required to achieve a detectable crack (typically 10% of the wall thickness). Preliminary prediction (Figure 4C) of the shroud head bolt cracking response (Ford & Andresen, 1994) also provides good agreement with the observation.

Table 1 also shows that high strength, nickel-base alloys components (Andresen et al., 1990; Hanninen & Aho-Mantila, 1988) have also experienced IASCC, with many incidents in lower radiation flux regions (e.g., where the end-of-life fluence is below $\approx 5 \times 10^{19} \text{ n/cm}^2$), such as cracking of Inconel X-750 jet pump beams in BWRs. Alloy X-750 PWR fuel hold-down springs, which have an end-of-life fluence of $1\text{--}10 \times 10^{21} \text{ n/cm}^2$, also cracked extensively. Cracking may have been aggravated by vibrational stresses. Compared with stainless steels, the effects of irradiation on IASCC in high strength, precipitation-hardened nickel-base alloy components have not been characterized as well.

The two most common examples of IASCC in the last fifteen years are BWR core shrouds (Andresen & Ford,

1995; Andresen et al., 1990; Ford & Andresen, 1994; Horn et al., 1997) and PWR baffle bolts (Heuze et al., 1998; Scott et al., 2000), although control blade components, fuel components, the BWR top guide, and other components are also susceptible. SCC in the BWR core shroud occurs almost exclusively near circumferential and vertical welds, and initiation is observed from both the inside (ID) and outside (OD) surfaces (the shroud separates the upward core flow from the downward recirculation flow that occurs in the annulus between the shroud and the pressure vessel). This large diameter welded “pipe” has little active (ΔP) loading, and its susceptibility arises primarily from weld residual stresses and weld shrinkage strains (Angeliu, Andresen, Sutliff, & Horn, 1999; Angeliu et al., 2000; Othon & Morra, 2008). Cracking is observed in both low fluence and moderate fluence areas, and the extent of the enhancement in SCC susceptibility by irradiation is limited because, while radiation hardening and RIS occur, radiation creep also relaxes the weld residual stress. SCC predictions for a BWR core shroud that account for the damaging effect of RIS and RH and the beneficial effect of radiation creep relaxation are shown in Figure 8, which illustrates the complexity of the interactions of these phenomena in the evolution of cracking. If SCC does not nucleate early in life (e.g., below 0.5 dpa), e.g., from high coolant impurity levels or severe surface grinding, predictions indicate that susceptibility will decrease with fluence in the shroud welds because of radiation-induced creep relaxation. Note that many shroud welds are in very low flux areas where relaxation and other irradiation effects are small.

Extensive failures of PWR baffle bolts (Heuze et al., 1998; Scott et al., 2000), with large plant-to-plant and heat-to-heat differences, have been observed in the last fifteen years. Most baffle bolts are fabricated from type 316 stainless steel cold worked to $\approx 15\%$ to increase their yield strength. The complex baffle former structure exists in a PWR because their fuel does not have a surrounding “channel”, so the baffle former structure must conform closely to the geometry of the fuel to provide well-distributed water flow. The baffle former plates are usually made from annealed material, typically type 304 stainless steel. Because of their proximity to the fuel, very high fluences can develop – up to ~ 80 dpa by the end of the original design life, and proportionately longer for 60–80 year extended life. The high gamma flux produces significant heating in the components, estimated at up to $+40^\circ\text{C}$, especially in designs where the PWR coolant does not have good access to the bolt shank. Although heat-to-heat variability is not understood, plants that load-follow (and therefore undergo power level changes and thermal cycles) are much more prone to baffle bolt cracking. Another aggravant is the thermal gradient and possible boiling (resulting in altered water chemistry) in the shank area of the bolt if the coolant access is restricted. However, a key element that must be factored into any understanding of baffle bolt cracking is the very sizeable stress relaxation that occurs in the first few dpa, followed by preferential radiation swelling of the annealed baffle plates compared to the cold worked baffle bolts, which will cause reloading. The dynamic equilibrium between swelling and radiation creep, which determines the “reloading” stress in the bolt, is likely a complex function of

many parameters, including local neutron flux, temperature, baffle plate geometry and composition.

Replaceable components often receive less attention than structural components. However, cracking that can impede the safe operation of the reactor is very important. Cracking of control rods and control rod blade handles can result in dissolution of B_4C (making it unavailable to absorb neutrons when needed) and potentially affect the insertion of the control rods. An NRC Event Notification Report on December 2, 2010 (U.S.NRC, 2010) describes significant cracking of control rods that was severe enough to raise concern for insertion.

Given the diversity of components that have cracked, and the low stresses of many, the incidence of IASCC can only be expected to grow. Strategies to mitigate IASCC (e.g., NobleChem™ (Andresen, 1995; Andresen et al., 2004; Hettiarachchi et al., 1995) and manage IASCC (e.g., by showing some IASCC can be tolerated, installing mechanical restraints to mitigate the impact of IASCC in BWR shrouds, or selectively inspecting and replacing baffle bolts) have been successful. Designs have been pushed to lower stress with, for example greater annular tolerance so that B_4C will not cause swelling of the stainless steel tubing in which it is contained. However, there is less success in understanding and identifying resistant materials, and it is likely that materials well outside of the historical 18Cr-8Ni austenitic stainless steels will have to be considered to improve IASCC resistance.

While there have been few indications of cracking in pressure vessel steels, and none in areas where radiation effects are involved, there are conceptual concerns for the effect of radiation on yield strength, and the effect of yield strength on SCC of low alloy steels. Structural components are attached to the pressure vessel via attachment pads that are welded directly onto the pressure vessel before the structure is heat treated (tempered). Attachment pads are generally Alloy 182 weld metal, which is susceptible to SCC, and as cracks propagate through the pad to the interface with the pressure vessel, they can be at high stress intensity factor. While the fluence is sufficient to affect the pressure vessel properties (especially yield strength and toughness), it is not sufficient to cause any consequential creep relaxation of weld residual stresses. As the pressure vessel fluence increases and its yield strength increases, its susceptibility to SCC (and its crack growth rate) increases. The discovery that ppb levels of chloride can increase the crack growth rate of low alloy steel by $>100\times$ (Figure 10) (Kumagai et al., 2009; Ritter & Seifert, 2004) adds to the concern.

IASCC field experience has led to the following trends and correlations:

2.1. Water impurities

Impurities, especially chloride and sulfate, strongly and similarly affect IASCC in BWR water (Figure 4). This correlation applies equally to low and high flux regions and to stainless steels (Figure 4A, B) and nickel-base alloys (Figure 4B) (Gordon & Brown, 1990), and is closely paralleled in out-of-core components (Andresen & Ford, 1988; Andresen et al., 1990; Brown & Gordon, 1988; Ford & Andresen, 1988, 1994; Ford et al., 1987, 1990) At higher levels, the

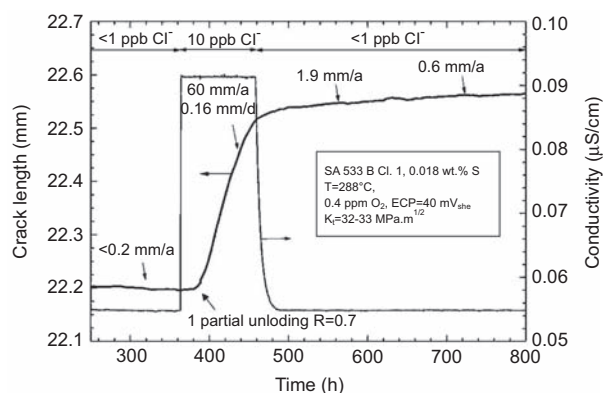


Figure 10 Example of the pronounced effect of 10 ppb Cl⁻ on the crack length vs. time response of low alloy steel.

same impurities can affect SCC in PWRs. If high corrosion potential conditions form in the PWR primary (where boron and lithium are present), the crack chemistry is dramatically altered, and high growth rates can result (Andresen & Was, 2008; Andresen et al., 2007).

2.2. Corrosion potential

Field and laboratory data demonstrate that corrosion potential is a very important parameter, with its effect being consistent from zero to low to high fluence, except in some high fluence materials and/or under high stress intensity factor conditions where high growth rates are always observed. Materials prone to high radiation-induced changes in grain boundary silicon level may exhibit a very limited effect of corrosion potential (Andresen & Morra, 2005a; Andresen et al., 2004). While corrosion potential usually affects crack initiation and growth, there is no evidence of *threshold* potential and, indeed, it is not fully accepted that irradiated materials exhibit IASCC in deaerated water.

2.3. Crevices

Cracking is enhanced by crevices, primarily because of their ability to create a more aggressive crevice chemistry from the gradient in corrosion potential (in BWRs) or in temperature (most relevant to PWRs), which in turn can accelerate crack initiation. Stress and strain concentration can also occur in crevices.

2.4. Temperature

Temperature increases both IASCC initiation and growth rate.

2.5. Fluence

IASCC of annealed stainless steel was once thought to occur only at fluences above $\approx 0.3 \times 10^{21}$ n/cm². But significant intergranular cracking in BWR core shrouds (which do not have thermal sensitization) occurs over a broad range

of fluences, showing that a true fluence threshold does not exist. (Andresen & Ford, 1995; Andresen et al., 1990) The observations of SCC in unirradiated, unsensitized stainless steel (with or without cold work) also undermine the concept of a *threshold* fluence below which no SCC occurs. Thresholds in corrosion potential, water impurities, temperature etc. (Andresen, 2001; Andresen, Angeliu, & Young, 2001; Ford & Andresen, 1994; Ford et al., 1990) have also been disproven.

2.6. Integration of fluence effects

Irradiation has a complex effect on SCC susceptibility. While RIS and radiation hardening increase susceptibility, radiation creep relaxation of constant displacement loads (e.g., bolts and welds) tends to reduce susceptibility. For these reasons, SCC in BWR shrouds and PWR baffle bolts does not always correlate strongly with fluence. SCC can only be interpreted and predicted by accounting for the conjoint effects of multiple factors.

2.7. Dynamic stress and strain

High stress and dynamic strains were responsible for the earliest incidents of IASCC, but cracking has since been observed at quite low stresses at high fluences during longer operating exposure. Laboratory and field data indicate that IASCC growth can occur at stress intensity factors well below 10 MPa√m, and initiation can occur at <20% of the irradiated yield stress.

2.8. Cold work

Bulk cold work, surface cold work (especially abusive surface grinding), and weld residual strain in the heat affected zone tend to exacerbate all forms of SCC, although it can also delay the onset of some radiation effects (especially swelling) by creating more sinks for interstitials and vacancies.

2.9. Grain boundary chromium depletion

Pre-existing grain boundary carbides and/or chromium depletion are not essential to IASCC, although furnace sensitized stainless steels are clearly highly susceptible to cracking in-core. Pre-existing chromium depletion, e.g., from thermal sensitization, will be magnified by irradiation, and increase IASCC susceptibility (primarily in pH-shifted environments, as can develop when potential or thermal gradients exist). In the absence of chromium depletion, the presence of grain boundary carbides or other particles is beneficial. The role of nitrogen, sulphur, phosphorous, and other grain boundary segregants is less clear.

2.10. Interdependencies

The fluence at which IASCC is observed depends on applied stress and strain, corrosion potential, solution conductivity, crevice geometry, cold work, prior sensitization, etc. At

sufficiently high conductivities, cracking has been observed in solution annealed stainless steel in the field (Gordon & Brown, 1990) and in the laboratory (Andresen & Ford, 1988; Andresen et al., 1990; Ford & Andresen, 1988, 1994; Ford et al., 1987, 1990). Thus, while potentially useful in an engineering context, the concept of a *threshold* fluence (or stress, corrosion potential, etc.) is scientifically misleading (Andresen, 2001; Andresen et al., 2001, 2004; Ford & Andresen, 1994). IASCC initiation and growth must be understood in terms of the inter-dependent effects of many parameters.

3. Laboratory data on IASCC

3.1. Crack initiation

Crack initiation in irradiated samples has been measured using two principal test types: the constant extension rate test (CERT), or the slow strain rate test (SSRT), and constant load tests (CLT). These tests are generally conducted on smooth specimens in the shape of tensile bars (CERT/SSRT), o-rings, c-rings or u-bends (CLT). While termed initiation tests, these experiments incorporate some amount of crack growth with the degree depending on the test type, how it is conducted and what is measured. Nevertheless, crack initiation is generally a significant part of the measured behavior.

CERT test data consistently shows that IASCC exhibits a “threshold type” behavior, in which cracking commences only after a minimum fluence in the range of 3×10^{20} n/cm² ($E > 1$ MeV) in BWR NWC, which corresponds roughly to about 0.5 dpa. However, observations of cracking in-core

indicate that this threshold value may be as high as 3 dpa in a PWR environment (Chopra & Rao, 2011; Gordon & Brown, 1990). Figure 11 shows an example of the severity of cracking as a function of fluence for tests conducted in the laboratory (post-irradiation) in high purity water containing high dissolved oxygen (8 ppm) water at $\sim 288^\circ\text{C}$. In these tests, the cracking severity is denoted by the percentage of the fracture surface that exhibits IG cracking. Note that, while there is significant scatter in the data, there is a strong trend of increased cracking severity with fluence. Further, while it appears that high purity 304 has greater susceptibility, the same trend with fluence applies to alloys CP304, HP304 and CP316. Figure 12 shows a complementary set of data for 15% cold worked 316 stainless steel from a thimble tube of a PWR and tested in the laboratory in PWR water conditions. Compared with the data in a BWR environment (Figure 11) in which the % IG reaches 100% by about 13 dpa, in the PWR environment, the IASCC severity continues to increase up through a dose of 33 dpa at which it saturates. The significance of these results is that they confirm that irradiation and straining do not need to be applied simultaneously for IASCC to occur, establishing that changes to the microstructure are the critical element in IASCC.

Constant load tests have been conducted on samples irradiated to doses as high as 76 dpa. Figure 13 shows the results of CLT tests in a PWR environment, in which the stress at fracture, expressed as a percent of irradiated yield stress, is plotted vs. dose in dpa. Note that cracking is observed to occur, at least in one case, at a stress that is only 40% of the irradiated yield stress. Note also that above about 95% of the irradiated yield stress, all samples exhibit fracture. As the database includes numerous alloys/conditions 304, 304L, 316, 316CW irradiated from several plants, it provides clear evidence that there is a strong dependence of cracking on the value of applied stress as a percentage of the yield stress.

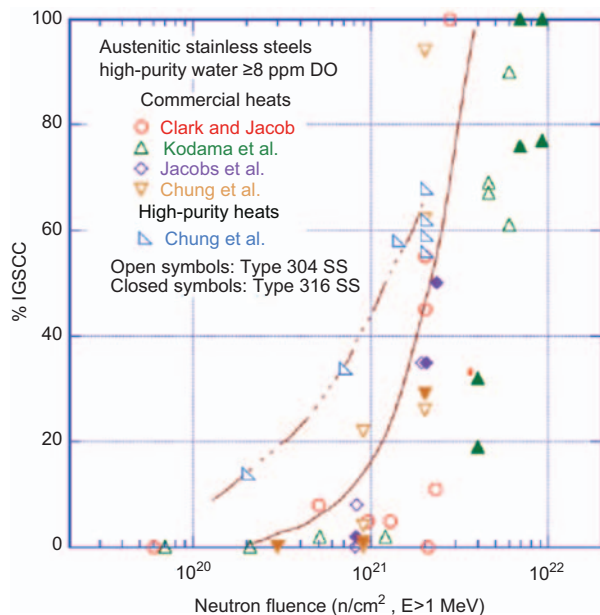


Figure 11 Susceptibility of irradiated austenitic stainless steels to IGSCC as a function of neutron fluence in high temperature water containing high dissolved oxygen content (≥ 8 ppm) (Chopra & Rao, 2011).

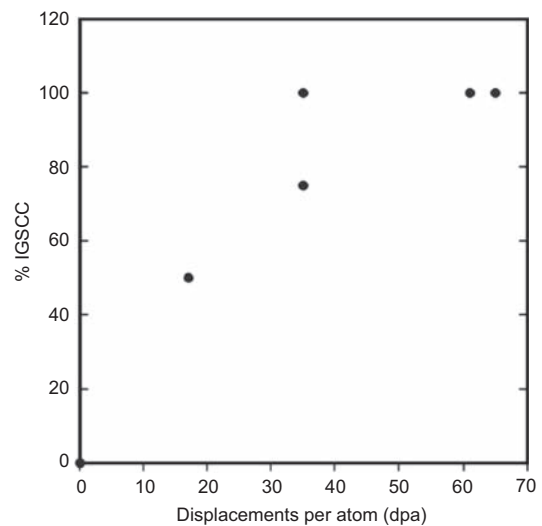


Figure 12 Dependence of IGSCC on displacement damage for 15% cold worked 316SS from a thimble tube in a PWR (Fujimoto et al., 2005).

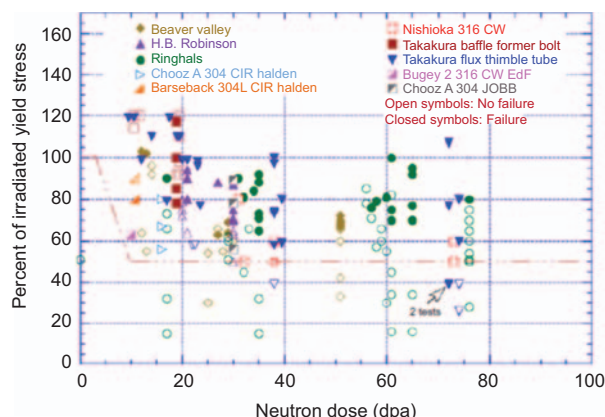


Figure 13 Stress as percent of irradiated yield stress vs. neutron dose for IASCC flaw initiation in austenitic stainless steels in a PWR environment (Chopra & Rao, 2011).

3.2. Crack growth

Crack growth data on unirradiated structural materials in BWR and PWR environments have been acquired since the 1970s, and have been extensively discussed by Andresen and many other researchers (Andresen, 1993, 2002, 2005a; Andresen, Chou, Morra, Nelson, & Rebak, 2009; Andresen, Emigh, & Morra, 2005; Andresen & Ford, 1994; Andresen & Morra, 2005a, 2007a, 2007b, 2008a, 2008b; Andresen et al., 2004a; Young, Andresen, & Angeliu, 2001). The first crack growth rate measurements on irradiated stainless steels in the laboratory (Andresen et al., 1996) and *in situ* in an operating BWR (Andresen & Ford, 1995; Andresen et al., 1990) were performed by GE in the mid-1980s, and subsequently many other IASCC growth rate studies have been performed. Figure 14 shows an example of crack length vs. time data obtained in the earliest studies (Andresen et al., 1996), which used careful intergranular transitioning techniques, along with small sulfate additions to promote well-behaved crack growth. In these initial studies, it was proposed, then demonstrated, that irradiation affects four primary parameters: grain boundary chemistry from RIS; yield strength from radiation hardening; elevated corrosion potential from radiolysis of water; and, reduction in stress from radiation creep relaxation. Other parameters, such as swelling (especially differential swelling that can lead to reloading) and perhaps phase formation and precipitate dissolution may occur and play a role in SCC at higher fluences.

Subsequent testing did not improve on the quality of data from these early tests, and problems related to poor IG transitioning, large errors in dcPd that resulted in much higher stress intensity factors than intended, specimen unloading during critical portions of the tests, failure to manage the test to obtain well-behaved data, etc. continue to affect the quality of some data. Summaries of the data (e.g., Figure 15 in BWR water) show extensive scatter, which makes it difficult to identify dependencies or even propose disposition lines for regulation with confidence.

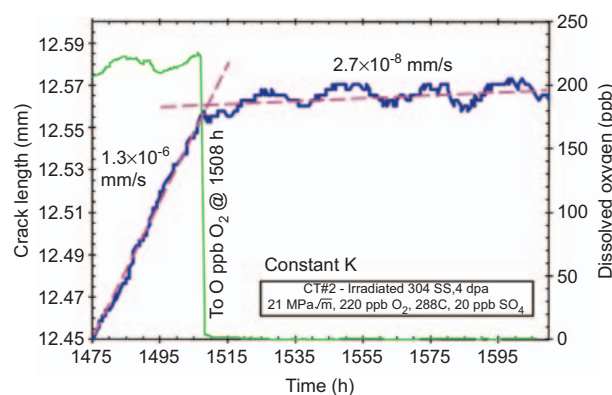


Figure 14 Crack length vs. time for Type 304 stainless steel irradiated to 4 dpa and tested in at 21 MPa/m in 288°C water. A large reduction in crack growth rate is observed as the dissolved O_2 and corrosion potential are decreased (Andresen et al., 1996).

In the last decade, a significant amount of new data has been obtained in various programs (Chatani et al., 2007; Chen, Chopra, Gruber, & Shack, 2010; Chopra, Alexandreanu, Gruber, Daum, & Shack, 2006; Chopra, Gruber, & Shack, 2003; Chopra & Shack, 2008; Davis & Andresen, 2002; Davis, Horn, Andresen, & Nelson, 2008; Dumbill & Vatter, 2001; Edwards, Oliver, Garner, & Bruemmer, 2002; Espeland, 2004; Forsyth, 2001; Horn, 2009; Jenssen & Espeland, 2002; Jenssen & Jakobsson, 2001; Jenssen & Stjarnsater, 2009; Jenssen, Gott, Efsing, Andersson, 2003; Jenssen, Sundberg, & Konig, 2007; Jenssen, Stjarnsater, Torimaru, & Kitsunai, 2009; Karlsen, 1996, 2009; Karlsen & Aaltonen, 2007; Karlsen & Horvath, 2004; Karlsen, Bennett, & Hogberg, 2005a; Karlsen, Espeland, & Helin, 2010; Massoud, Dubuisson, Scott, & Chamardine, 2005; Nakamura et al., 2007; Nakano, Karlsen, & Espeland, 2007, 2008; Ooki et al., 2005; Sumiya et al., 2007; Valo, 1993; Yamashita, 2008), including in the EPRI coordinated and sponsored Cooperative IASCC Research (CIR) program (Pathania, Gott, & Scott, 2007; Scott, 2003) in which Studsvik (Jenssen & Stjarnsater, 2009; Jenssen et al., 2007), NRI (Ernestova & Burda, 2009) and SK-CEN (Van Dyck, 2009) conducted CGR tests on fast reactor (Bor-60) irradiated commercial and tailored stainless steels. In addition, in-core or ex-core CGR tests of pre-irradiated specimens were conducted at the Halden reactor in Norway (Chopra & Rao, 2011; Karlsen, Espeland, & Horvath, 2005b) and at the Japan Materials Testing Reactor (JMTR) in Japan (Chatani et al., 2007; Kaji et al., 2007, 2009; Takakura et al., 2009).

A key evolution in the understanding of SCC growth rate response is that immunity and thresholds apparently do not exist as a function of any parameter apart from stress intensity factor, where it is agreed that no growth occurs at zero. However, the concepts of threshold corrosion potentials, threshold water purity, threshold sensitization of Cr depletion, threshold cold work, threshold fluence, threshold K (apart from zero), heat treatments or compositions that provide immunity, etc. have eroded as improved test techniques

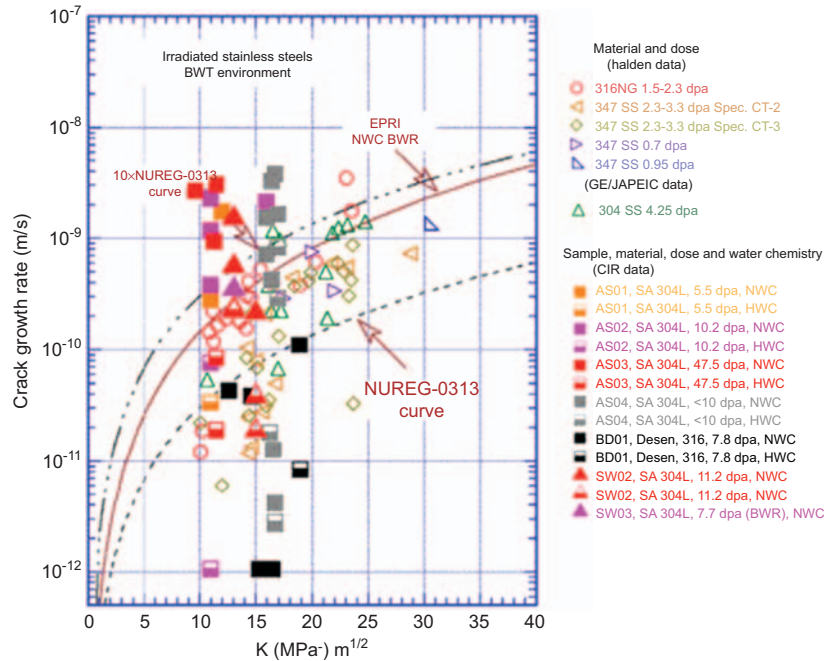


Figure 15 Crack growth rates in irradiated stainless steels tested in BWR water. Data include CGRs in NWC generated by Halden project (As cited in Chopra & Rao, 2011), CGRs in NWC reported by GE/JAPEIC (As cited in Chopra & Rao, 2011), and CGRs in NWC and HWC obtained in CIR program (Ernestova & Burda, 2009; Jenssen & Stjarnsater, 2009).

and crack length resolution have evolved (Andresen et al., 2001).

3.2.1. CGRs in unirradiated and irradiated stainless steels Many papers (Andresen, 2002; Pathania et al., 2007; Was & Andresen, 2007) compare the CGRs of irradiated stainless steels with unirradiated stainless steels. Increased CGRs in cold and warm work hardened 304L and 316 SS were observed in both BWR and PWR water, with the growth rate at a yield strength of >800 MPa being about an order of magnitude higher than in solution annealed materials. Sensitization is not a necessary condition for IGSCC growth, but it accelerates cracking at high corrosion potential.

Andresen (2002) summarized data that showed radiation hardening and cold work have similar effects on enhancing yield strength and also SCC growth rate. Figure 16 compares the CGRs in cold worked, sensitized and irradiated stainless steels (Was & Andresen, 2007). Unirradiated and irradiated alloys with similar yield strength show similar CGRs at low corrosion potential. The similarity in CGR response of unirradiated and irradiated materials can also be seen in their similarity in K dependence. As reported by Andresen (2002), a $K^{2.3}$ dependence was observed for unirradiated cold worked stainless steel tested over a wide K range (Figure 17), and irradiated stainless steel showed a similar K dependence (Figure 18).

In PWR water, Fukuya, Nakano, Fujii, and Torimaru (2004c) reported that the apparent CGR acquired from SSRT tests in deaerated water increased with increasing dissolved hydrogen, although the enhancement may be

associated with the near-surface (initiation) phase of crack-ing. However, the effect of hydrogen on CGR was less significant in high dose CW316 (50 dpa). PWR-irradiated pipe specimens were used for this study. The crack growth behavior in PWR water needs to be examined with additional CGR testing.

3.2.2. Key factors affecting IASCC growth rate The scatter in the CGR data, whether measured in post-irradiation laboratory tests or in-reactor, is significant. Discerning the effect of key parameters is generally not possible by statistical evaluation of large collections of (scattered) data – a parameter that affects CGR by a factor of 10 or 20 is impossible to identify with high statistical confidence in a set of data exhibiting a 1000X scatter (Andresen, Gott, & Nelson, 1999). Similar to SCC of unirradiated materials, the dependencies are best identified in specific, well-controlled studies, especially studies where changes are made on-the-fly and with no other test perturbations.

3.2.2.1. Specimen size and K -size validity A key factor in testing irradiated materials is the issue of K -size validity. In the first CGR tests on irradiated stainless steel, Andresen et al. (Andresen, 2003; Kumagai et al., 2009) considered the issue of strain softening, whereby the first dislocations to pass through an irradiated structure clear out obstacles, creating channels where subsequent dislocations preferentially moved. The importance of this in testing irradiated materials is not easily predicted, and observations showed that a large increase in CGR occurred at much

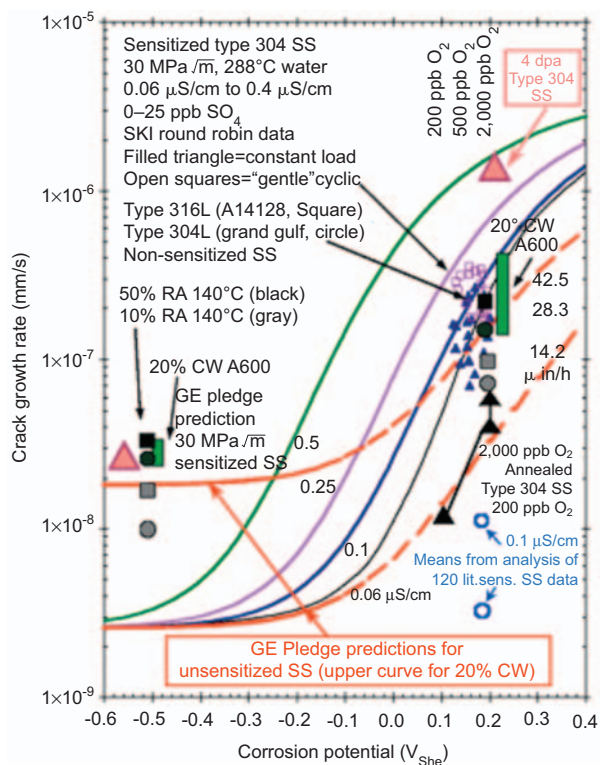


Figure 16 Observed and predicted (GE Pledge) relationships of crack growth rate vs. corrosion potential for furnace-sensitized Type 304 stainless steel at a constant K of $\approx 27.5 \text{ MPa}\sqrt{\text{m}}$. The observed data were obtained in water of conductivity between $0.1 \text{ }\mu\text{S/cm}$ and $0.3 \text{ }\mu\text{S/cm}$. The predicted relationships show the sensitivity of the crack growth rate to changes in combinations of corrosion potential and water purity ($0.1 \text{ }\mu\text{S/cm}$ – $0.5 \text{ }\mu\text{S/cm}$). The large triangles were obtained on Type 304 stainless steel irradiated to 4 dpa.

lower K values than predicted from standard ASTM K -size criteria based on unirradiated materials. They proposed, and in loose collaboration with Jenssen et al. (2003) modified, a criteria that discounted the effect of irradiation hardening. In particular, the recommended effective yield

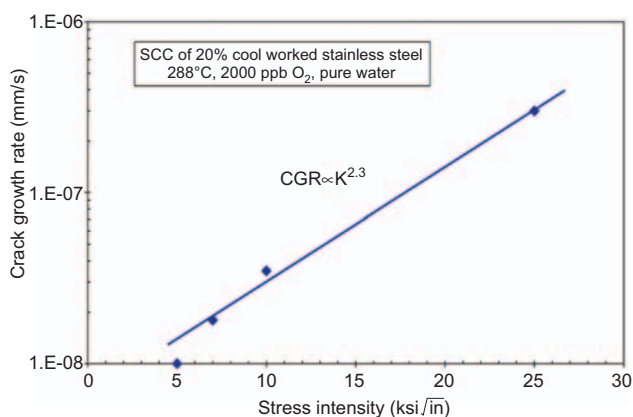


Figure 17 SCC growth rate vs. stress intensity factor for 20% cold worked 316L stainless steel tested in 288°C pure water containing 2 ppm O_2 .

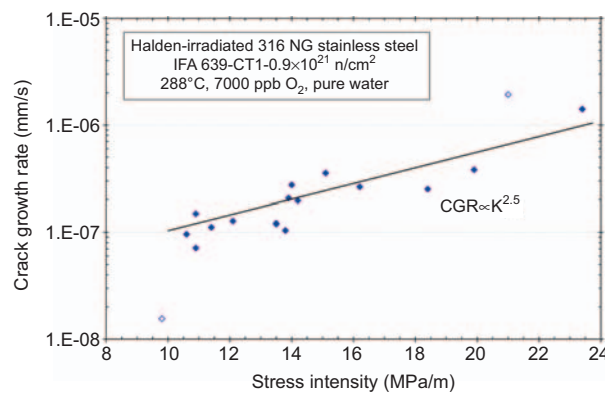


Figure 18 SCC growth rate vs. stress intensity factor for irradiated 316NG stainless steel tested in 288°C pure water containing 7 ppm O_2 .

strength used in K -size criteria is the unirradiated yield strength plus one-third of the increase in yield strength from irradiation hardening. For a 288°C yield strength of 200 MPa , and an irradiated yield strength of 800 MPa (an increase of 600 MPa), the effective yield strength would be $200 + 600/3 = 400 \text{ MPa}$.

The origin of the observed increase in CGR at lower K than expected is unclear. Nakamura et al. (2007) argued that an effective yield strength should not be used to calculate K validity criterion for 3.5 dpa irradiated 304 SS – rather, they argued for the use of the measured yield strength. The FEM calculations of Sumiya et al. (2007) indicated that the plastic zone in an irradiated specimen was smaller than that in an unirradiated specimen and it increases with K , as expected. As the plastic zone size increases and extends to a greater extent on the edges of the specimen, strain softening and dislocation channeling suggest that the plastic zone will rapidly grow. However, ultimately, K -size validity is not primarily about yield strength or strain softening, it is about similitude – that is, the CGR remains well-behaved as the specimen size is changed. Extensive data on irradiated materials indicate that

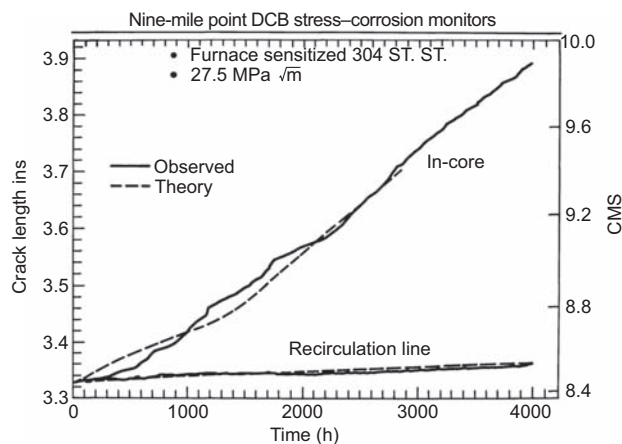


Figure 19 Crack length vs. time for sensitized type 304 stainless steel, wedge-loaded DCB specimens tested at the Nine Mile Point 1 boiling water reactor.

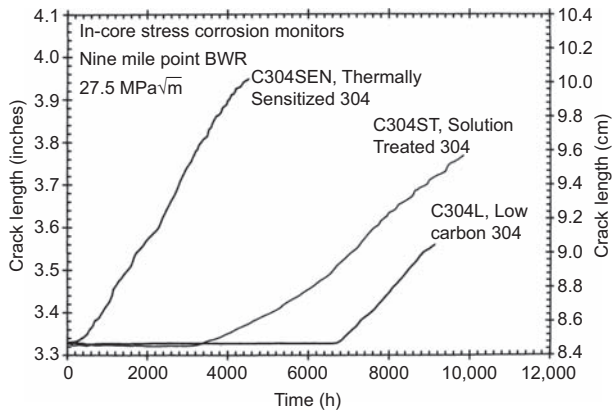


Figure 20 Crack length vs. time for sensitized and unsensitized type 304 and 304L stainless steel, wedge-loaded DCB specimens tested at the Nine Mile Point 1 boiling water reactor.

a very large increase in CGR occurs in a narrow range of K values that is well below that estimated from the measured yield strength.

3.2.2.2. Effect of irradiation dose Double cantilever beam (DCB) specimens were exposed in the recirculation system and in-core at a BWR plant, and the evolution in CGR was observed as dose was accumulated. For sensitized stainless steel, the in-core specimen grew faster (Figure 19), consistent with the more oxidizing conditions from radiolysis.

While some differences in the dose at which cracking was first observed varied for 304 vs. 304L stainless steel (Figure 20), the observed growth rates were essentially identical vs. fluence once cracking occurred. The evolution in CGR vs. fluence was evaluated, which led to algorithms to account for the effect of fluence on CGR (Andresen & Ford, 1995; Andresen et al. 1990, 1996). These actively-monitored SCC specimens covered a wider range of fluence than any other in-reactor test. The CGR evolution vs. fluence is consistent with the known effects of RIS and radiation hardening, which begin to have a notable effect at ~ 0.1 dpa and substantially saturate at several dpa.

As shown in Figure 15, the CGR of solution annealed AS heat 304L SS in NWC slightly increased with increasing irradiation dose in the range of 5.5–47.5 dpa. The CGR are a factor of 10X above the disposition line NUREG-0313. However, no obvious trend was observed in HWC because of the data scatter. Chopra and Rao (2011) reported a dose level of 0.5 dpa for NWC and 3 dpa for HWC, below which the CGRs did not exceed NUREG-0313.

3.2.2.3. Effect of irradiation source In the CIR program, limited data suggested that CGRs of commercial 304L SS in NWC did not vary between BWR irradiation and fast reactor irradiation. Was et al. (2002) observed very similar SCC behavior in proton irradiation as had been observed for LWR neutron irradiated materials. Bruemmer, Cole, Carter, and Was, (1993), Bruemmer et al., (1999) observed that the individual radiation damage

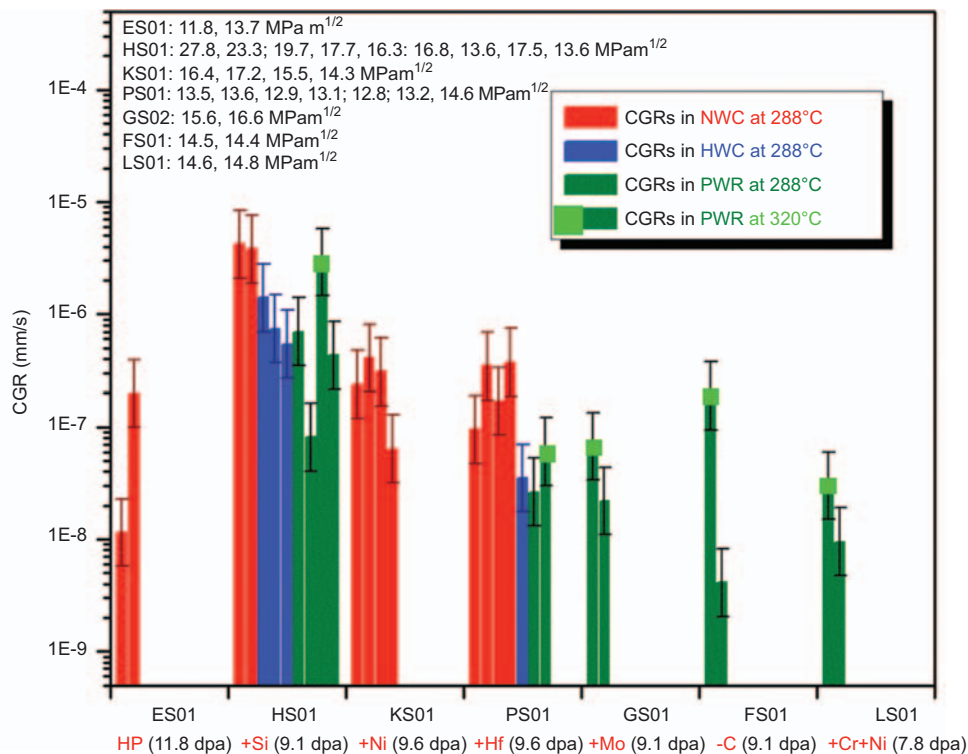


Figure 21 CGR as a function of solute addition for the solute addition alloys tested at 288°C and 320°C by NRI and SCK-CEN in CIR program.

phenomena – especially RIS, hardening and creep relaxation – were similar for various types of radiation. One must account for the effect of radiation dose rate, which is generally done by increasing temperature when irradiating a higher dose rate.

A fourth major factor is radiolysis of water, which is important only in tests involving an active radiation flux. Radiolysis is addressed in other sections.

3.2.2.4. Effect of solute addition Figure 21 shows the effect of solute addition on CGR. Seven alloys were tested at K between 11 and 28 MPa $\sqrt{m}$. Compared with the high purity 304 SS without solute addition (ES01), the addition of silicon (HS01, Si: 1.05 wt%, 9.1 dpa) significantly promoted crack growth in HWC and PWR water. The CGRs were not directly comparable in NWC because of the difference in K . Miwa et al. (1996) reported lower radiation hardening in HP+Si (0.69 wt%) than in HP alloy. It is believed that the segregation of silicon at the grain boundaries caused the increase in CGR. It has been reported (Li, Kaneshima, & Shoji, 2000) that CGR of solution annealed unirradiated stainless steels in PWR water at 325°C increased significantly with silicon content in the range of 0.5–2.7 wt%.

Besides silicon, the addition of chromium and nickel improved the resistance to IASCC in PWR water compared to additions of molybdenum or hafnium. Not enough data exist to compare the CGRs for other element additions.

3.2.2.5. Effect of chromium depletion Grain boundary Cr depletion has been recognized as deleterious to corrosion and stress corrosion resistance for decades. Like cold work, sensitization affects the SCC susceptibility before neutron fluence has accumulated, and there is an evolution in response during irradiation. Unlike cold work, where the characteristic microstructure is converted to a purely irradiated microstructure after ~5 dpa, irradiation enhances the extent of chromium depletion but does not eliminate the prior, wider chromium profile from thermal sensitization. In

most cases, thermal sensitization results in a 30–70 nm width chromium depletion profile near grain boundaries, while radiation segregation at <350°C produces a 3–5 nm wide chromium depletion profile, which therefore adds to, rather than substitutes, for the pre-existing chromium depletion profile.

Andresen (1994) and Brummer, Arey, and Charlot (1994) separately evaluated the effect of very narrow chromium depletion profiles using multi-step heat treatments that nucleated grain boundary carbides, then eliminated the (wide) chromium depletion profile (by heat treating near the Cr-Cr₂₃C₆ equilibrium temperature of ~950°C for several hours), then water quenching and heat treating at 400–450°C. This produced a very narrow chromium depletion profile, although also a deep profile, because at ~400°C the chromium, in equilibrium with the growing carbide, is only ~5%. However, in high temperature water, SCC responded to the minimum in chromium in the grain boundary (chromium diffuses rapidly along the grain boundary, and so the entire grain boundary is close to the chromium content at the interface of the growing carbide). This is true whether the minimum chromium is 5%, 10%, or 15% – that is, there is no threshold chromium level at which SCC response suddenly changes. In contrast, sensitization tests including electrochemical potentiokinetic repassivation (EPR, whether *single loop* or *double loop*), respond to a threshold chromium concentration in the vicinity of 14–15%, and often are influenced strongly by the width of the chromium profile at this threshold chromium level, not by the depth of the chromium profile. Thus, care must be used when interpreting such measures of sensitization.

The impact of chromium depletion is larger in more aggressive water chemistries, such as BWR water at high corrosion potential (especially when chloride or sulfate is also present), or even deaerated low pH water. There are some effects of chromium depletion in deaerated neutral water (Yonezawa, Iwamura, Fujimoto, & Ajiki, 1999), but it is not as pronounced as in aggressive water chemistry.

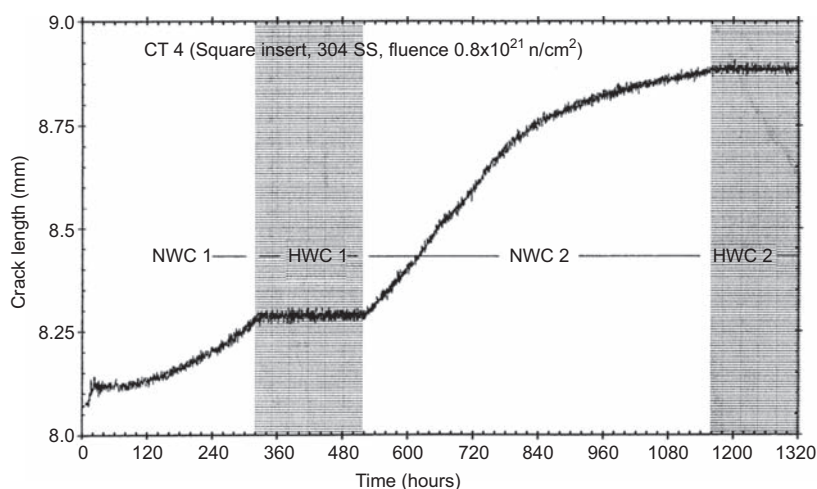


Figure 22 Crack length vs. time for unsensitized type 304 stainless steel, actively-loaded DCB specimens tested at the Halden test reactor.

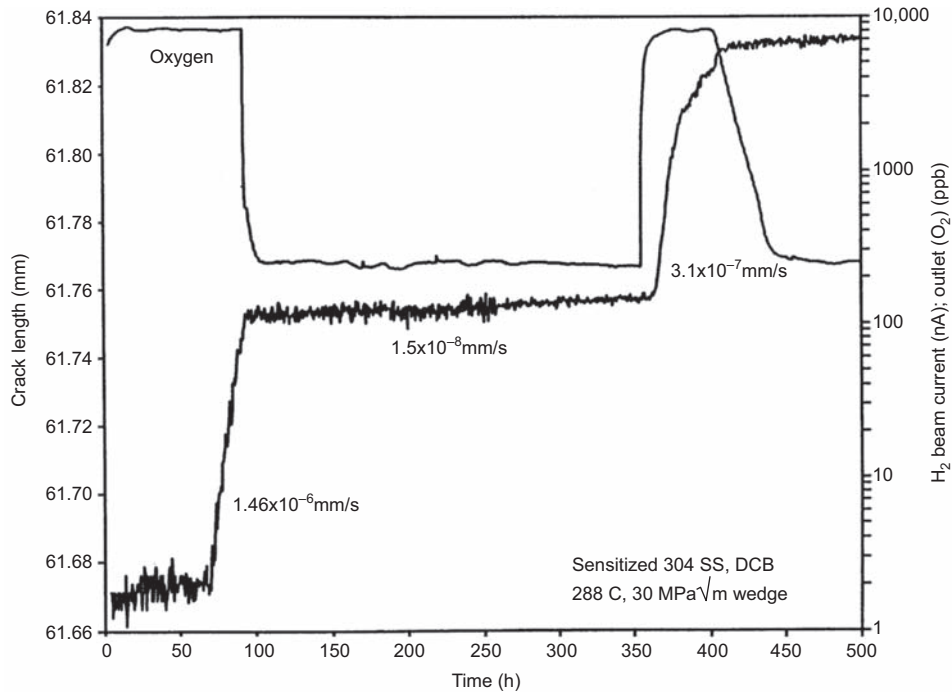


Figure 23 Crack length vs. time for sensitized type 304 stainless steel, actively-loaded DCB specimens tested at the Harwell Variable Energy Cyclotron under proton irradiation.

3.2.2.6. Effect of water chemistry Figure 14 shows an example of an on-the-fly change in corrosion potential or water purity on irradiated SS. Such changes are very compelling because *nothing* else changes but the corrosion potential of the water. There are dozens of similar examples from both in-reactor (Figures 22 and 23) and laboratory (Figures 14 and 24) tests. In most cases, large changes in crack growth rate accompany large changes in corrosion potential, i.e.,

associated with changes from 2 ppm oxygen ($\sim 0.15\text{--}0.20 V_{\text{she}}$) to 0.02–0.20 ppm hydrogen ($\sim -0.50\text{--}0.54 V_{\text{she}}$). However, with increasing fluence, sometimes starting as low as 1 dpa, some materials exhibit little or no change in CGR with decreasing corrosion potential. This is aggravated by testing at high K (Figure 25; Toloczko, Andresen, & Brummer, 2007) and by high silicon content (Figure 26) (Andresen & Morra, 2005a). Fortunately, radiation creep relaxes constant displacement

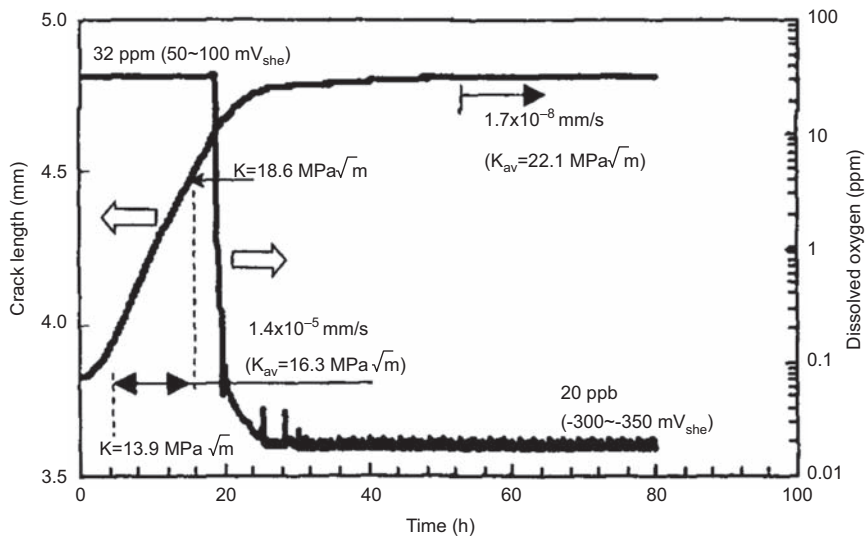


Figure 24 Crack length vs. time for a small round CT of type 304L stainless steel irradiated to $2 \times 10^{21} \text{ n/cm}^2$ and tested in 288°C water showing a strong effect of corrosion potential.

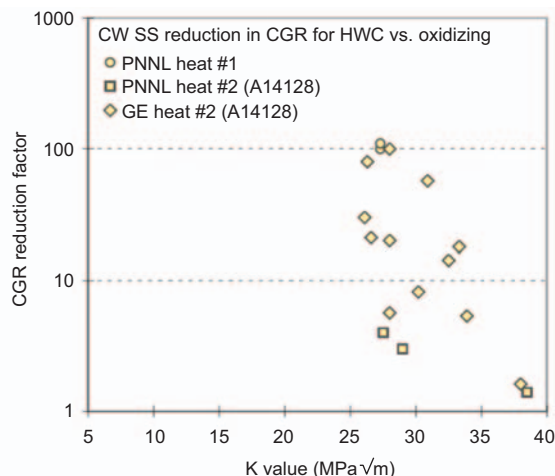


Figure 25 Factor of improvement (crack growth rate reduction factor) vs. stress intensity factor for two heats and several tests in 288°C water.

stresses (e.g., in welds and bolts) and thus the combination of high K and high fluence is not as common as it might be. However, swelling of B_4C in control rods and differential swelling of baffle former plates and bolts can produce straining and stressing with increasing fluence.

The effects of water purity are not always as pronounced as believed, and its effect varies with corrosion potential, chromium depletion and loading, etc. In fact, some data obtained in supposedly “pure water” had an outlet conductivity from the autoclave system that exceeded $1 \mu S/cm$, when the average conductivity in BWRs has been $<0.11 \mu S/cm$ for nearly 20 years. However, numerous investigators have found that the use of 5–20 ppb sulfate or chloride provides much

more reproducible and well-behaved data than pure water, while biasing the CGR of high alloy (e.g., $>12\%$ chromium) materials by less than 2–3X.

3.2.2.7. Effect of temperature In unirradiated testing, the effect of temperature on the CGR of stainless steels (~ 90 kJ/mole) is lower than for nickel alloys (~ 130 kJ/mole). This translates to a change in CGR between 290 and 340°C of $\sim 4.8X$ vs. $\sim 9.2X$, respectively. The available data on irradiated stainless steels broadly corroborates this, e.g., Figure 21, where CGRs in solute addition alloys, HS01, PS01, GS02, FS01 and LS01 were higher at 320°C than at 288°C.

3.2.2.8. Effect of stress intensity factor K Figure 27 shows the dependence of CGR on K of solute additions alloys tested in NWC, HWC, and PWR water. The CGR increased from 1×10^{-8} mm/s to 4×10^{-6} mm/s in the K range of 12–28 $MPa\sqrt{m}$. In overall sets of data, the effect of K on CGR is clouded by the scatter, as well as violations in the K -size validity recommended by Andresen (2003) and Jenssen et al. (2003). In specific data sets, the effect of K is very clearly observed (Was et al., 2010). An analysis of laboratory (Figure 17) and in-reactor data from the Halden test (Figure 18) reactor showed a dependence of $\sim K^{2.2}$.

3.2.2.9. Effect of cold work Crack growth has been observed in annealed materials, but it increases with cold work, typically by a factor of 10–20X (Figure 28), both in aerated and deaerated water. The response tends to saturate with increase yield strength, so the difference between, for example, 800 and 1000 MPa, is not as consequential as between 200 and 400 MPa. The increase in crack growth rate makes it much easier to perform crack

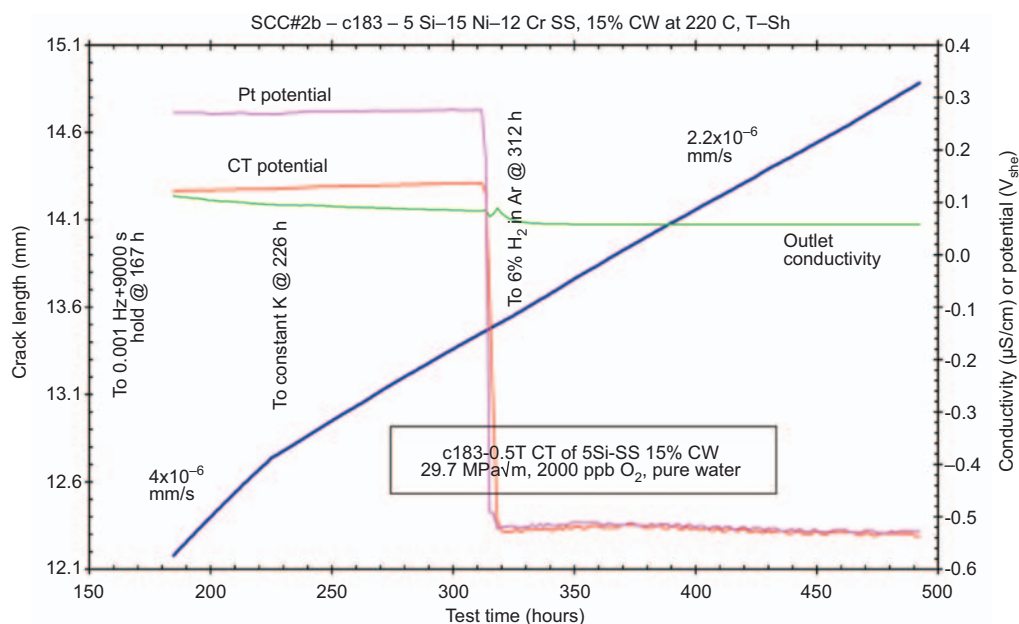


Figure 26 SCC response for an unsensitized model “stainless steel” with 5% Si and 15% cold work at 220°C and tested in 288°C water.

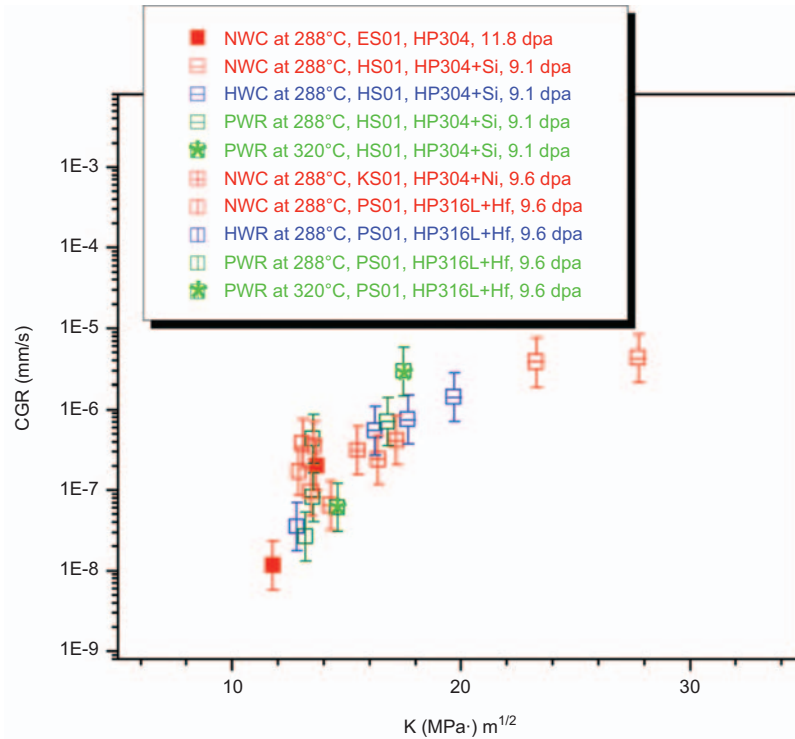


Figure 27 The K dependence of solute addition alloys tested in NWC, HWC, and PWR water (CIR).

growth rate experiments, so observations of crack growth are much more common at elevated yield strength. While few materials are used in the cold worked condition, it is common for cold work to exist in the surface layers of many components. More importantly, weld residual *strains* exist (along with weld residual stresses) in the heat affected zone of weld, and the equivalent room temperature tensile strain is often in the range of 20% and sometimes as high as 30%. Unsurprisingly, most SCC in stainless steels occurs in the heat affected zone near the weld fusion line, where the residual strain is highest (Andresen, 2010b; Angelu et al., 1999, 2000).

Another source of increased yield strength is irradiation, which begins to enhance the yield strength of austenitic materials at ~0.2 dpa, and it saturates at ~5 dpa. The mechanism of hardening is quite different than precipitation hardening or the dislocation structures that form during cold work. The very small vacancy and interstitial loops that form during irradiation impede dislocation motion, but once the first few dislocations move through the structure, subsequent dislocations move through the channel readily. The reason these different mechanisms of hardening all produce a similar effect on crack growth rate is attributed to the size of the plastic zone (smaller in high yield strength materials) and the resultant steeper strain profile at the crack tip.

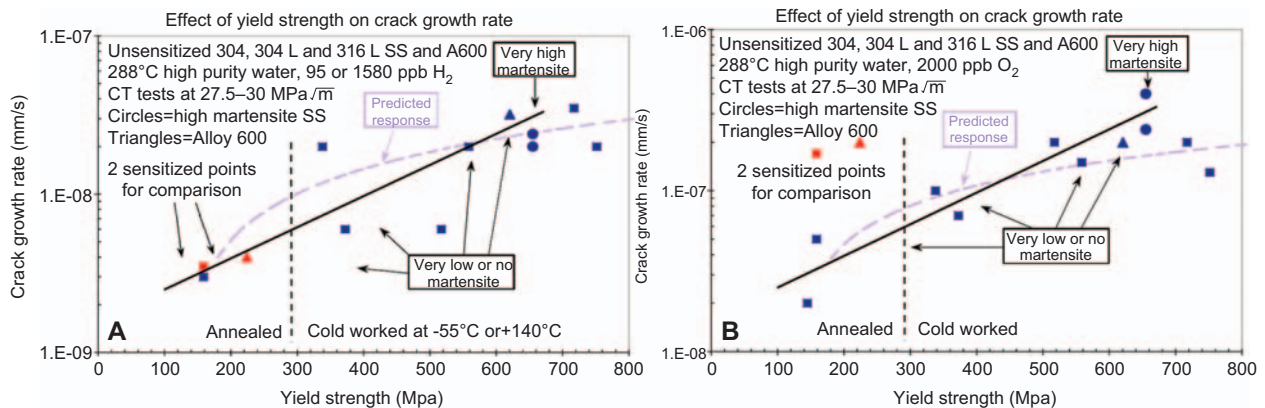


Figure 28 Effect of yield strength and martensite on the stress corrosion crack growth rate on stainless steel and alloy 600 in 288°C, high purity water (<0.10 μS/cm outlet) at (A) low and (B) high potential.

As the crack advances, more redistribution (which occurs by dislocation motion) occurs when the strain profiles are steep, and thus more strain impinges on the crack tip. As noted elsewhere, evidence points to the underlying importance of passivity and dynamic strain on environmental crack advance.

The effect of cold work during subsequent irradiation is complicated by the conversion of the cold worked microstructure to a purely irradiated microstructure after ~ 5 dpa. That is, after ~ 5 dpa, annealed and cold worked materials exhibit a similar microstructure. It is unclear how best to handle cold worked materials between ~ 0.5 and ~ 5 dpa, where there is normally significant irradiation hardening (in annealed materials) and the prior history of the cold work structure is slowly being converted to a purely irradiated structure. Perhaps the simplest approach is to assume the yield strength slowly increases from its value in the initial cold worked state to the irradiated yield strength, which tends to saturate after ~ 5 dpa.

4. Effects of irradiation

IASCC can be categorized into radiation effects on the water chemistry (radiolysis) and on the material, and the accepted definition of IASCC encompasses cases where either factor contributes (low fluence materials tested in water undergoing radiolysis, or pre-irradiated materials tested without an active radiation flux). Radiation dose rate in Rads/h is often used in radiolysis, and in neutrons/cm² (n/cm²) or displacements per atom (dpa) in materials. In light water reactors (LWR) within or near the core, 1 dpa corresponds to $\sim 7 \times 10^{20}$ n/cm² when counting neutrons with $E > 1$ MeV, or $\sim 1.5 \times 10^{21}$ n/cm² for $E > 0.1$ MeV.

The primary effects of radiation on materials (Andresen & Ford, 1995; Andresen et al., 1990; Bruemmer et al., 1999; Ford & Andresen, 1994; Hanninen & Aho-Mantila, 1988; Jacobs & Wozadlo, 1985; Scott, 1994; Was & Andresen, 1992a, 2007) include microcompositional effects (grain boundary chemistry) and microstructural changes (formation of dislocation loops, voids, precipitates and the resulting changes in hardening and deformation mode). In terms of their effect on IASCC, the primary effects of radiation are:

- *Radiolysis of water*, in which a variety of short- and long-lived radicals and species are produced. There is no evidence that the specific species formed are important in IASCC, and indeed their effect on cracking appears to be fully captured by their overall effect on the corrosion potential of the material.
- RIS, which produces an enrichment in some species (e.g., nickel and silicon) at grain boundaries and other defect sinks, and a depletion in other species (e.g., chromium). Even though the distance over which RIS occurs is very limited (a few nm), studies of unirradiated materials have shown that the narrow profiles can affect SCC (Andresen, 1994; Bruemmer et al., 1994).
- *Radiation hardening (RH) and localized deformation (LD)*, which results from radiation damage and the creation of vacancy and interstitial loops, which impede dislocation motion. Once a few dislocations move along a given slip plane, they clear the “channel” of most of these

barriers, and subsequent dislocation occurs primarily in these channels, resulting in localized deformation.

- *Radiation creep relaxation*, which reduces constant displacement stresses, such as in bolts or associated with weld residual stress. During active irradiation, radiation creep can promote dynamic strain, and thereby SCC.
- *Swelling*, which occurs to a limited extent at temperatures between ~ 300 and 360°C , but can be sufficient to produce re-loading of components such as PWR baffle former bolts. Swelling occurs differently in different materials, and is delayed in cold worked materials. Stresses owing to swelling are balanced by radiation creep relaxation, but the resulting stress can be sufficient to cause IASCC.
- *Other microstructural changes*, such as precipitation or dissolution of phases in materials. While there is no clear evidence that such changes affect IASCC response, this may only reflect the limited characterization and IASCC studies that has been performed on high fluence materials.
- *Transmutation* of various elements in the alloy result in the production of helium and hydrogen. There is no evidence that either transmutant affects IASCC directly, but rather indirectly through changes in the microstructure or the hardening and fracture behavior of the alloy.

The individual (segregation, hardening and creep, Figure 29) and composite (SCC, Figures 5–9) effects of radiation increase with dose in much the same manner, which makes the isolation of, and attribution to, individual contributions difficult. The dislocation loop microstructure is closely tied to radiation hardening and both increase with dose until saturation occurs by ~ 5 dpa. RIS also increases with dose and tends to saturate by ~ 5 dpa. Although dependent on both metallurgical and environmental parameters, IASCC generally occurs at doses between 0.5 dpa (for BWRs) and 2–3 dpa (for PWRs),

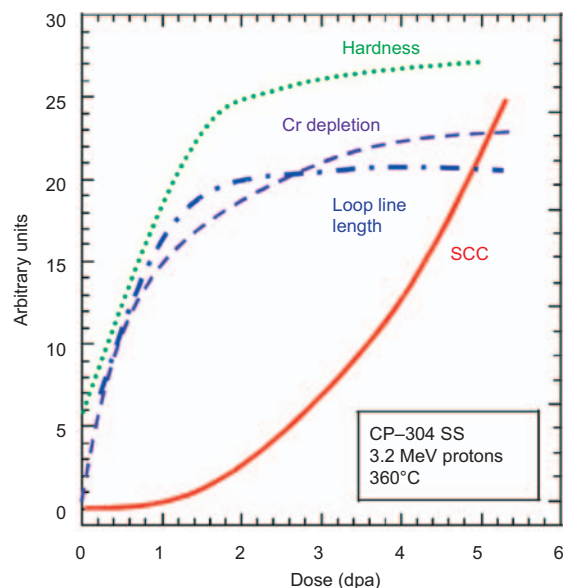


Figure 29 Schematic diagram showing the dose dependence of key irradiated microstructure features (RIS and dislocation microstructure) and radiation hardening along with SCC susceptibility.

which encompasses the steeply rising portion of the curves in Figure 4 that describe the changes in materials properties with irradiation. This section will focus on the effects of irradiation on the water chemistry and on the material. Section 5 will then investigate working hypotheses for the mechanism(s) of IASCC, incorporating many of the effects discussed next.

4.1. Water chemistry

4.1.1. Radiolysis and its effect on corrosion potential Corrosion potential, not oxidant and reductant concentrations *per se*, control SCC susceptibility (Andresen & Young, 1995; Andresen et al., 1990; Ford & Andresen, 1994). The lower hydrogen concentration in BWRs (typically 10–40 ppb H_2) distinguishes BWRs from PWRs (typically 3 ppm H_2) because it permits the radiolytic formation of oxidants. Above ≈ 500 ppb (5.6 cc/kg) H_2 , radiolytic formation of oxidants is effectively suppressed and the corrosion potential remains close to its thermodynamic minimum (which is a function of temperature, H_2 fugacity, and pH). BWRs cannot achieve this H_2 level in the core because H_2 partitions to the steam phase, which begins to form about a quarter of the way up the fuel rods. Thus, this section is primarily relevant to BWRs, although there is some thought to reducing H_2 in PWRs, which would be dangerous if radiolysis were to occur.

When considering the effect of radiolysis and oxidizing species, many sequential and non-linear dependencies must be considered, e.g., radiation flux produces oxidizing and reducing species, the corrosion potential is controlled by multiple reactions involving these species, the crack chemistry is non-linearly dependent on the corrosion potential and the aqueous ions present, and the SCC growth rate is a non-linear function of the crack chemistry (Andresen & Young, 1995; Ford & Andresen, 1994). The relationship between radiation flux and SCC cannot easily be determined empirically, but rather requires a fundamental understanding of each sub-process.

Ionizing radiation decomposes water into a variety of primary species (Burns & Moore, 1976; Cohen, 1969; Lin, 1986), including radicals (e.g., e_{aq}^- , H , OH , HO_2) and molecules (e.g., H_2O_2 , H_2 , O_2), which can be oxidizing (e.g., O_2 , H_2O_2 , HO_2) or reducing (e.g., e_{aq}^- , H , H_2). The predominant species that are stable after a few seconds are H_2O_2 and H_2 , with O_2 forming primarily from the decomposition of H_2O_2 . Because H_2 partitions to the steam phase and H_2O_2 is not volatile, the $\approx 87\%$ of the water that is recirculated in a BWR (≈ 11 – 14% of the core flow becomes steam) is oxidant rich. H_2 is introduced in the feed water, which mixes with the recirculated water near the top of the annulus (the region of down-flow between the core shroud and pressure vessel). So the annulus, recirculation piping, lower plenum (below the core) and lower part of the core are H_2 -rich.

The concentrations of radiolytic species are roughly proportional to the square root of the radiation flux in pure water. The radiation energy-intensity spectrum influences the concentration of each radiolytic species, which is described in terms of a yield, or G value, which is the number of molecules produced per 100 eV absorbed by water. In LWRs, the

G values for most species are within a factor of ≈ 3 for fast neutron vs. gamma radiation; however, fast neutrons have a much bigger impact than gamma radiation primarily because the energy deposition rate, or mean linear energy transfer (LET), is greater [40 eV/nm for fast neutrons vs. 0.01 eV/nm for gamma radiation (Burns & Moore, 1976)]. Also, the neutron flux in LWRs (e.g., $\approx 1.03 \times 10^9$ Rad/h core average and $\approx 1.68 \times 10^9$ Rad/h peak in a BWR4 of 51 W/cc power density) is also higher than the gamma flux ($\approx 0.34 \times 10^9$ Rad/h). The contribution of thermal neutrons and beta particles to radiolysis in LWRs is small.

Gamma radiation in a certain flux range can be beneficial. The intermediate gamma level in the downcomer of the annulus of a BWR core promotes recombination of hydrogen and various oxidants (Lin, 1986; *Proc. Water Chemistry of Nuclear Reactor Systems, No.5*, October 23–27, 1989). BWRs that have a wider annulus and lower gamma flux near the pressure vessel respond less to a given H_2 addition; this is a key element in the effectiveness of HWC.

The integrated effects of various oxidants and reductants on environmental cracking is best described via changes in corrosion potential, which controls the thermodynamics and influences the kinetics of most reactions. Electrochemical potentials are logarithmically dependent on local oxidant, reductant and ionic concentrations via the Nernst relationship, $\phi = \phi_o + RT/nF \ln(\text{products/reactants})$, where ϕ is electrochemical potential, R is the gas constant, T is temperature in Kelvin, n is the number of moles and F is Faraday's constant. Radiation may increase the concentration of various species by many orders of magnitude but have comparatively small effects on the corrosion potential in hot water. Further, corrosion potentials are mixed potentials involving a balance of anodic and cathodic reactions on the metal surface, and radiolysis produces a stoichiometric balance of oxidizing and reducing species. At low oxidant concentrations, the rapid drop in corrosion potential to $\approx -0.5 V_{she}$ (Figure 30A) results from mass transport limited kinetics of oxidants to the metal surface.

The effect of radiation type and flux on the relationship between dissolved oxygen and corrosion potential in hot water is shown in Figure 30A, in which the connected points represent data obtained in controlled, radiation on/off experiments. The data from connected points are shown in Figure 30B in terms of a radiation-induced shift in potential. The curves in Figure 30A represent the scatter band for the data obtained under unirradiated conditions. Similar scatter also exists in the irradiated corrosion potential data in Figure 30A and is comprised of contributions from both real effects and experimental error. High radiation flux experiments were performed by Taylor (1990) and Andresen et al. (1996) using multiple fundamentally different types of radiation-hardened reference electrodes. Indig (Gordon, 1983; Head, Indig, & Andresen, 1989) employed multiple, radiation-qualified silver chloride reference electrodes to measure the corrosion potential in-reactor.

In high temperature water, little elevation in corrosion potential results from irradiation sources that do not include neutrons or simulate their contribution, e.g., using high

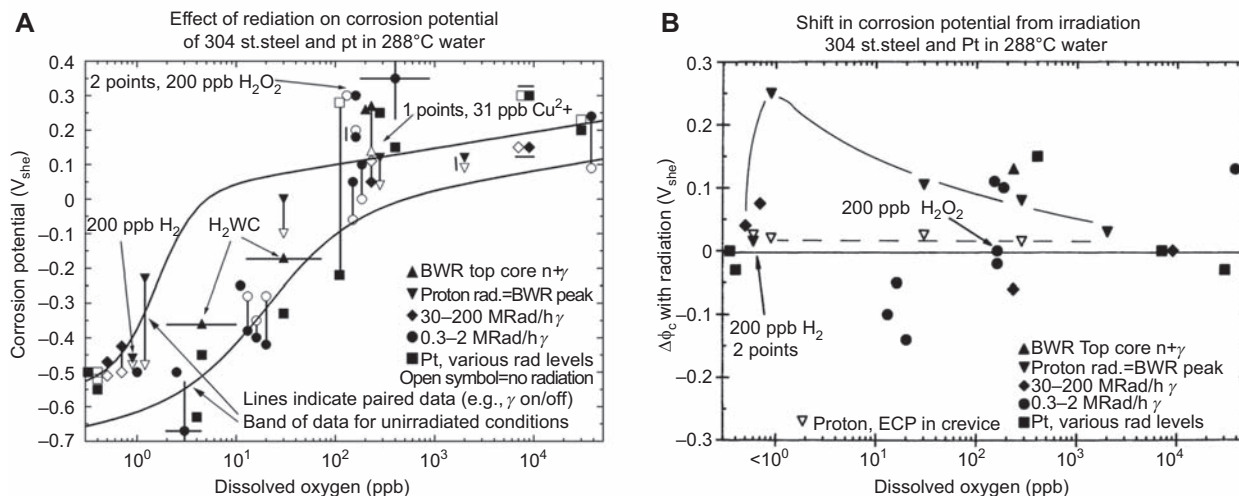


Figure 30 (A) Effect of radiation on the corrosion potential of type 304 stainless steel in 288°C water. The curves denote the range of typical values in the unirradiated corrosion potential data. (B) Effect of radiation on the shift in corrosion potential from the value under unirradiated conditions for type 304 stainless steel in 288°C water. With the exception of the BWR measurements, all data were obtained in controlled radiation on/off experiments. Curves in (B) show the trends in the proton irradiated data, where the effects of radiation (on/off and over a range of fluxes) were evaluated for a variety of dissolved O_2 and H_2 concentrations under otherwise identical conditions.

energy protons (Andresen et al., 1996) (Figures 31A and B). Some studies using gamma radiation (Andresen et al., 1990; Taylor, 1990) showed a significant decrease in corrosion potential, especially in the intermediate (e.g., 10–200 ppb) range of dissolved oxygen. This is consistent with enhanced recombination of oxidizing and reducing species, as occurs in the downcomer region of BWRs (*Proc. Water Chemistry of Nuclear Reactor Systems*, No.5, October 23–27, 1989). In instances where neutrons or protons have been used, a consistent, significant elevation in corrosion potential is observed. This is more pronounced at low dissolved oxygen concentrations with no dissolved hydrogen (Figure 30B), where increases of over +0.25 V occur. At higher inlet oxygen concentrations (e.g., ≈ 200 ppb), the data still show a shift (typically +0.1–0.15 V) in corrosion potential for radiation conditions representative of peak LWR core fluxes (Figure 30B). Less increase is observed for higher oxygen concentrations associated, e.g., with air saturation (≈ 8.8 ppm O_2) or oxygen saturation (≈ 42 ppm O_2 at STP). A similar elevation in corrosion potential is observed for additions of H_2O_2 (200 ppb, Figure 30A), which suggests that H_2O_2 may be a major factor in increasing the corrosion potential under irradiated conditions.

BWR in-core measurements show a corrosion potential of $\approx +0.2$ to $+0.25 V_{\text{she}}$ in normal water chemistry (NWC), which can be decreased by >0.5 V by sufficient additions of dissolved hydrogen in a BWR (Gordon, 1983). This is corroborated by other measurements (Andresen et al., 1996) (Figure 30B), which show very little effect of radiation on corrosion potential when water contains a moderate level of dissolved H_2 (e.g., >200 ppb H_2 , 0 ppb O_2). At high H_2 levels, the core becomes reducing, and the small concentration of ^{16}N

(transmuted from ^{16}O) changes from soluble NO_3^- to volatile NO_x and NH_3 , causing a large increase in radiation level in

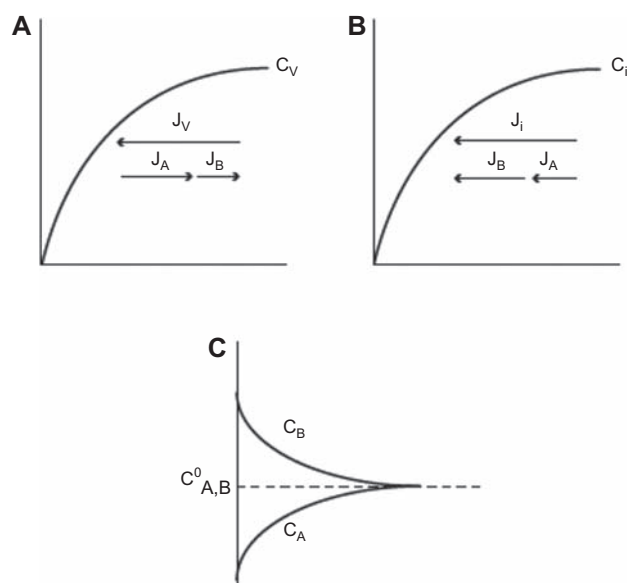


Figure 31 Schematic illustration of RIS in a binary, 50A–50B system showing. A) the development of the vacancy concentration profile by the flow of vacancies to the grain boundary balanced by an equal and opposite flow of A and B atoms, but not necessarily in equal numbers, B) the development of the interstitial concentration profile by the flow of interstitials to the grain boundary balanced by an equal and opposite flow of A and B atoms migrating as interstitials, but not necessarily in equal numbers, and C) the resulting concentration profiles for A and B.

the steam lines and turbine. The half-life of ^{16}N is 7.1 s, so it usually decays in the pressure vessel.

Because residence time affects radiolysis of water, the question of whether the corrosion potential might be higher *within* a crack or crevice (than at the crack mouth) was examined. In the absence of radiation, measurements in high temperature water in artificial crevices (e.g., tubing) (Combrade, Foucault, & Slama, 1985; Taylor & Caramihas, 1983), at the tip of growing cracks (Gabetta & Buzzanaca, 1985), and of short crack growth behavior (Andresen et al., 1990) show that the corrosion potential remains low (i.e., $-0.5 \pm 0.1 V_{\text{she}}$ in 288°C pure water) for all bulk oxygen concentrations, indicating that complete oxygen consumption occurs within the crack. Measurements of radiation effects in crevices (Andresen et al., 1996) show that the elevation in corrosion potential is limited to $<0.05 V$ ($\phi_c < -0.45 V_{\text{she}}$) in-core; this is consistent with interpretation of available corrosion potential data on free surfaces (Andresen & Ford, 1995; Andresen et al., 1990; Ford & Andresen, 1994). These small changes will not significantly affect the $\approx 0.75 V$ ($+0.25 V_{\text{she}}$ minus $-0.5 V_{\text{she}}$) potential difference in the crack under irradiated normal BWR water chemistry conditions. The potential difference, along with other factors, controls the enhancement mechanism that can lead to an increased anion activity and altered pH at the crack tip, and the associated elevation in crack growth rate (Andresen & Ford, 1988; Ford & Andresen, 1988; Ford et al., 1987, 1990).

4.1.2. Effects of corrosion potential on IASCC Jacobs et al. (1986) evaluated pre-irradiated stainless steels using SSRT tests (Jacobs et al., 1986; Ljungberg, personal communication) in hot water containing oxygen and/or H_2O_2 to elevate the corrosion potential and simulate the effect of radiation. At $\approx 3 \times 10^{21} \text{ n/cm}^2$, the amount of intergranular cracking was strongly affected by dissolved oxygen (and, by inference, corrosion potential) (Figure 5B). Ljungberg also evaluated pre-irradiated materials in SSRT tests and observed lower average crack growth rates with decreasing corrosion potential. Crack growth rate tests (discussed later) and other tests confirm that IASCC decreases but does not vanish at low corrosion potential (Figures 5 and 9).

Fracture mechanics specimens were installed in the recirculation piping and core of Nine Mile Point Unit 1 BWR, with continuous crack monitoring using DC potential drop. Identical furnace sensitized type 304 stainless steel specimens installed in both locations showed higher growth rates in the core where the corrosion potential was higher (Figures 19 and 20). Specimens of annealed type 304 and 304L stainless steel showed growth that initiated at different fluences, but the growth rates at a given fluence were identical. The delay time (fluence) and differences in delay time were attributed to the need to transition from a transgranular fatigue precrack to IG SCC.

The first laboratory SCC growth rate tests ever performed on irradiated (4 dpa) type 304 SS are shown in Figures 2 and 14. The SCC response was very well behaved, and the higher rates at high corrosion potential decreased as the corrosion potential was lowered. There is some evidence that

the benefit of corrosion potential and stress intensity factor can decrease at high fluence in some stainless steels; this has been observed in unirradiated alloys possessing high silicon (Andresen & Morra, 2005a; Andresen et al., 2004). The detrimental effect of silicon is likely associated with its ability to oxidize at all LWR-relevant potentials, coupled with its relatively high solubility in hot water – that is, it does not form a protective oxide like Cr_2O_3 or iron/nickel oxides and spinels.

Figure 2 compares these data with other irradiated and unirradiated data, using simultaneous measurements of corrosion potential and crack growth rate in fracture mechanics specimens. Also shown are curves that represent model predictions (Andresen & Ford, 1988, 1989a, 1989b, 1995; Andresen et al., 1990; Ford & Andresen, 1994; Ford et al., 1987, 1990). Clearly the *in situ* data compare favorably with the spectrum of unirradiated data and data obtained on a fracture mechanics specimen of furnace sensitized type 304 stainless steel using high energy proton irradiation to simulate the mix of neutron and gamma radiation present in power reactors.

4.2. Metallurgy and microstructure

4.2.1. RIS RIS describes the redistribution of major alloying elements and impurity elements at point defect sinks (Allen, Busby, Was, & Kenik, 1998; Hindmarsh, 1972; Jenssen et al., 1998; Lam, Kumar, & Wiedersich, 1982; Okamoto & Rehn, 1979; Okamoto & Wiedersich, 1974; Perks, Marwick, & English, 1987; Simonen (personal communication); Was, 2007; Was et al., 1999a; Wiedersich, Okamoto, & Lam, 1979). As IASCC is intergranular, the sinks of greatest interest are the grain boundaries. RIS is driven by the flux of radiation-produced defects to sinks, and is therefore fundamentally different from thermal segregation or elemental depletion because of grain boundary precipitation. Vacancies and interstitials are the basic defects produced by irradiation and can reach concentrations that are orders of magnitude higher than those at thermal equilibrium, resulting in large increases in diffusion rates. If the relative participation of alloying elements in the defect fluxes is not the same as their relative concentration in the alloy, then a net transport of the constituents to or from the grain boundary will occur (Figure 31). This unequal participation of solutes in the vacancy and/or interstitial fluxes to sinks results in either enrichment or depletion of an alloying element at the grain boundary. The species that diffuse more slowly by the vacancy diffusion mechanism are enriched and the faster diffusers become depleted.

Enrichment and depletion can also occur by association of the solute with the interstitial flux. The undersized species will enrich and the oversized species will deplete (Okamoto & Wiedersich, 1974). The magnitude of the build-up/depletion is dependent upon several factors, such as whether a constituent migrates more rapidly by one defect mechanism or another, the binding energy between solutes and defects, the dose, dose rate and the temperature. RIS profiles are also characterized by their narrowness, often confined to within

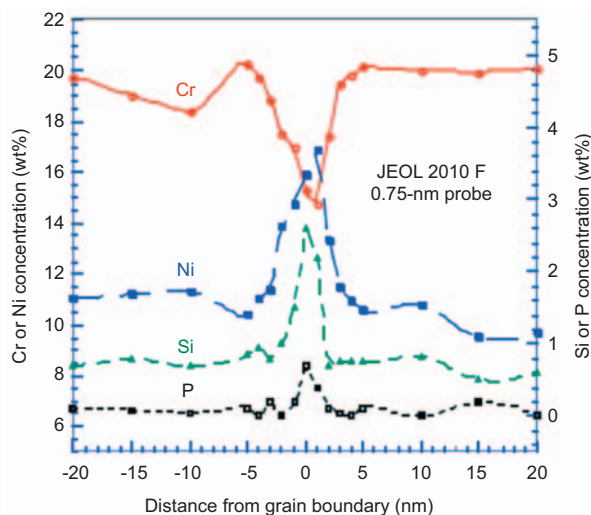


Figure 32 Compositional profiles across grain boundaries obtained by dedicated STEM from a low strain, high purity 348 stainless steel swelling tube specimen irradiated to 3.4×10^{21} n/cm² at 288°C in a BWR (Jacobs, Clausen, Miller & Shepherd, 1990). Composition profiles were measured using a field emission gun scanning transmission electron microscope (FEGSTEM).

5–10 nm of the grain boundary, as shown in Figure 32 for an irradiated stainless steel.

Segregation is a strong function of irradiation temperature, dose and dose rate (Figure 33). Segregation peaks at intermediate temperatures since a lack of mobility suppresses the process at low temperatures, and back-diffusion of segregants minimize segregation at high temperature (where defect concentrations approach their thermal equilibrium values). For a given dose, a lower dose rate results in a greater amount of segregation. At high dose rates, the high defect population results in increased recombination that reduces the number of defects that are able to diffuse to the grain boundary. Figure 33 shows the interplay between temperature and dose rate for an austenitic stainless steel. RIS occurs in the intermediate temperature range, and this range rises along the temperature scale with increasing dose rate to compensate for the higher recombination rate.

In iron-chromium-nickel alloys, the vacancy exchange (inverse Kirkendall) mechanism successfully explains the observed major element segregation (Allen et al., 1998; Was et al., 1999b). Studies have shown that nickel segregates to grain boundaries while chromium and iron deplete. The directions of segregation are consistent with an atomic volume effect in which the subsized solute migrates preferentially with the interstitial flux, and the oversized solute participates preferentially in the vacancy flux. The results are also consistent with the diffusivity of the solutes in iron-chromium-nickel, in which nickel is the slow diffuser, chromium is the fast diffuser and iron is intermediate. In commercial austenitic stainless steels (Figure 32), chromium depletes at grain boundaries and nickel enriches, while iron can either deplete or enrich according to the magnitude of the diffusion coefficient relative to the other solutes (Was & Allen, 1994).

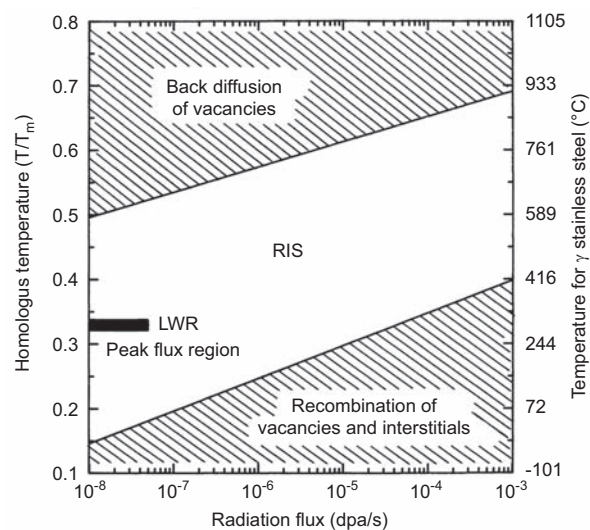


Figure 33 Dependence of RIS on homologous temperature and dose rate for austenitic stainless steels.

RIS increases with neutron dose in LWRs and saturates after several (~ 5) dpa in the 300°C temperature range. Figure 34 shows grain boundary chromium depletion for austenitic stainless steels as a function of dose (Asano, Fukuya, Nakata, & Kodama, 1992; Carter et al., 1994; Jacobs, 1995; Jacobs, Clausen, Miller, & Shepherd, 1990; Jacobs et al., 1993; Kenik, 1992; Nakahigashi et al., 1992; Walmsley, Spellward, Fisher, & Jenssen, 1997; Was et al., 2002). As the slowest diffusing element, nickel, becomes enriched at the grain boundary. As iron depletes in 304 and 316 stainless steels, the nickel enrichment makes up for both chromium and iron depletion and can reach very high levels, up to ~ 30 wt%.

Minor alloying elements and impurities also segregate and have been implicated in the IASCC process. Manganese and molybdenum deplete at the grain boundary under irradiation (Bruegger et al., 1999). Minor alloying or impurity elements such as silicon and phosphorus also segregate under irradiation. Silicon strongly enriches at the grain boundary to as much as 10 times the bulk (0.7–2.0 at%) composition in the alloy (Bruegger, 2002), and can be important in IASCC. Phosphorus is present at much lower concentrations and is only modestly enriched at the grain boundary as a result of irradiation (Bruegger et al., 1999; Jenssen et al., 1998; Jiao & Was, 2011a). Phosphorus tends to segregate to the grain boundary following thermal treatment, which reduces the amount of additional segregation to the grain boundary during irradiation making the contribution caused by irradiation difficult to detect (Bruegger et al., 1999). Undersized solutes such as carbon, boron and nitrogen should also segregate, but there is little evidence of RIS, owing, in part, to the difficulty of measurement. Another potential segregant is helium, produced by the transmutation of ^{10}B . The mobility of helium is low at LWR core temperatures, but the opportunity for accumulation at the grain boundary is increased by segregation of boron to the boundary. Overall, the behavior of these minor elements under irradiation is not well understood.

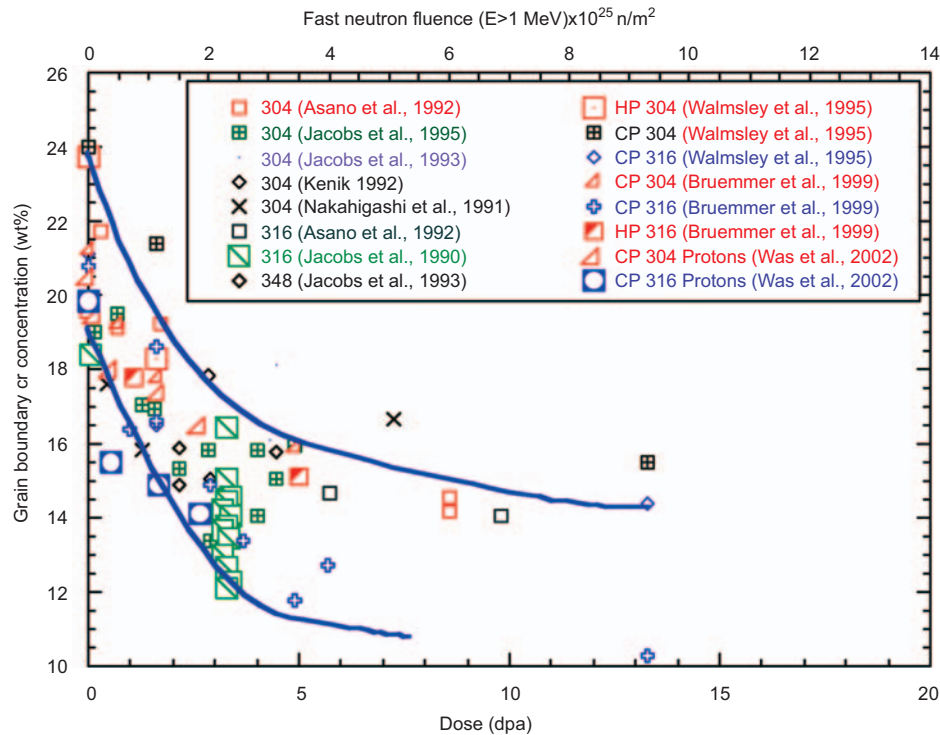


Figure 34 Dose dependence of grain boundary chromium concentration for several 300-series austenitic stainless steels irradiated at a temperature of about 300°C.

Over-size solutes can affect the microchemistry or microstructure of the alloy, thereby altering the IASCC susceptibility. They are believed to affect RIS by acting as vacancy traps, thereby increasing the effective recombination of vacancies and interstitials and thus reducing RIS. Kato, Takahashi, and Izumiya (1992) conducted electron irradiations of several stainless steels at temperatures of 400–500°C up to 10 dpa. Results showed that some solutes (zirconium and hafnium) consistently produced a large suppression of radiation-induced chromium depletion while others resulted in less suppression or suppression at only certain temperatures. Fournier, Sencer, Was, Simonen, and Bruemmer (2003) conducted irradiation of 316 containing hafnium or platinum using three MeV protons (400°C) and 5 MeV nickel ions (500°C). Nickel irradiations showed little effect of the over-size impurity in reducing grain boundary chromium depletion (chromium depletion increased in the case of hafnium), but proton irradiation showed a significant suppression of RIS of chromium at low dose (2.5 dpa) with the effect diminishing at higher (5.0 dpa) dose. In zirconium-doped 304SS there were no consistent results of suppression of grain boundary chromium after 3.2 MeV proton irradiation to 1.0 dpa at 400°C (Gan et al., 2004). Recent work involving neutron irradiations suggested that zirconium had no effect on RIS, even at 0.38 dpa (Allen, Was, Bruemmer, Gian, & Ukai, 2006). Hackett and Was conducted the most comprehensive study of oversized solutes and concluded that size was not the most important issue, but regardless, the benefit was temporary, as higher doses resulted in the resumption of

chromium depletion that was delayed at lower dose (Hackett, Busby, Miller, & Was, 2009a; Hackett, Najafabadi, & Was, 2009). Titanium and niobium similarly produced little change in the grain boundary chromium concentration after irradiation with 3.2 MeV protons to 5.5 dpa at 360°C. Neutron irradiation at very low dose (0.5 dpa) shows a small effect of titanium and niobium on grain boundary chromium (Dumbill & Hanks, 1993). In all, the data on the effect of over-size solutes on RIS of chromium is inconsistent.

4.2.2. Irradiated microstructure The microstructure of austenitic stainless steels under irradiation changes rapidly at LWR service temperatures. Point defect clusters (called “black dot damage” when electron optics are unable to resolve the details) begin to form at very low dose, dislocation loops and network dislocation densities evolve with dose over several dpa, and the possibility exists for the formation and growth of helium-filled bubbles, voids and precipitates in core components in locations exposed to higher dose and temperature (Edwards, Simonen, & Bruemmer, 2003a; Edwards et al., 2002, 2003b; Garner, 1993; Maziasz, 1993; Maziasz & Mchargue, 1987; Zinkle, Maziasz, & Stoller, 1993). Below 300°C, the microstructure is dominated by small clusters and dislocation loops. Near 300°C, the microstructure contains larger faulted loops plus network dislocations from loop unfaulting and cavities at higher doses.

The primary defect structures in LWRs are vacancy and interstitial clusters and Frank dislocation loops. The clusters

are formed during the collapse of the damage cascade associated with primary and secondary atom collisions after interaction with a high energy particle. The larger, faulted dislocation loops nucleate and grow as a result of the high mobility of interstitials. The loop population grows in size and number density until absorption of vacancies and interstitials equalize, at which point the population has saturated. Figure 35 shows the evolution of loop density and loop size as a function of irradiation dose during LWR irradiation at 280°C. Note that saturation of loop number density occurs very quickly, by ~1 dpa, while loop size continues to evolve up to ~5 dpa. The specific number density and size are dependent on irradiation conditions and alloying elements, but the loop size rarely exceeds 20 nm and densities are of the order of $1 \times 10^{23} \text{ m}^{-3}$.

The hardening process and IASCC susceptibility are influenced by small defects. The traditional view that small defect clusters are predominantly faulted interstitial loops and vacancy clusters (Stoller, 1996) may be inaccurate. Analysis of recent post-irradiation annealing experiments by Busby, Kenik, & Was (2005) suggests that there are at least two types of defects with different annealing characteristics; vacancy and interstitial faulted loops, each with different annealing kinetics. The step change in hardness as a function of annealing time suggests that the density of vacancy loops is perhaps much higher than previously believed, and higher than the density of interstitial loops (Busby, Sowa, Was, & Simonen, 2005).

Above 300°C, voids and bubbles may begin to form, aided by the increased mobility of point defects at the higher temperature. The dislocation structure will evolve into a network structure as larger Frank loops unfault. The reduction in the sink strength of the dislocation loops aids in the growth of voids and bubbles. While their size and number density increase with temperature, the dislocation microstructure continues to be the dominant microstructure component over the temperature range expected for LWR components (<350°C).

Irradiation can also accelerate or retard the growth of second phases, modify existing phases or produce new phases. In stainless steels, the principle second phase is chromium carbides, which are relatively stable under irradiation. A key factor in phase formation in austenitic stainless steels under LWR operating conditions is RIS, which can induce the formation of phases by exceeding the local solubility limit. Precipitation of γ' was observed in cold worked 316 stainless steel baffle bolts irradiated to 7 dpa at 299°C (Edwards, Simonen, & Bruemmer, 2007). Gamma prime is a coherent precipitate that can significantly strengthen the matrix and has the potential to alter the deformation behavior in the unirradiated and irradiated conditions. Was et al. (Busby, Kenik, & Was, in press) irradiated a high purity stainless steel containing 1 wt% silicon with 3.2 MeV protons to 5.5 dpa at 360°C, and observed the formation of γ' (Ni_3Si) in the matrix. In addition to γ' , G phase (nickel/silicon rich) is frequently observed in irradiated stainless steel alloys (Edwards & Bruemmer, 2008). In high strength nickel-base alloys, second phases can undergo several types of transformations: γ' can dissolve, γ

can dissolve and re-precipitate, and Laves phase can become amorphous.

Oversize solutes can also affect the irradiated microstructure by mechanisms similar to RIS. Proton and nickel ion irradiations show that the addition of hafnium to a 316SS base alloy increased loop density, decreased loop size and eliminated voids (Fournier et al., 2003). Platinum addition to 316SS resulted in no change in loop density and a small increase in loop size, but increased void size and density. The good agreement between proton and nickel ion irradiation results indicates that the major effect of the oversized solute is not caused by the cascade (where there are large differences between proton and nickel ion irradiation), but rather is caused by the post-cascade defect partitioning to the microstructure evolution. Electron irradiation experiments by Watanabe, Muroga, and Yoshida (1996) and proton irradiation experiments by Was et al. (Busby et al., in press) showed that stainless steel with titanium additions had slightly lower dislocation loop densities and larger sizes compared to the base alloy. Niobium addition only increased loop size. In contrast to the base alloy, neither the titanium- nor the niobium-doped alloys formed voids under the conditions tested. Zirconium addition to 304 SS resulted in reduced hardness, decreased loop density and no change in loop size in proton irradiation to 1.0 dpa at 400°C and compared to the base alloy (Hackett et al., 2009b). Zirconium-containing samples also had a lower void density with no change in void size compared with the base alloy.

4.2.3. Swelling and creep Irradiation can also result in the creation of stress and enhanced plastic flow through processes such as swelling, stress relaxation and creep. While swelling is more prevalent at high doses (>30 dpa) and high temperatures (>350°C), there are locations in LWRs, such as the baffle bolt and baffle former plates, that experience high temperatures (from gamma heating) and high doses. Further, the lower dose rate compared with fast reactors enhances the nucleation of voids and the production of helium through transmutation reactions that can stabilize void embryos. Thus, swelling can occur in core internal components in LWRs. The result of swelling is an increase in volume of the component, which translates into an applied stress. For example, swelling of baffle former plates bolted together with baffle former bolts will produce a tensile stress along the axis of the bolt. Circumferential cracking in baffle former bolts has been observed at the intersection of the head and shank, and the sources of stress may well be swelling of the baffle plates.

At LWR temperatures radiation creep results from diffusion of the radiation-produced vacancies and interstitial atoms to dislocations, enhancing the climb-to-glide process that controls time-dependent deformation. Radiation creep can be both beneficial and detrimental. Benefits accrue because of relaxation of constant displacement stresses, e.g., weld residual stress and in loaded bolts and springs. However, under these conditions – and more so under constant load conditions – radiation creep also induces elevated creep rates, including grain boundary sliding, that may help

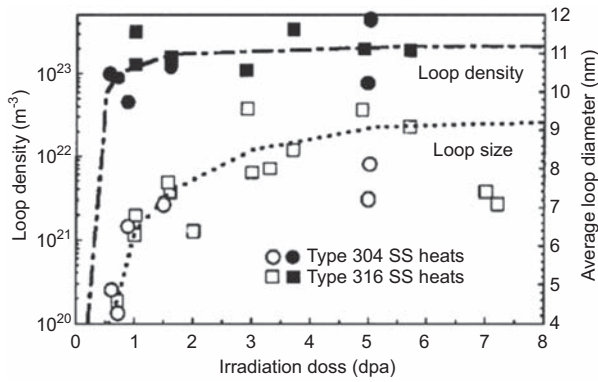


Figure 35 Measured change in density and size of interstitial loops as a function of dose during LWR irradiation of 300-series stainless steels at 275°C–290°C (Brummer et al., 1999).

to initiate and sustain SCC. Further, while the *average* magnitude of stress may be reduced by creep-induced stress relaxation, constraint in the polycrystal aggregate may result in high local stresses that could initiate a stress corrosion crack directly or cause large amounts of local grain boundary deformation that can rupture the oxide film locally, creating conditions for SCC initiation.

Figures 36A and B show examples of load relaxation under constant displacement conditions, a process that is quite reproducible over a wide range of materials and loading modes, and generally produces sizeable (>50%) load relaxation within a few dpa. Thus, for example, in areas of the BWR shroud that receive a moderate neutron flux, if SCC initiation does not occur early in life (e.g., by 1 dpa), the relaxation in residual stress should diminish the likelihood of cracking later in life. Because the effect of relaxation is significant, it tends to offset the detrimental effects of RIS and RH. Thus, it is not surprising that the incidence of SCC in BWR shroud welds, where the neutron flux can vary by two orders of magnitude because

of the varying proximity of the fuel, does not show a strong correlation with fluence.

Radiation creep relaxation also affects PWR baffle bolts, which are subject to large variations in fluence and temperature (Heuze et al., 1998; Scott et al., 2000). Baffle bolts in high flux regions can accumulate more than 3 dpa per year, and thus the preload will rapidly decrease during the first several years. Thus, SCC probably initiates early in life (before significant radiation creep relaxation occurs) or later in life when reloading occurs from differential swelling in the (annealed) baffle plates relative to the (cold worked) baffle bolts.

While difficult to prove, the elevated and sustained deformation rates associated with radiation creep can also accentuate susceptibility to SCC. Estimates of crack tip deformation rates (Andresen et al., 1990) indicate that radiation creep is not a large contributor in actively growing cracks, but rather it is expected to promote crack initiation and to sustain crack growth (or promote crack re-initiation, if an existing crack does arrest). It is obviously very important to factor radiation creep relaxation into initial component design and subsequent SCC analysis. Its impact is large, and occurs in the same fluence range as RIS and radiation hardening.

4.2.4. Hydrogen and helium generation Transmutation of various elements causes an increase in the helium content of irradiated stainless steel and nickel alloys. At low fluence, transmutation of boron produces helium ($^{10}\text{B} + n \rightarrow ^6\text{Li} + ^4\text{He}$); the ^{10}B is consumed (burned out) within a few dpa. While B segregates strongly to grain boundaries, the (n, α) reaction is energetic, and helium is distributed throughout a region of $\sim 10 \mu\text{m}$ around the grain boundary. A more prolific source of helium occurs at higher fluence associated with $^{58}\text{Ni}(n, \gamma)^{59}\text{Ni}$ and $^{59}\text{Ni}(n, p)$ and (n, α) reactions. Because ^{59}Ni must be produced first, the production of helium from this reaction is limited until 5–10 dpa. There is no evidence that helium directly affects SCC behavior, but it can produce an increase in hardness and yield strength, and may affect fracture toughness. It is known to have a large impact on weld repairs

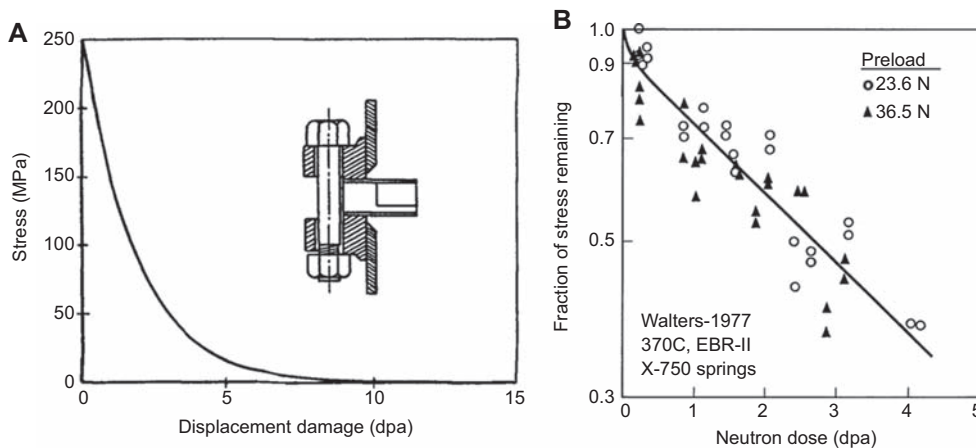


Figure 36 (A) The effects of radiation-induced creep on load relaxation of stainless steel at 288°C (Van Duysen, Todeschini & Zacharie, 1994), and (B) radiation creep relaxation of X-750 springs at 370°C (Walters & Ruther, 1977).

because the thermal transient permits migration of helium and the formation of bubbles along the grain boundaries. This can be managed to a certain extent by using low heat input processes, such as laser welding. Helium also stabilized void embryos, resulting in a shortening of the nucleation stage of voids. This also reduces the dose to reach steady state (breakaway) swelling. Swelling can impact IASCC indirectly by causing isotropic strains, resulting in stresses in welded or bolted components, especially when swelling occurs at different rates in the fastened structures.

Hydrogen is also produced by transmutation and proton injection during radiolysis of water, but these are small contributions in iron- and nickel-base alloys compared to the flux of hydrogen that occurs from exposure to the coolant. Unlike helium, hydrogen is quite mobile, and readily permeates through the material, entering and leaving voids and other microstructural features. Figure 37 shows that the permeation rate in stainless steel is proportional to the square root of the coolant H_2 fugacity, and that H_2 inside a tube can dissociate and permeate back out of the tube into the coolant if the coolant H_2 level is reduced. Elevated hydrogen levels in metals are not evidence of extreme H fugacities in the metal, only that there are more storage or trapping sites as it permeates through the metal.

5. IASCC mechanisms

The processes by which the effects of irradiation on water chemistry and the alloy can cause IASCC are discussed in this section. This is done by exploring various potential mechanisms for IASCC, starting with the crack tip strain rate model describing IASCC.

5.1. Radiolysis and the crack tip strain rate – slip oxidation model

The mechanisms by which radiolysis can affect IASCC are well established. The production of radiolytic species from

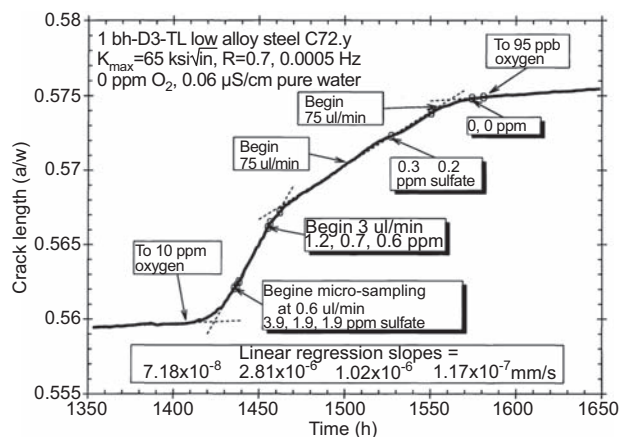


Figure 38 The effect of flow rate within the crack (solution being drawn out of a sampling tube at the crack tip) on crack growth rate in low alloy steel. As the sampling (withdrawal) rate is increased, the occluded crack chemistry becomes more dilute, and eventually the bulk aerated pure water exists at the crack tip. Despite the high corrosion potential and high availability of O_2 for cathodic reduction, the growth rate becomes very low.

water is reasonably well established and, more importantly, their effect on corrosion potential has been measured under a wide variety of conditions – as summarized in the section “Effects of irradiation on water chemistry”.

The effects of corrosion potential on crack chemistry and crack growth rate has also been described in detail (Andresen, 2005b, 2010a; Andresen & Young, 1995). High corrosion potential does not directly affect crack growth rate – tests in which water is rapidly drawn from the crack tip (Figure 38), and tests performed at very high flow rate (Figure 39) show a reduced growth rate. High corrosion potential is damaging because oxidants are consumed by metal corrosion and reaction with H_2 near the mouth of the geometrically restricted crack (Figure 40). With the

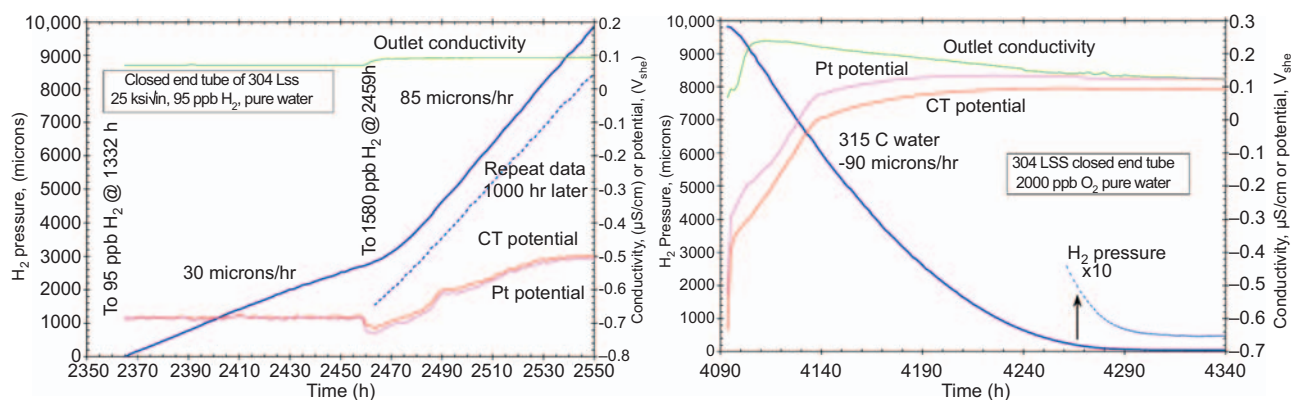


Figure 37 Hydrogen permeation vs. time and coolant H_2 fugacity unsensitized Type 304L stainless steel. The permeation rate is proportional to the square root of the coolant H_2 fugacity, and H_2 that permeates into the sealed tube readily dissociates and permeates into the coolant with the coolant H_2 is lowered.

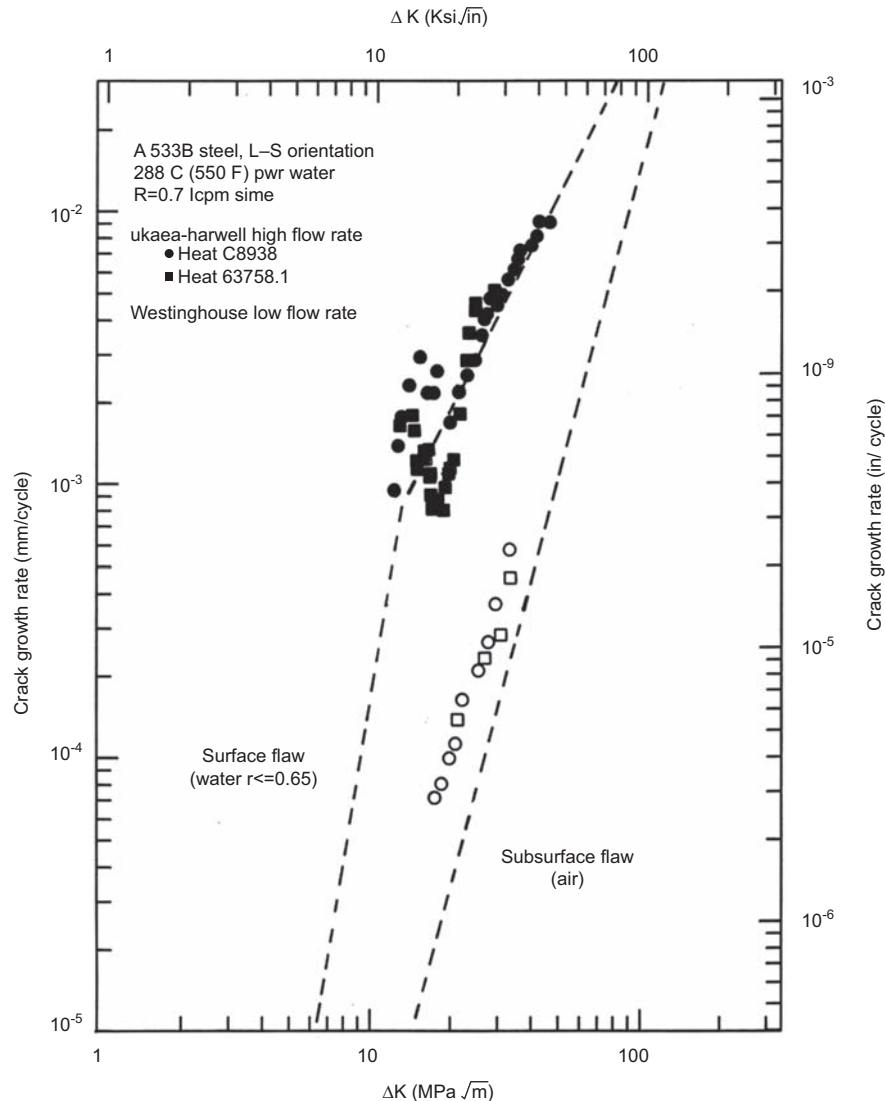


Figure 39 Very high recirculation flow rate (open symbols) causes a flushing of the crack chemistry in low alloy steels, and a reduction in growth rate. In this instance, the aggressive crack chemistry forms from the dissolution of MnS inclusion in the crack.

gradient in oxidant concentration comes a gradient in corrosion potential, and that produces a driving force for migration of ions, such as anions like Cl^- migrating into the crack and cations like Na^+ migrating out of the crack. The change in crack chemistry can produce a change in crack growth, and this has been proven using deaerated water and (larger) additions of H_2SO_4 (Figure 41). Thus, elevated corrosion potential *per se* is not damaging; rather, it is the resulting effect on crack chemistry.

The damaging element is not corrosion potential *per se*, it is the gradient in potential, or a gradient in temperature. The corrosion potential in PWR primary water is much lower than in BWR water, but this is a simple reflection of a higher pH at temperature, and a higher H_2 fugacity (Figure 42). Neither results in a potential gradient, nor in a shift in crack chemistry. Indeed, because the metal-metal oxide phase boundaries are parallel to the H_2 - H_2O line (Figure 42), changes in pH in the

near-neutral regime do not change the stability of metal or metal oxide phases.

Dynamic strain is recognized as important in SCC in hot water, and it is widely considered to be a fundamental parameter, both by experimental observation and conceptually because the slip offsets caused by plastic strain disrupt the surface passive films that in turn impart corrosion resistance. Experimentally, laboratory (Andresen, 2008a, 2008b) and in-reactor data (Bennett & Torill Karlsen, 2003; Scott, 2010) show that there is a strong effect of very slow straining or test perturbation on crack nucleation, short crack coalescence and crack growth. Analysis of field data on PWR baffle former bolts (Scott et al., 2000) showed a major influence of load following (daily reactor power changes) on incidence of SCC in bolts made from the same materials and used in multiple plants.

If cyclic loading is present, the dynamic strain at the crack tip, or crack tip strain rate, results from the reversed

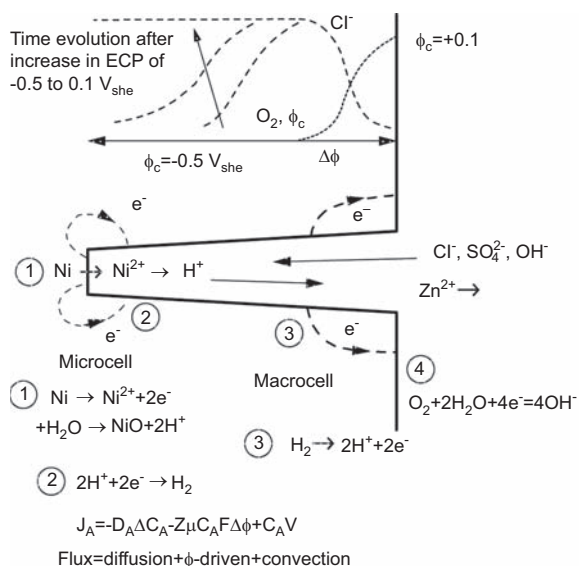


Figure 40 Schematic of crack chemistry transport processes in high temperature water with O_2 .

slip processes that cause (inert) fatigue crack advance. In slow strain rate tests, the crack tip strain rate results from the applied strain rate, which is partitioned among the number of growing cracks. At constant load (or constant K), the dynamic strain results from the crack advance process itself, although thermal and irradiation creep can play a role. As the crack advances, the stress/strain field at the crack tip must be redistributed, which requires dislocation motion. This creates a circularity: dynamic strain causes crack advances which sustains the dynamic strain. This is not non-physical, but it does

make SCC much more difficult to study than, e.g., corrosion fatigue.

The basis on which crack tip strain rates are calculated is strong for tests with fatigue loading or continuous straining, but is not as well defined for constant load (or K) conditions. A simplified formulation that produces reasonably accurate predictions has been proposed by Ford and Andresen (Ford & Andresen, 1994; Ford et al., 1987):

$$\dot{\epsilon}_{ct} = CK^4$$

where C is the proportionality constant that accounts for the material and its yield strength, temperature and other factors. These formulations for fatigue, slow strain rate and constant load provide well-behaved and accurate prediction continuum across these different loading conditions (Figure 43). A more complex version has been proposed by Gao et al. (Gao & Hwang, 1981; Gao, Zhang, & Hwang, 1983) for growing cracks in strain hardening materials and developed by (Shoji, 2003; Shoji, Lu, & Murakami, 2010) and Young et al. (2001). These formulations recognize that strain redistribution occurs during crack advance, and also addresses yield strength, strain hardening and other factors. While such formulations provide some basis for quantifying the important parameters that control crack tip strain rate, until verified there is only a limited basis for confidence in their accuracy. This is especially true of formulations based on continuum mechanics, when the crack tip deformation processes are in fact localized in space and time, and strongly affected by inhomogeneities in grain boundaries, relative grain orientation and slip systems. Estimations of the effect of thermal and irradiation creep (Andresen et al., 1990) suggest that their role during active crack advance is small, although their role in crack initiation or in reactivating cracks that have stopped growing may be large.

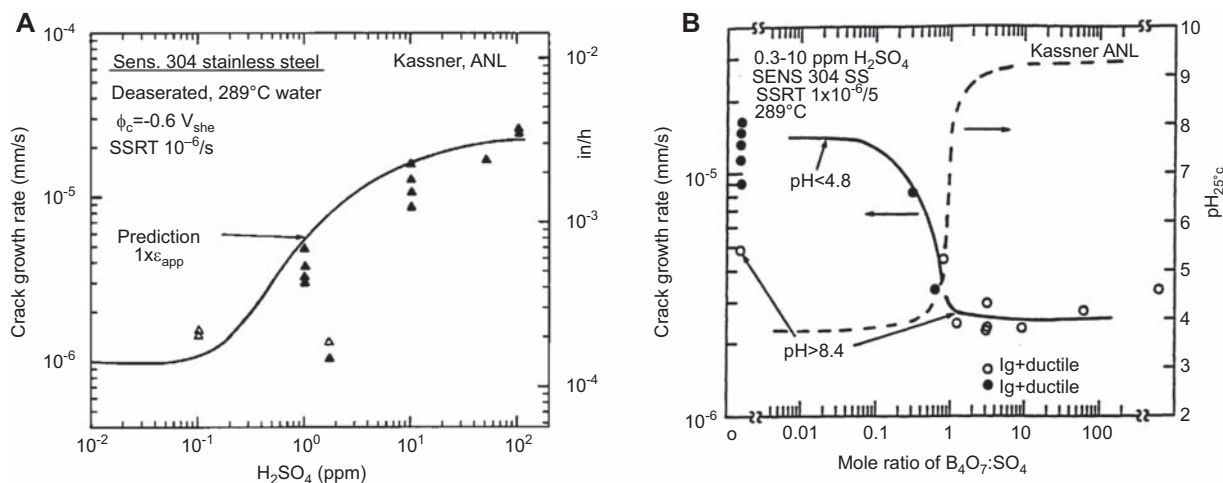


Figure 41 (A) The effect of H_2SO_4 addition to deaerated water on SCC growth rate in slow strain rate tests of sensitized type 304 stainless steel. Despite the low corrosion potential, the addition of H_2SO_4 produces a large increase in growth rate. (B) The primary crack chemistry effect may be related to pH, although at sufficiently high pH, the crack growth does increase.

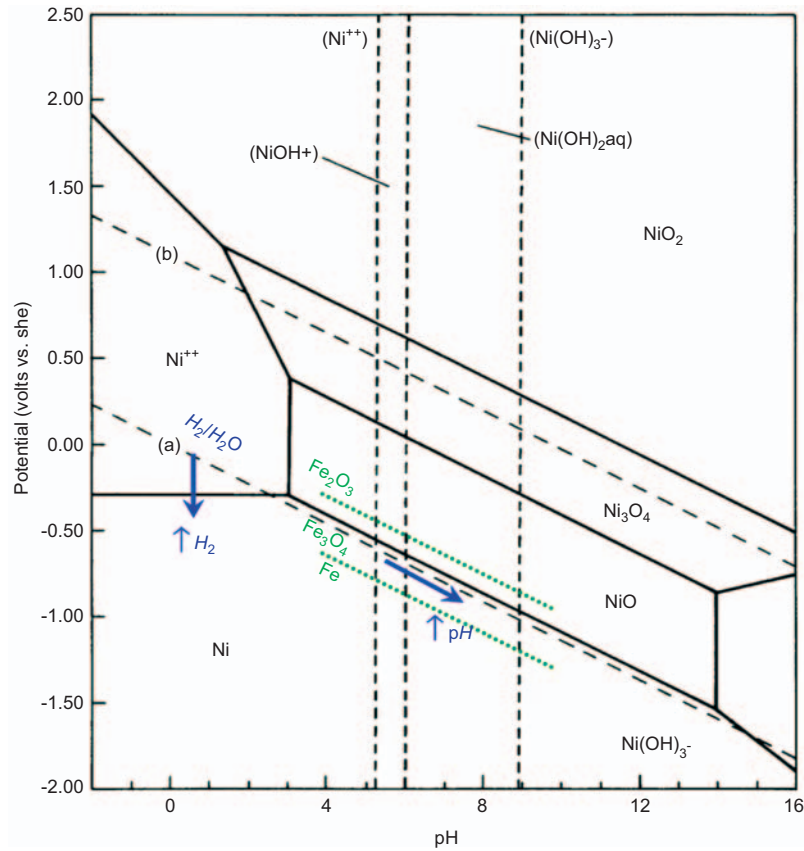


Figure 42 Ni-H₂O Pourbaix diagram at 300°C.

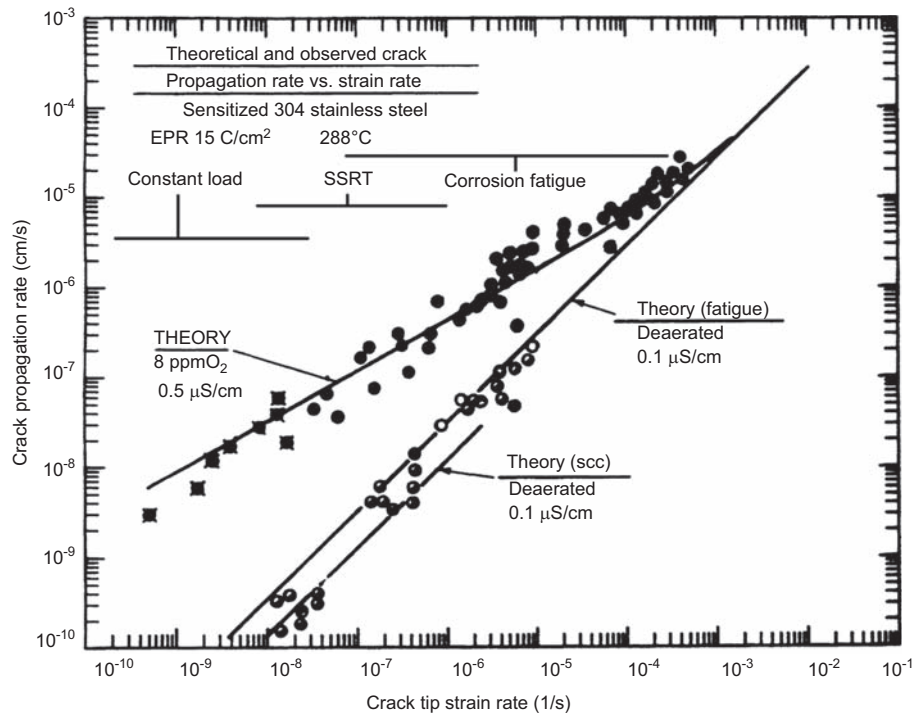


Figure 43 SCC growth rate vs. crack tip strain rate (for various loading conditions) for sensitized stainless steel illustrating the ability of crack tip strain rate formulations to account for the spectrum of loading from corrosion fatigue to slow strain rate to constant K.

5.2. Grain boundary chromium depletion

Because of the similarity to thermally-induced sensitization for creating a chromium depletion zone adjacent to the grain boundary, chromium depletion arising from RIS has been implicated as a likely cause of IASCC, especially in oxidizing environments. This is understandable because of the documented effect of thermal sensitization in extensive data from laboratory and plant operational experience (Andresen & Ford, 1995; Andresen et al., 1990; Brown & Gordon, 1988; Bruemmer et al., 1999; Gordon & Brown, 1990; Nelson & Andresen, 1992; Scott, 1994; Was & Andresen, 1992b, 2007; Was et al., 1999b). One distinct difference between chromium depletion because of RIS and that caused by chromium carbide precipitation is the width of the depleted zone. While RIS produces depleted zones that are typically 5–10 nm at half maximum, chromium depletion because of chromium carbide precipitation produces depleted zones 100x that width, resulting in a much higher volume of affected material adjacent to the boundary.

Figure 44A shows a correlation between grain boundary chromium level and IGSCC susceptibility in unirradiated stainless steels where the grain boundary depletion is caused by thermal sensitization (Bruemmer & Was, 1994). Much data have been accumulated to support the role of chromium depletion as an agent in IGSCC of austenitic alloys in oxidizing conditions. Numerous studies show that as the grain boundary chromium level decreases, intergranular SCC increases. As shown in Figure 34, grain boundary chromium depletion during irradiation can be severe.

Figure 44B shows a similar correlation between grain boundary chromium level and IASCC susceptibility, as measured by the % IG cracking on the fracture surface during SSRT experiments. There is also a significant amount of scatter in the data of Figure 44B that makes a direct correlation difficult to support. In particular, at low grain boundary chromium levels, the % IG ranges from 0 to 100. However, as the grain boundary chromium content increases, the data

becomes less “noisy”, suggesting that a higher grain boundary chromium concentration is a strong indicator of resistance to IASCC. Also, there is a trend in the maximum level of % IG with grain boundary chromium content. Further, the nature of the water chemistry appears to play a large role in whether chromium depletion correlates with cracking. Fukuya, Nakano, Fujii, and Torimaru (2004b) measured chromium depletion in 304L, 316 and 316L after irradiation to 0.8 dpa at 400°C, and then assessed SCC susceptibility via SSRT tests in pure water containing either dissolved hydrogen (2.8 ppm) at 320°C or dissolved oxygen (3 ppb or 32 ppm) at 288°C. Chromium depletion was greatest for 304L, followed by 316L and then 316. Cracking was greatest for 304L 32 ppm DO water, for 316L in 3 ppm DO water, and in 2.8 ppm DH, none of the samples exhibited cracking. These data indicate that chromium depletion is not likely a major factor in IASCC in reducing environments.

In fact, even in oxidizing environments, the data do not clearly support chromium depletion as the major cause of IASCC. Busby, Was, and Kenik (2002) showed that the different rates at which RIS and dislocation loops anneal during a post-irradiation heat treatment creates a window in time-temperature space in which annealing of the dislocation loop structure can occur to a great extent with little change in the grain boundary composition profile (Figure 45). Several studies have been conducted on the post-irradiation annealing of irradiated stainless steels and the assessment of the IASCC susceptibility as a function of annealing time and temperature (Busby, Was, and Kenik, 2002; Cole, Allen, Was, & Kenik, 2004; Fukuya, Nakano, Fujii, Torimaru, & Kitsunai, 2004a; Ishiyama et al., 1996; Jacobs, Bell, Shepherd, & Wozadlo, 1992; Katsura et al., 1998), and all show that the grain boundary chromium concentration remains at the as-irradiated level well beyond the point where the IASCC susceptibility has been substantially reduced. This occurrence is again an indication that while RIS may be an important factor in IASCC, it does not alone explain IASCC. Hence, grain boundary chromium

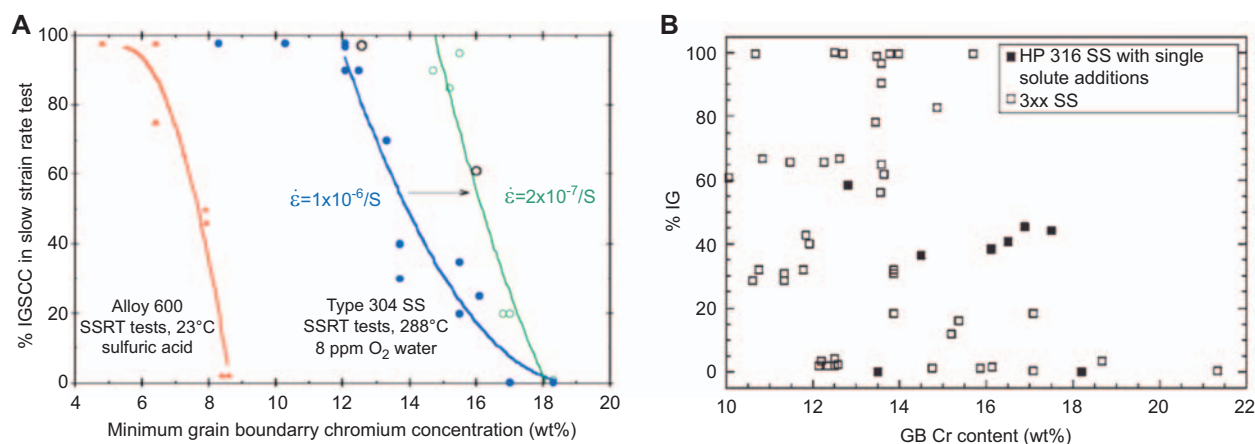


Figure 44 Effect of grain boundary Cr content on IGSCC for: (A) sensitized stainless steel and Alloy 600 and (B) irradiated stainless steels (Bruemmer & Was, 1994). SSRT tests are slow strain rate tests in which the specimen is monotonically strained vs. time (Was, 2003).

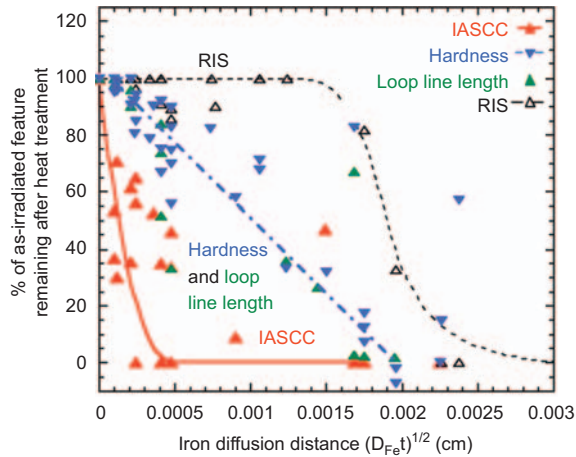


Figure 45 Removal of RIS and dislocation microstructure as measured by loop line length and hardness with extent of annealing as measured by $\sqrt{(Dt)}$ for iron, where D is the diffusivity of iron and t is time. This relationship accounts for annealing at different times and temperatures. The effect on IASCC in slow strain rates is also shown (Was, 2003).

depletion appears to be an *important*, but *insufficient* condition for IASCC.

5.3. Irradiation hardening

The dislocation loop microstructure formed during irradiation hardens the alloy, which correlates with increased SCC susceptibility. Under applied stress, the dislocation network interacts elastically with dislocation loops, producing an increase in the yield strength of the alloy, which can be detected in tensile tests or indentation hardness measurements. This is accompanied by a decrease in elongation, which is affected more than reduction in area because necking occurs relatively early in most tensile tests of irradiated stainless steel. Fracture toughness also decreases with fluence.

The increase in yield strength with a dose in 300 series stainless steels irradiated around 300°C is shown in Figure 46. The yield strength can reach values up to five times the unirradiated value by about 5 dpa, and its increase follows a square root dependence on dose. Both the source hardening model (Singh, Foreman, & Trinkaus, 1997) and the dispersed barrier hardening model (Seeger, 1958) provide reasonable correlations between hardening and the dislocation loop microstructure. In the dispersed barrier hardening model, the increase in hardness is proportional to $(N_{\text{loop}} \times d_{\text{loop}})^{1/2}$, where N_{loop} is the loop number density and d_{loop} is the loop diameter.

Hardening has been cited as a key factor in IASCC susceptibility. Figure 47A shows that increasing cold work and yield strength increases the crack growth rate in non-sensitized austenitic stainless steel tested in 288°C BWR NWC (Speidel & Magdowski, 1999). This dependence carries over to irradiated material. Figure 47B shows a correlation between yield strength and susceptibility to IASCC in SSRT tests, although the correlation is complicated by other

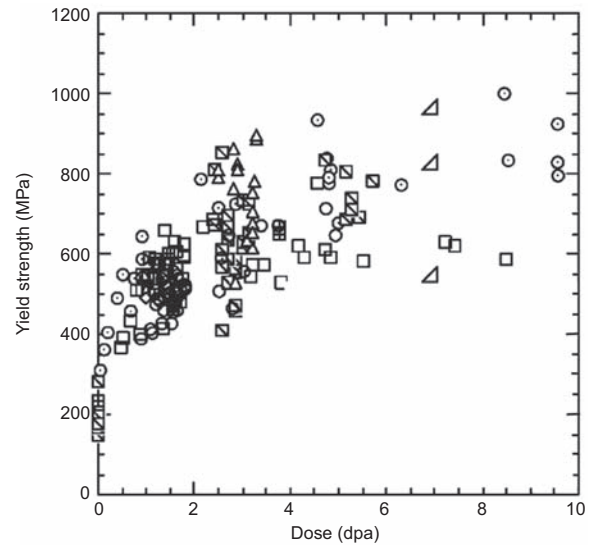


Figure 46 Irradiation dose effects on measured tensile yield strength for several 300-series stainless steels, irradiated and tested at a temperature of about 300°C (Was, 2003).

radiation induced changes, especially RIS. The correlation is better at high values of yield strength (>800 MPa) and at very low values (<400 MPa), with more scatter at intermediate values (400–800 MPa).

Hardening alone is not sufficient to explain IASCC. Annealing experiments tracked the change in hardness and dislocation loop microstructure vs. annealing condition (Was, 2003). Figure 45 shows the change in hardness and the change in the dislocation loop line length ($N_{\text{avg}} \times d_{\text{avg}}$), along with the change in IASCC susceptibility in SSRT tests as annealing progresses (Edwards et al., 2003; Jacobs, Wozadlo, & Gordon, 1995; Katsura et al., 1998). Except for very short annealing times, the loop line length and the hardness closely track each other, as expected if the loop structure is controlling the hardening. At very small values of $(Dt)^{1/2}$, the hardening remains flat before decreasing with annealing time. While both hardening and cracking are reduced with increased annealing, the behavior of the hardness does not fully explain the response.

Busby et al. (2005) postulated that the removal of very small defects at short annealing times may account for the IASCC behavior. Short times at high temperature (500°C) may preferentially remove the small defect clusters either by annihilation or by spontaneous dissociation. The dislocation loops may absorb the free vacancies and interstitials, thus adding to their line length. The loss of the small defect clusters will be offset by the growth of Frank loops producing no net change in measured hardness or yield strength. Despite a lack of hardness change, this process may alter the deformation mode at the local level by removing the small obstacles to dislocation motion, thus changing the character of localized deformation.

Hash, Wang, Busby, and Was (2004) showed that hardening is not the sole factor in IASCC by testing a series of samples of commercial purity 304 SS with nominally the same hardness, but with different contributions from cold

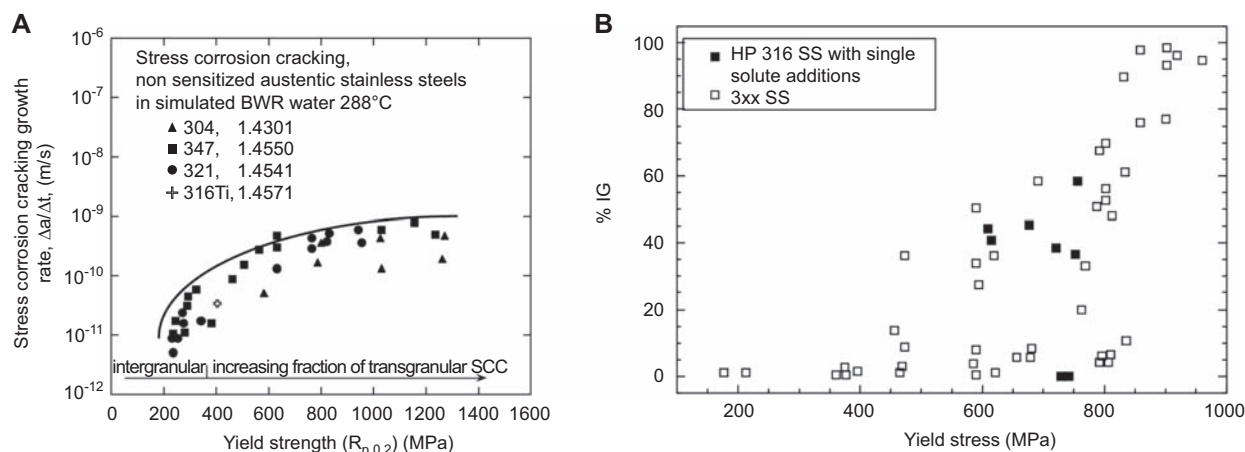


Figure 47 Effect of yield strength on IGSCC. (A) crack growth rate of cold worked, unirradiated 300-series stainless steels tested in 288°C simulated BWR water (Speidel & Magdowski, 1999), and (B) % IGSCC in SSRT tests on 300-series stainless steels where hardening is by irradiation (Was, 2003).

work and proton irradiation. At the extremes were a sample that was cold rolled to a 35% reduction in thickness and no irradiation and one irradiated to 1.67 dpa and no cold work. Three samples had varying amounts of cold work (10%, 20%, 25%) and corresponding amounts of irradiation dose (0.55, 0.25 and 0.09 dpa) to give a hardness level that was within 5% over all samples. SCC susceptibility was measured by the amount of IG cracking in a SSRT test in 288°C BWR NWC. IASCC susceptibility was not constant, as would be expected if hardness were the only factor, with cracking observed in only the two highest dose samples (0.55 dpa with 10% cold work and 1.67 dpa with no cold work), irrespective of their hardness (Figure 48). The amount of cracking in the lower dose sample was higher than in a companion sample at the same dose but without cold work, indicating that cold work can enhance the IASCC susceptibility. These results also suggest that radiation hardening may promote crack initiation, as it does crack growth. Combined with the annealing results, they suggest that other factors besides the hardness level and yield strength – such as RIS and deformation mode – play a role in the IASCC process.

5.4. Localized deformation

Deformation changes dramatically with radiation. Homogeneous deformation at low dose is replaced by heterogeneous deformation at higher doses as the defect microstructure begins to impede the motion of dislocations. Plasticity becomes localized to narrow channels that have been cleared of defects by preceding dislocations, providing a preferred path for subsequent dislocation motion. The channels are very narrow (<100 nm) and closely spaced (<1 μ m) and typically run the full length of a grain, terminating at the grain boundaries (Bruemmer et al., 1993). Dislocation channeling can cause localized necking and a sharp reduction in uniform elongation (Bloom, 1975).

Results of IASCC experiments on proton-irradiated samples over a wide range of doses and alloys have consistently showed that high nickel alloys have high IASCC resistance in SSRT tests. In particular, nickel concentrations >18 wt% are highly resistant to IASCC compared with 304 SS with 8 wt% nickel. Kodama, Fukuya, and Kayano (1994) showed that there is a good, but not perfect, correlation between nickel equivalent and IASCC. Notable exceptions are alloy 800 (Jenssen, Ljungberg, Walmsley, & Fisher, 1996, 1998) and a Fe-20Cr-25Ni-1Nb alloy used in AGR that experienced IGSCC (Norris, Baker, & Titchmarsh, 1987b). Nickel may affect IASCC directly through a change in composition, or indirectly through changing the slip character. Higher nickel content in stainless steel increases the stacking fault energy (SFE) significantly, producing a change in slip from planar to wavy. Swann et al. (1963) studied the effect of SFE on slip behavior using a series of Fe-18Cr-xNi alloys where $8 < x < 23$. They showed that for the 8% nickel alloy, the slip was entirely planar and, as nickel increases, cross-slip increases. By 20% nickel, there was no evidence of planar slip and the deforma-

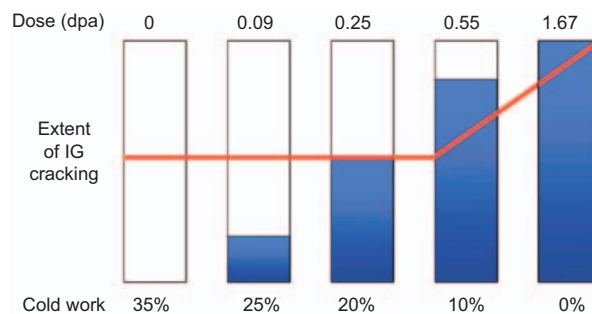


Figure 48 Degree of IASCC in Type 304 stainless steel samples with the same hardness but with different combinations of hardening by cold work and irradiation using 3.2 MeV protons at 360°C.

tion microstructure consisted of a web of dislocation tangles, evidence of wavy slip in a high SFE material.

Stacking fault energy has been linked to SCC resistance in stainless steels by Thompson and Bernstein who found that increasing SFE correlates well with increased reduction in area and decreased SCC susceptibility (Thompson & Bernstein, 1980). The IASCC susceptibility of several irradiated stainless steels shows that there is a good correlation between SFE and IASCC (Was, 2003).

Microstructure can also influence the deformation mode. Farrell, Byun, and Hashimoto (2002) conducted neutron irradiation of 316 stainless steel at temperatures between 65 and 100°C to doses of <1 dpa, then characterized the deformation behavior. With increasing dose, the propensity for dislocation channeling increased. The volume of material occupied by channels increases rapidly with dose and saturates at <0.5 dpa. These data show that the defect microstructure created by irradiation can induce planar deformation in the form of narrow dislocation channels.

The importance of slip localization in IASCC may be in the way in which the dislocations interact with the grain boundary. In planar slip, well defined and separated slip bands or dislocation channels (for irradiated materials) transmit dislocations during plastic deformation. These slip bands or channels terminate at grain boundaries where dislocations are fed into the grain boundary region vs. forming a tangled dislocation network within the grain. In the more traditional view of dislocation infusion into grain boundaries, the pile-up of dislocations in the intersecting channel creates progressively higher stresses at the grain boundary at the head of the pileup. If the stress exceeds a critical value, separation of the grain boundary could occur according to a Stroh (wedge) cracking type of mechanism (Stroh, 1957). This cracking process could occur regardless of the environment and may in fact be the mechanism that occurs in some of the IG cracking observed in very highly irradiated steels in inert environment (Bloom, 1975). At lower fluences, the stress at the grain boundary may promote rupture of the oxide film, leading to exposure of the metal to the solution and subsequent corrosion and IASCC.

Alternatively, deformation could occur in the boundary plane, which would rupture the oxide film and promote IASCC. Alexandreanu and Was (2003) showed that dislocation absorption by grain boundaries in nickel-base alloys leads to deformation in the grain boundary, usually by boundary sliding, and promotes IGSCC. They observed that random grain boundaries exhibited more sliding, and boundaries where sliding occurred were four times more susceptible to IGSCC in primary water at 360°C than boundaries that did not slide. This same process could explain how dislocation injection into grain boundaries by planar slip or dislocation channeling can result in IASCC.

The concept that slip planarity controls the deformation mode and IASCC is consistent with observations of IASCC in 304 SS and 316 SS. 316 SS has, in some cases (Jenssen et al., 1998) been shown to be more resistant to IASCC than is 304 SS, which is consistent with its higher SFE. Observations by Bailat et al. (2000) on neutron irradiated

samples and by Busby, Sowa, and Was (2004) on proton irradiated samples show clear dislocation channel patterns on the surface of 304 SS samples that cracked in SSRT tests, but an absence of those patterns on the 316 SS samples that did not crack. In addition, SSRT tests on high nickel alloys (with SFE of ~ 40 mJ/m²) showed no IG cracking and also no dislocation channeling. Further support for the influence of slip planarity on IASCC comes from the annealing studies cited earlier. On samples that underwent either a short, low temperature anneal, or were tested in the as-irradiated condition, IG cracking was accompanied by a high density of dislocation slip bands on the surface of ~ 65 /mm². Annealing at 500°C for 45 min. resulted in the elimination of IG cracking and a reduction in the surface slip band density to ~ 25 /mm².

In the annealing experiments described earlier, hardening by irradiation was much more effective in initiating IGSCC than was cold work, where the dislocation structure will likely be more cellular than planar. Thomas and Bruegger (1999) showed a correlation between slip band intersection with the walls of a growing crack and the accompanying steps on the oxide on the walls. Figure 49A shows a TEM micrograph of SCC in a cold worked 316 stainless steel baffle bolt. The slip bands intersecting the walls of the narrow crack at 45° to the crack growth direction. The slip band-crack wall intersections are also coincident with steps in the oxide as shown schematically in Figure 49B. This suggests that the flow of dislocations along the slip steps and into the grain boundary may have been responsible for discontinuous growth of the crack along the grain boundary.

Recent experiments have been conducted (Jiao & Was, 2010, 2011a) to measure the degree of localized deformation and correlate it with the susceptibility to IASCC. In this study, seven austenitic alloys were irradiated to 1 and 5 dpa at 360°C using 2–3.2 MeV protons and were tested both in a simulated BWR environment and in argon. Cracking susceptibility was evaluated at both 1% and 3% strain intervals using crack length per unit area. SFE, hardness, RIS and localized deformation were characterized and the strengths of their correlations with cracking were quantified on a scale from 0 (none) to 1 (perfect correlation). Correlation strengths of SFE and hardness with cracking were both around 0.5, indicating that both contributed to cracking but neither was the dominant factor. The correlation strength of RIS with cracking was zero in the grain boundary chromium content range of 12–19 wt%. RIS did not play an important role in this study. The correlation strength of localized deformation, which was characterized using the weighted average channel height, with IASCC was found to be significantly higher (0.88) than the others (Figure 50) implying that localized deformation is probably the most important factor in IASCC.

5.5. RIS of minor elements

Among the minor alloy elements, only silicon is known to segregate to high levels and silicon content has been shown to correlate with IASCC. Experiments by Busby et al. (Busby & Was, 2003; Busby et al., 2002) on a high purity 316 base alloy doped with 1 wt% silicon showed severe IASCC in NWC and

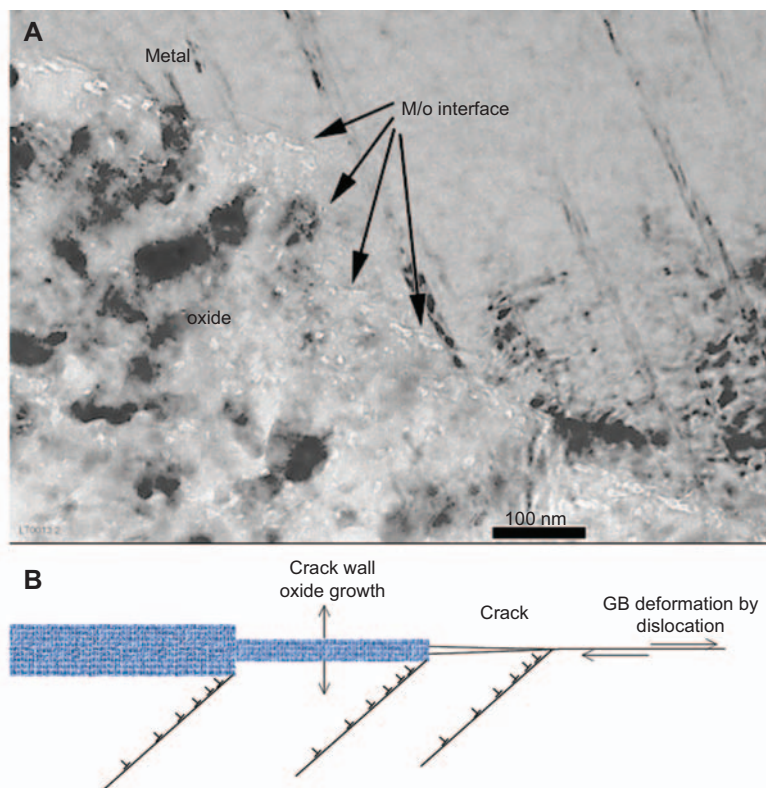


Figure 49 Micrograph of (A) deformation bands intersecting a crack in a baffle bolt (Thomas & Bruemmer, 1999) and (B) a schematic of the role that the deformation bands may be playing in IASCC.

in primary water after irradiation to 5.5 dpa at 360°C. STEM measurements of grain boundary silicon confirm levels up to 6 wt%. Past studies comparing AES and STEM results have shown that the actual concentration of silicon at the grain boundary plane may be as high as 15–20 wt% as a result of under-prediction of grain boundary RIS because of electron

probe spreading (Carter et al., 1993, 1994). Researchers have provided extensive evidence to show that increased silicon in stainless steel results in increased IGSCC in alloys tailored to imitate the composition of grain boundaries under irradiation (Fujimoto, Yonezawa, Iwamura, Ajiki, & Urata, 2000; Li et al., 2000; Yonezawa, Fujimoto, Iwamura, & Nishida, 2000a; Yonezawa et al., 1999, 2000b). However, it should be noted that these alloys were not irradiated and this difference may be important in the relevance of such experiments to IASCC. Using 1.5–5% silicon stainless steels of both standard (e.g., 304L) base composition and synthetic irradiated grain boundary composition, Andresen has observed significantly increased growth rates, reduction in the benefit of lowering corrosion potential, and very little effect of stress intensity factor between 27 and 13 ksi $\sqrt{\text{in}}$.

The data on impurity segregation effects on IASCC remains inconclusive. Extensive experiments have been conducted to isolate the effect of particular impurities such as sulphur, phosphorus, carbon, nitrogen and boron in IASCC, but none have yielded unambiguous results. Sulfur has not been found to segregate under irradiation, and while phosphorus thermally segregates to a significant extent; irradiation-induced phosphorus segregation is small in comparison (Jiao & Was, 2011b).

Carbon, nitrogen and boron cannot be measured in STEM; nitrogen and boron are very difficult to identify in AES, and carbon is a common contaminant. However, atom probe tomography has shown that boron enriches to high levels at

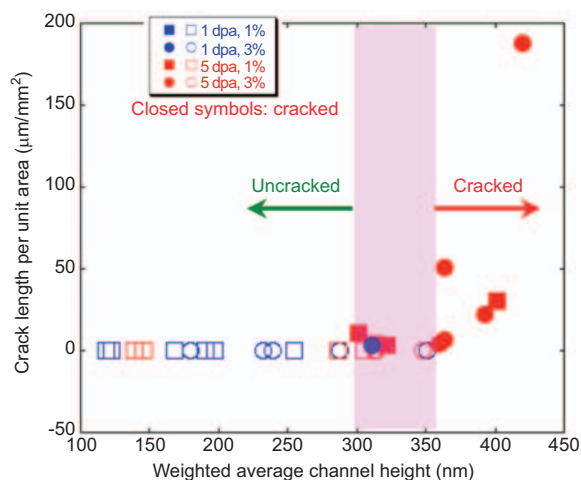


Figure 50 The contribution of localized deformation as measured by the weighted average channel height to cracking. The correlation strength of localized deformation with IASCC, $S_{\text{IASCC}}(\text{LD})=0.88$.

grain boundaries during alloy processing, and the concentration does not change following irradiation. Conversely, carbon is observed to deplete at the grain boundaries during irradiation (Jiao & Was, 2011b). The loss of carbon at grain boundaries may be significant, given the importance of carbon in promoting grain boundary cohesion and resistance to IGSCC in nickel-base alloys (Hertzberg & Was, 1998). Overall, establishing a link between impurity element segregation and IASCC in austenitic stainless steels has not been definitive.

6. IASCC-resistant alloys

The fundamental result of irradiation that leads to *all* of the changes in microstructure (RIS, loops, voids, precipitates), that then lead to *all* of the changes in materials properties (hardening, embrittlement, IG corrosion, IG cracking, creep, growth, swelling, etc.) is the production of vacancies and interstitials in excess of their thermodynamic equilibrium values. The strategy to eliminate or minimize the effect of irradiation is then quite simple: reduce the vacancy and interstitial concentrations in the solid.

Vacancy and interstitial concentrations are determined by the reactions responsible for their creation and annihilation. Vacancies and interstitials are created by the impingement of neutrons on the lattice and cannot be avoided in a reactor environment. They are destroyed by either mutual annihilation or by diffusing to sinks where they are lost from the matrix. Therefore, the strategy is to develop materials that greatly enhance mutual annihilation and loss to sinks.

Annihilation occurs when vacancies and interstitials encounter each other. The rate of loss by mutual annihilation is governed by the concentrations of the defects with higher concentrations leading to greater annihilation rates. However, the concentrations at which annihilation becomes effective is already far above the equilibrium values, so this is not a sensible strategy. Annihilation can be “assisted” or “augmented” by providing sites at which defects become trapped until their complement arrives and annihilation can occur. Such sites are coherent precipitates and oversized solutes.

Increasing the trap density is the obvious path for keeping defect concentrations low. The trap density must be high enough such that the distance between traps is small compared to that between defects, which will prevent defects from building up to high levels. Strategies to accomplish this include providing interfaces or providing other sinks such as dislocations. This is the idea behind ODS alloys, in which the very high density of closely spaced, nm-sized ppts provides enormous surface area per unit volume to facilitate defect annihilation, preventing the build-up of defects to high levels. Cold work serves the same purpose but is not stable at high doses. Grain boundaries can perform the same function but would require grain sizes on the nm-scale, resulting in such high levels of internal energy that the microstructure would likely be unstable against recrystallization at high temperature.

IASCC resistance can thus be achieved by eliminating the vacancy and interstitial super-saturation as a result of

irradiation. In considering what an IASCC-resistant alloy may look like, the following characteristics are noted. While not complete, the following is intended to provide a starting point in thinking about the attributes of an alloy that is resistant to IASCC.

6.1. High nickel content

Experience with high nickel alloys, such as alloy 800 indicate that higher nickel content is effective in reducing IASCC susceptibility. The mechanism by which nickel imparts increased resistance to IASCC is not well understood, but it could be linked, in part, to an increase in stacking fault energy. High stacking fault energy produces wavy slip, which could be important in avoiding high levels of localized deformation, which, as shown in section 5.4, contributes to IASCC susceptibility.

6.2. High chromium+nickel content

While not all IASCC results can be explained by chromium depletion at grain boundaries because of RIS, there is a correlation in oxidizing environments. Therefore, a higher chromium content will make it more difficult for IASCC to occur as the grain boundary chromium level will not fall as much because of RIS and, presumably, remain above some critical level. However, higher chromium contents in nickel base alloy 690 vs. 600 has been shown to provide significantly higher resistance to IGSCC in PWR environments as well, in which the potential is very low. Thus, high chromium+nickel has the potential for further increases in IASCC resistance above either addition alone.

6.3. Low silicon

As discussed earlier, there is growing evidence that high silicon content is detrimental to both crack initiation and crack growth in irradiated alloys in water environments. The formation of γ' or G-phase may be a part of the detrimental effect of silicon as these precipitates could serve to increase hardness and enhance localized deformation, both of which are detrimental to IASCC resistance. Thus, keeping the silicon content low in prototype alloys should reduce IASCC susceptibility.

6.4. High CSL fraction (grain boundary engineering)

Data on the effect of grain boundary engineering is mixed. As opposed to just increasing the number or fraction of special boundaries, it is probably more important to break up the high angle grain boundary (HAB) network. However, both phenomena generally occur during grain boundary engineering. There exists a sizeable set of data indicating that alteration of the grain boundary structure in a way that replaces HABs with CSLs and breaks up the connectivity of the HABs is quite beneficial for IG corrosion, and less so for IGSCC (Gertsman & Bruemmer, 2001; Lehockey, Palumbo, & Lin, 1998; Palumbo, King, & Aust, 1991; Pan, Adams, Olson, & Panayotou, 1996). The application of a stress also brings in

other factors, such as the orientation of the boundary with respect to the tensile stress and, thus, the resolved shear stress (and normal stress) on that boundary. Nevertheless, while the improvement may be small, it is unlikely to be detrimental and, if additive with other measures, it could contribute to improved IASCC resistance.

6.5. Low inclusion density

Studies by both Cookson and Was (1998) and Chung, Ruther, Sanecki, and Kassner (1992) showed that oxide inclusions were excellent sites for nucleation of IASCC cracks. IASCC is promoted by the fracture of brittle oxides that intersect grain boundaries, leading to crack extension along the grain boundary. Hard carbide particles such as TiC or TiN can also contribute in a similar way. Thus, it is clear that avoiding hard, brittle inclusions will be a key feature of an IASCC resistant alloy.

6.6. Grain boundary carbides

Grain boundary carbides have been shown to be an effective means of reducing cracking in nickel-base alloys by what is generally agreed upon as a “mechanical” mechanism (Alexandeanu, Capell, & Was, 2001; Norring, Engstrom, & Norberg, 1988). Grain boundary carbides have also been shown to homogenize slip (Bruemmer & Henager Jr., 1986). So, both by improving the mechanical response of grain boundaries, as well as by homogenizing deformation, grain boundary carbides should positively impact IASCC resistance.

6.7. Other microstructure features

There are other changes to the microstructure that could be made to affect IASCC, but neither the direction nor magnitude of the effect is well enough understood. Grain size is one such factor, cold work is another and the addition of titanium is a third. Definitive data on grain size effects are hampered by the difficulty in controlling just that one feature. Cold work appears to increase crack growth rate, but its effect on crack initiation is not well understood and good data is lacking. Finally, while titanium addition is helpful in preventing weld sensitization, experiments in the CIR program (Scott, 2010) showed that the crack growth rate of cold-worked 316 containing titanium was significantly higher than that without titanium. Therefore, there is no clear advantage of adding titanium to cold worked stainless steel.

6.8. Ferritic or Ferritic-Martensitic steels

Ferritic-Martensitic (F-M) steels are very attractive for use in fast reactor and fusion reactor systems for several reasons, including excellent dimensional stability (low swelling), high strength, high thermal conductivity, and low neutron activation making repairs less costly. Relative to LWR conditions, F-M steels exhibit high resistance to stress corrosion cracking,

but suffer from higher general corrosion rates and radiation embrittlement manifest in an increase in the ductile-to-brittle temperature. For high dose applications in LWRs, the key challenges for austenitic steels are IASCC and swelling, both of which are advantages of F-M steels. Hence, the challenge of F-M steels for LWR applications is whether the corrosion rates are low enough to be acceptable for water chemistry control, and whether radiation embrittlement is an issue for the operating temperatures of LWR cores.

7. Summary

Cracking of in-core reactor components exposed to both irradiation and high temperature water indicate that there is a significant role of irradiation in intergranular SCC. The effects of irradiation on SCC occur through changes in the water chemistry and in the alloy microstructure. It is the “persistent” radiation effects on the microstructure that are responsible for the “threshold-like” behavior of IASCC. The principal effect of irradiation on water chemistry is through radiolysis, which forms a variety of oxidizing and reducing molecules and radicals. Their net effect is captured by the change in corrosion potential, which can be considerable when the dissolved H₂ level is below ~500 ppb. High corrosion potentials strongly correlate with severity of cracking in both laboratory tests of pre-irradiated specimens and in core components in operating power reactors.

Irradiation causes a significant change in local composition near grain boundaries and other defect sinks. The enrichment of nickel and silicon, and the depletion of chromium can affect the susceptibility to IASCC, especially under oxidizing conditions. Irradiation also alters the microstructure, and under LWR conditions, faulted dislocation loops represent the primary irradiation-induced microstructure defect. The loops impede the motion of dislocations, resulting in an increase in the yield strength by factor of up to five. Radiation hardening correlates with IASCC propensity, and also induces highly localized deformation in the form of dislocation channels, which could contribute to IASCC. Irradiation also induces creep that can relax macroscopic stresses, and can also enhance local dynamic deformation. Other factors, such as swelling and formation of new phases, may accentuate IASCC at high fluence. With the many effects of irradiation, which overlap spatially and temporally, more work is needed to identify their roles in the mechanism(s) of IASCC and develop a comprehensive prediction methodology.

In considering the attributes of an IASCC-resistant alloy, the following are likely beneficial: high nickel and chromium contents, low silicon content, absence of brittle oxide and nitride inclusions, high CSL fraction, low connectivity of high angle grain boundaries and grain boundary coverage by chromium carbides. Finally, ferritic or ferritic-martensitic alloys should be considered because of their inherently greater resistance to radiation effects and IGSCC compared to iron- and nickel-base austenitic alloys.

References

- Alexandrescu B, Was GS. Grain boundary deformation-induced intergranular stress corrosion cracking of Ni-16Cr-9Fe in 360 degrees C water. *Corrosion* 2003; 59: 705–720.
- Alexandrescu B, Capell B, Was GS. combined effect of special grain boundaries and grain boundary carbides on IGSCC of Ni-16Cr-9Fe-xC alloys. *Mater Sci Engin A* 2001; 300: 94–104.
- Allen TR, Busby JT, Was GS, Kenik EA. On the mechanism of radiation-induced segregation in austenitic Fe-Cr-Ni alloys. *J Nucl Mater* 1998; 255: 44–58.
- Allen T, Was GS, Bruemmer S, Gian J, Ukai S. Design of Radiation-Tolerant Structural Alloys for Gen IV Nuclear Energy Systems. DE-FG07-03ID14542, Project No. 02–110. DOE, 2006.
- Andresen PL. Effects of temperature on crack growth rate in sensitized type 304 stainless steel and alloy 600. *Corrosion* 1993; 49: 714–725.
- Andresen PL. Effect of noble metal coating and alloying on the stress corrosion crack growth rate of stainless steel in 288°C water. In: *Proc 6th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. AIME, 1994: 245–253.
- Andresen PL. Application of noble metal technology for mitigation of stress corrosion cracking in BWRs. In: *Proc 7th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Houston, TX: NACE, 1995: 563–578.
- Andresen PL. Conceptual similarities and common predictive approaches for SCC in high temperature water systems. In: *Corrosion/96*. NACE, 1996: Paper 96258.
- Andresen PL. Perspective and direction of stress corrosion cracking in hot water. In: *Proc 10th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. NACE, 2001.
- Andresen PL. Similarity of cold work and radiation hardening in enhancing yield strength and SCC growth of stainless steel in hot water. In: *Corrosion 2002*. Houston, TX: NACE, 2002: Paper No. 02509.
- Andresen PL. K/Size effects on SCC in irradiated, cold worked and unirradiated stainless steel. In: *Proc 11th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Stevenson, WA: LaGrange Park, IL: American Nuclear Society, 2003: 995–1014.
- Andresen PL. Critical processes to model in predicting SCC response in hot water. In: *Corrosion 2005*. Houston, TX: NACE, 2005a: Paper No. 05470.
- Andresen PL. Critical processes to model in predicting SCC response in hot water. *Power Plant Chemistry* 2005b; 7: 541–560.
- Andresen PL. Short crack response of alloy 182 in PWR primary water. Report 1016545. Palo Alto, CA: EPRI, 2008a.
- Andresen PL. Scaling stress corrosion crack growth data and understanding down through short crack behavior toward initiation. In: *Workshop on detection, avoidance, mechanisms, modeling, and prediction of SCC Initiation in water-cooled nuclear reactor plants*. France: EPRI/Areva, Beaune, 2008b.
- Andresen PL. Stress corrosion cracking in austenitic stainless steels. In: *Tipping PG, editor. Understanding and mitigating ageing in nuclear power plants*. Woodhead Publishing, 2010a.
- Andresen PL. SCC of stainless steels in hot water. In: *Proc. Fontevraud 7*. Avignon, France: SFEN, 2010b.
- Andresen PL, Ford FP. Life prediction by mechanistic modeling and system monitoring of environmental cracking of iron and nickel-alloys in aqueous systems. *Mat Sci Eng a-Struct* 1988; 103: 167–184.
- Andresen PL, Ford FP. Modeling of irradiation effects on stress corrosion crack growth in laboratory and field data. In: *Int. Cooperative Group on Irradiation Assisted Stress Corrosion Cracking (ICG-IASCC) minutes*. 1989a.
- Andresen PL, Ford FP. Modeling of irradiation effects on stress corrosion crack growth rates. In: *CORROSION/89*. Houston, TX: NACE, 1989b: Paper No. 497.
- Andresen P, Ford F. Fundamental modeling of environmental cracking for improved design and lifetime evaluation in BWRs. *Int J Pres Ves Pip* 1994; 59: 61–70.
- Andresen PL, Ford FP. Modeling and prediction of irradiation assisted stress corrosion cracking. In: *Proc 7th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Houston, TX: NACE, 1995: 893–908.
- Andresen PL, Morra MM. Effects of Si on SCC of irradiated and unirradiated stainless steels and nickel alloys. In: *Corrosion 2005*. Houston, TX: NACE, 2005a: Paper No. 05591.
- Andresen PL, Morra MM. Effects of Si on SCC of irradiated and unirradiated stainless steels and nickel alloys. In: *Proc 12th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Snowbird: TMS, 2005b.
- Andresen PL, Morra MM. Effect of rising and falling K profiles on SCC growth rates in high temperature water. *Transaction of the ASME* 2007a; 129: 488–506.
- Andresen PL, Morra MM. Effects of dK/da on SCC growth rates. In: *Corrosion 2007*. Houston, TX: NACE, 2007b: Paper No. 07488.
- Andresen PL, Morra MM. Stress corrosion cracking of stainless steels and nickel alloys in high-temperature water. *Corrosion* 2008a; 64: 15–29.
- Andresen PL, Morra MM. IGSCC of non-sensitized stainless steels in high temperature water. *J Nucl Mater* 2008b; 383: 97–111.
- Andresen PL, Was GS. SCC of unirradiated stainless steels and nickel alloys in hot water. In: *17th International Corrosion Congress*. Houston, TX: NACE, 2008.
- Andresen PL, Young LM. Characterization of the roles of electrochemistry, convection and crack chemistry in stress corrosion cracking. In: *Proc 7th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. NACE, 1995: 579–596.
- Andresen PL, Ford FP, Murphy SM, Perks JM. State of knowledge of radiation effects on environmental cracking in light water reactor core materials. In: *Proc 4th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Houston, TX: NACE International, 1990a: 1–83.
- Andresen PL, Vasatis IP, Ford FP. Behavior of short cracks in stainless steels and Inconels in 288°C water. In: *CORROSION/90*. Houston, TX: NACE, 1990b: Paper No. 495.
- Andresen PL, Ford FP, Higgins JP, Suzuki I, Koyama M, Akiyama M, Okubo T, Mishima Y, Hattori S, Anzai H, Chujo H, Kanazawa Y. Life prediction of boiling water reactor internals. In: *Proc ICONE-4 Conf.*: New York, NY: ASME International, 1996.
- Andresen PL, Gott K, Nelson JL. Stress corrosion cracking of sensitized type 304 stainless steel in 288°C water: a five laboratory round Robin. In: *Proc 9th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. AIME, 1999.
- Andresen PL, Angeliu TM, Young LM. Immunity, thresholds, and other SCC fiction. In: *Proc. Staehle Symp. on Chemistry and Electrochemistry of Corrosion and SCC*. Warrendale, PA: The Minerals, Metals, and Materials Society (TMS), 2001.
- Andresen PL, Diaz TP, Hettiarachchi S. Resolving electrocatalytic SCC mitigation issues in high temperature water. In: *Corrosion/2004*. Houston, TX: NACE, 2004a: Paper No. 04668.

- Andresen PL, Morra MM, Catlin WR. Effects of yield strength, corrosion potential, composition and stress intensity factor in SCC of stainless steels. In: Corrosion/2004. Houston, TX: NACE, 2004b: Paper No. 04678.
- Andresen PL, Emigh PW, Morra MM. Effects of PWR primary water chemistry and deaerated water on SCC. In: Corrosion 2005. Houston, TX: NACE, 2005: Paper No. 05592.
- Andresen PL, Hickling J, Ahluwalia KS, Wilson JA. Effects of PWR primary water chemistry on PWSCC of Ni alloys. In: Proc 13th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. Canadian Nuclear Soc., 2007.
- Andresen PL, Chou PH, Morra MM, Nelson LJ, Rebak RB. Microstructural and stress corrosion cracking characteristics of austenitic stainless steels containing silicon. Metall Mater Trans A 2009; 40: 2824–2836.
- Angeliu TM, Andresen PL, Sutliff JA, Horn RM. Intergranular stress corrosion cracking of unsensitized stainless steels in BWR environments. In: Proc 9th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. The American Institute of Mining, Metallurgical, and Petroleum Engineers (AIME), 1999: 311–318.
- Angeliu TM, Andresen PL, Hall E, Sutliff JA, Sitzman S, Yamamoto M, Kuniya J. Strain and microstructure characterization of austenitic stainless steel weld HAZs. In: Corrosion 2000. NACE, 2000: Paper No. 00186.
- Armijo JS, Low JR, Wolff UE. Radiation effects on mechanical properties and microstructure of type-304 stainless steel. Nuc Appl 1965; 1: 462.
- Asano K, Fukuya K, Nakata K, Kodama M. Changes in grain boundary composition induced by neutron irradiation on austenitic stainless steels. In: Proc 5th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. LaGrange, IL: American Nuclear Society, 1992: 838.
- Bailat C, Almazouzi A, Baluc N, Schäublin R, Gröschel F, Victoria M. The effects of irradiation and testing temperature on tensile behaviour of stainless steels. J Nucl Mater 2000; 283–287: 446–450.
- Bennett P, Torill Karlsen T. Recent developments in instrumentation for in-core materials degradation studies in the halden reactor. In: Proc 11th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. Stevenson, WA: ANS, 2003.
- Bloom EE. Irradiation strengthening and embrittlement. In: Radiation damage in metals. Materials Park, OH: ASM International, 1975: 295.
- Brown Jr. JB, Storhok BW, Gates JE. Trans. Am Nucl Soc 1967; 10: 668–669.
- Brown KS, Gordon GM. Effects of BWR coolant chemistry on the propensity for IGSCC initiation and growth in creviced reactor internals components. In: Proc 3rd Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. New York, NY: The American Institute of Mining, Metallurgical, and Petroleum Engineers (AIME), 1988: 243–248.
- Bruemmer SM. New issues concerning radiation induced materials changes and irradiation assisted stress corrosion cracking in light water reactors. In: Proc 10th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. Houston, TX: NACE, 2002.
- Bruemmer SM, Henager Jr. CH. An overview of radiation chemistry in reactor coolants. In: Proc 2nd Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. LaGrange Park, IL: American Nuclear Society (ANS), 1986: 296.
- Bruemmer SM, Was GS. microstructural and microchemical mechanisms controlling intergranular stress-corrosion cracking in light-water-reactor systems. J Nucl Mater 1994; 216: 348–363.
- Bruemmer SM. Ed. Proc. Parkins Symp. on Fundamental Aspects of Stress Corrosion Cracking. AIME, 1992.
- Bruemmer SM, Cole J, Carter R, Was GS. Radiation hardening effects on localized deformation and Stress Corrosion Cracking of Stainless Steels. In: Proc 6th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. Warrendale, PA: The Minerals, Metals, and Materials Society (TMS), 1993: 537.
- Bruemmer SM, Arey BW, Charlot LA. Grain boundary chromium concentration effects on the IGSCC and IASCC of austenitic stainless steels. In: Proc 6th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. AIME (TMS), 1994: 277–285.
- Bruemmer SM, Simonen EP, Scott PM, Andresen PL, Was GS, Nelson JL. Radiation-induced material changes and susceptibility to intergranular failure of light-water-reactor core internals. J Nucl Mater 1999; 274: 299–314.
- Burns WG, Moore PB. Water radiolysis and its effect upon in-reactor zircaloy corrosion. Radiation Effects and Defects in Solids 1976; 30: 233–242.
- Busby JT, Was GS. Irradiation assisted stress corrosion cracking in model austenitic alloys with solute additions. In: Proc 11th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. Stevenson, WA: LaGrange Park, IL: American Nuclear Society, 2003: 995–1014.
- Busby JT, Kenik EA, Was GS. Effect of single solute additions on radiation induced segregation and microstructure of model austenitic alloys. J Nucl Mater (in press).
- Busby JT, Was GS, Kenik EA. Isolating the effect of radiation-induced segregation in irradiation-assisted stress corrosion cracking of austenitic stainless steels. J Nucl Mater 2002; 302: 20–40.
- Busby JT, Sowa MM, Was GS. The role of fine defect clusters in irradiation assisted stress corrosion cracking of proton irradiated 304 stainless steel. In: Effects of Radiation on Material: 21st Int. Symp. (ASTM STP1447). West Conshohocken, PA: ASTM International, 2004: 78–91.
- Busby JT, Sowa MM, Was GS, Simonen EP. Post-irradiation annealing of small defect clusters. Philos Mag 2005; 85: 609–617.
- Carter RD, Damcott D, Atzmon M, Was GS, Bruemmer S, Kenik EA. Capabilities and limitations of analytical methods used to measure radiation-induced grain boundary segregation. In: Proc 6th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. Warrendale, PA: The Minerals, Metals, and Materials Society (TMS), 1993: 501–510.
- Carter RD, Damcott DL, Atzmon M, Was GS, Bruemmer SM, Kenik EA. Quantitative-analysis of radiation-induced grain-boundary segregation measurements. J Nucl Mater 1994; 211: 70–84.
- Chatani K, Takakura K, Ando M, Nakata K, Tanaka S, Ishiyama Y, Hishida M, Kaji Y. IASCC crack growth rate of neutron irradiated low carbon austenitic stainless steels in simulated BWR condition. In: Proc 13th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. Whistler, B.C. Canada: CNS, 2007.
- Chen Y, Chopra OK, Gruber EE, Shack WJ. Irradiation-assisted stress corrosion cracking of austenitic stainless steels in BWR environments. Argonne Report ANL-09/17, NUREG/CR-7018. 2010.
- Cheng CF. Corrosion aspects of iron austenitic stainless steel+Inconel in high temperature water. Corrosion 1964; 20: 341.

- Cheng CF. Intergranular corrosion cracking of type-304 stainless steel in water-cooled reactors. *React Technol* 1970; 13: 310.
- Cheng CF. Intergranular stress-assisted corrosion cracking of austenitic alloys in water-cooled nuclear-reactors. *J Nucl Mater* 1975; 57: 11–33.
- Chopra OK, Rao AS. A review of irradiation effects on LWR core internal materials-IASCC susceptibility and crack growth rates of austenitic stainless steels. *J Nucl Mater* 2011; 409: 235–256.
- Chopra OK, Shack WJ. Crack growth rates and fracture toughness of irradiated austenitic stainless steels in bwr environments. Argonne Report ANL-06/58, NUREG/CR-6960. 2008.
- Chopra OK, Gruber EE, Shack WJ. Fracture toughness and crack growth rates of irradiated austenitic stainless steels. Argonne Report ANL-03/22, NUREG/CR-6826. 2003.
- Chopra OK, Alexandreanu B, Gruber EE, Daum RS, Shack WJ. Crack growth rates of irradiated austenitic stainless steel weld heat affected zone in BWR environments. Argonne Report ANL-04/20, NUREG/CR-6891. 2006.
- Chung HM, Ruther WE, Sanecki JE, Kassner TF. In: *Proc 5th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Monterey CA: LaGrange Park, IL: American Nuclear Society, 1992: 795.
- Cohen P. Water Coolant technology of power reactors. New York, NY: Gordon and Breach Science Publishers, 1969.
- Cole JJ, Allen TR, Was GS, Kenik EA. The influence of pre-irradiation heat treatments on thermal non-equilibrium and radiation-induced segregation behavior in model austenitic stainless steel alloys. In: *Effects of Radiation on Materials: 21st International Symposium*. American Society for Testing and Materials, West Conshohocken, 2004.
- Combrade P, Foucault M, Slama G. About the crack tip environment chemistry in pressure vessel steel exposed to primary PWR coolant. In: *Proc Atomic Energy Agency Specialists Meeting on Subcritical Crack Growth*, NUREG/CP0067, Vol. 2. Sendai, Japan: Nuclear Regulatory Agency, 1985: 201–218.
- Cookson J, Was GS. Oxide-induced initiation of stress corrosion cracking in irradiated stainless steel. *Corrosion* 1998; 54: 299–312.
- Cowan RL, Gordon GM. Intergranular stress corrosion cracking and grain boundary composition of Fe-Ni-Cr alloys. In: *Stress corrosion cracking and hydrogen embrittlement of iron-base alloys*. Firminy, France: NACE, Houston, 1977: 1063–1065.
- Davis RB, Andresen PL. Evaluation of fundamental linkage among stress corrosion cracking (SCC) Phenomena. Report 1007378. Palo Alto, CA: EPRI, 2002.
- Davis RB, Horn RM, Andresen PL, Nelson JL. BWRVIP-99-A: crack growth rates in irradiated stainless steels in BWR internal components. Report 1016566. Palo Alto, CA: EPRI, 2008.
- Dumbill S, Hanks M. Strategies for the moderation of chromium depletion at grain boundaries in irradiated steels. In: *Proc 6th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Warrendale, PA: The Minerals, Metals, and Materials Society (TMS), 1993: 521.
- Dumbill S, Vatter A. Microstructural characterisation of irradiated CT material. Halden Project Report HWR-684. 2001.
- Duncan RN, Arlt WH, Williams He, Baroch CJ, Hoffmann JP, Pashos TJ. Stainless-steel-clad fuel rod failures. *Nuc Appl* 1965; 1: 413.
- Edwards D, Bruemmer S. Characterization of Wurgassen Top Guide Materials. VG Presentation at Halden IASCC Meeting. 2002.
- Edwards D, Bruemmer SM. Final report EP-P19021/C9406. Palo Alto, CA: EPRI, 2008.
- Edwards DJ, Oliver BA, Garner FA, Bruemmer SM. Microstructural evaluation of a cold worked 316SS baffle bolt irradiated in a commercial PWR. In: *Proc. 10th Int. Symp. on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Houston, TX: NACE, 2002.
- Edwards DJ, Simonen EP, Bruemmer SM. Evolution of fine-scale defects in stainless steels neutron-irradiated at 275 degrees C. *J Nucl Mater* 2003a; 317: 13–31.
- Edwards DJ, Simonen EP, Garner FA, Greenwood LR, Oliver BM, Bruemmer SM. Influence of irradiation temperature and dose gradients on the microstructural evolution in neutron-irradiated 316SS. *J Nucl Mater* 2003b; 317: 32–45.
- Edwards DJ, Simonen EP, Bruemmer SM. Radiation-induced segregation behavior in austenitic stainless steels: fast reactor versus light water reactor irradiations. In: *Proc 13th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Canadian Nuclear Soc 2007.
- Ernestova M, Burda J. Crack growth testing of tailored alloys from Boris 6 and 7 irradiations. Report No. 1021235. Palo Alto, CA: EPRI, 2009.
- Espeland M. PIE of four compact tension (CT) specimens from IFA-657. Institute for Energy Technology Report IFE/KR/F-2004/115. 2004.
- Farrell K, Byun TS, Hashimoto N. Mapping flow localization processes in deformation of irradiated reactor structural alloys. Report ORNL/TM- 2002/66. ORNL, 2002.
- Ford FP, Andresen PL. Development and use of a predictive model of crack propagation in 304/316L, A533B/A508 and Inconel 600/182 Alloys in 288°C Water. In: *Proc 3rd Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. New York, NY: The Metallurgical Society of The American Institute of Mining, Metallurgical, and Petroleum Engineers (AIME), 1988: 789.
- Ford FP, Andresen PL. Corrosion in nuclear systems: environmentally assisted cracking in light water reactors. In: *Corrosion mechanisms*. New York, NY: Marcel Dekker, 1994: 501–546.
- Ford FP, Taylor DF, Andresen PL, Ballinger RG. Environmentally controlled cracking of stainless and low alloy steels in LWR environments. NP-5064M RP2006-6. Electric Power Research Institute (EPRI), 1987.
- Ford FP, Andresen PL, Solomon HD, Gordon GM, Ranganath S, Weinstein D, Pathania R. Application of water chemistry control, on-line monitoring, and crack growth models for improved BWR materials performance. In: *Proc 4th Int. Conf. on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. NACE, 1990: 4–26 to 24–51.
- Forsyth DR. Materials reliability program: hot cell testing of baffle/former bolts removed from two lead PWR plants (MRP-51). Report 1003069. Palo Alto, CA: EPRI, 2001.
- Fournier L, Sencer BH, Was GS, Simonen EP, Bruemmer SM. The influence of oversized solute additions on radiation-induced changes and post-irradiation intergranular stress corrosion cracking behavior in high-purity 316 stainless steels. *J Nucl Mater* 2003; 321: 192–209.
- Fujimoto K, Yonezawa T, Iwamura T, Ajiki K, Urata S. *Corros Eng* 2000; 49: 701.
- Fukuya K, Nakano M, Fujii K, Torimaru T, Kitsunai Y. Separation of microstructural and microchemical effects in irradiation assisted stress corrosion cracking using post-irradiation annealing. *J Nucl Sci Technol* 2004; 41: 1218.
- Fukuya K, Nakano M, Fujii K, Torimaru T. Role of radiation-induced grain boundary segregation in irradiation assisted

- stress corrosion cracking. *J Nucl Sci Technol* 2004a; 41: 594–600.
- Fukuya K, Nakano M, Fujii K, Torimaru T. IASCC susceptibility and slow tensile properties of highly-irradiated 316 stainless steels. *J Nucl Sci Technol* 2004b; 41: 673–681.
- Gabetta G, Buzzanaca G. Measurement of corrosion potential inside and outside growing crack during environmental fatigue tests. In: *Proc Atomic Energy Agency Specialists Meeting on Subcritical Crack Growth*, NUREG/CP0067, Vol. 2. Sendai, Japan: Nuclear Regulatory Agency, 1985: 219–230.
- Gan J, Simonen EP, Bruemmer SM, Fournier L, Sencer BH, Was GS. The effect of oversized solute additions on the microstructure of 316SS irradiated with 5 MeV Ni²⁺ ions or 3.2 MeV protons. *J Nucl Mater* 2004; 325: 94–106.
- Gao YC, Hwang KC. Elastic plastic fields in steady crack growth in a strain-hardening material. In: *5th Int Conf on Fracture*. 1981: 669–682.
- Gao YC, Zhang XT, Hwang KC. The asymptotic near-tip solution for mode III crack in steady growth in power hardening media. *Int J Fract* 1983; 21: 301–331.
- Garner FA. Evolution of microstructure in face-centered cubic metals during irradiation. *J Nucl Mater* 1993; 205: 98–117.
- Garzarolli F, Rubel H, Steinberg E. Behavior of water reactor core materials with respect to corrosion attack. In: *Proc Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Houston, TX: NACE, 1984: 1–24.
- Garzarolli F, Alter D, Dewes P. Deformability of austenitic stainless steels and nickel-based alloys in the core of a boiling and a pressurized water reactor. In: *Proc 2nd Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. LaGrange Park, IL: American Nuclear Society (ANS), 1986: 131–138.
- Garzarolli F, Alter D, Dewes P, Nelson JL. Deformability of austenitic stainless steels and nickel-based alloys in the core of a boiling and a pressurized water reactor. In: *Proc 3rd Int Symp. on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. New York, NY: The American Institute of Mining, Metallurgical, and Petroleum Engineers (AIME), 1988: 657–664.
- Gertsman VI, Bruemmer SM. *Acta Mater* 2001; 49: 1589.
- Gordon BM. Hydrogen water chemistry for BWR, Task 27 – materials and environmental monitoring within the duane Arnold BWR. Contract RP1930-1. Palo Alto, CA: EPRI, 1983.
- Gordon GM, Brown KS. Dependence of Creviced BWR Component IGSCC behavior on coolant chemistry. In: *Proc 4th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Houston, TX: NACE, 1990: 14–46 to 14–62.
- Hackett MJ, Busby JT, Miller MK, Was GS. Effects of oversized solutes on radiation-induced segregation in austenitic stainless steels. *J Nucl Mater* 2009a; 389: 265–278.
- Hackett MJ, Najafabadi R, Was GS. Modeling solute-vacancy trapping at oversized solutes and its effect on radiation-induced segregation in Fe-Cr-Ni alloys. *J Nucl Mater* 2009b; 389: 279–287.
- Hanninen H, Aho-Mantila I. Environment sensitive cracking of reactor internals. In: *Proc 3rd Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. New York, NY: The American Institute of Mining, Metallurgical, and Petroleum Engineers (AIME), 1988: 77–92.
- Hash MC, Wang LM, Busby JT, Was GS. The effect of hardening source in proton irradiation assisted stress corrosion cracking of cold worked type 304 stainless steel. In: *Effects of Radiation on Materials: 21st Int. Symp. (ASTM STP1447)*. West Conshohocken, PA: ASTM International, 2004: 92–104.
- Head RA, Indig ME, Andresen PL. Measurement of in-core and recirculation system responses to hydrogen water chemistry at nine mile point unit 1 BWR. Contract RP2680-5 (Final Report). Palo Alto: EPRI, 1989.
- Hertzberg J, Was GS. Role of carbon in the deformation and fracture of Ni-16Cr-9Fe alloys in 360°C water. *Metall Trans A* 1998; 29A: 1035–1046.
- Hettiarachchi S, Wozadlo GP, Diaz TP, Andresen PL, Cowan RL. The concept of noble metal addition technology for IGSCC mitigation of nuclear materials. In: *Proc 7th Int. Symp. on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Houston, TX: NACE, 1995: 735–746.
- Heuze PA, Goltrant O, Cauvin R. Analysis of the partition – reinforcement connecting screws of the Tihange 1 power plant. In: *Proc Int Symp Fontevraud IV*. Paris, France: French Society of Nuclear Energy, 1998: 195 [in French].
- Hindmarsh AC. GEAR: ordinary differential equation system solver. Report UCID-30001. Lawrence Livermore Laboratory, 1972.
- Horn R. BWRVIP-232: highly irradiated stainless steel crack growth program on general electric control rod blade materials. Report 1019078. Palo Alto, CA: EPRI, 2009.
- Horn RM, Gordon GM, Ford FP, Cowan RL. Experience and assessment of stress corrosion cracking in L-grade stainless steel BWR internals. *Nucl Eng Des* 1997; 174: 313–325.
- Ishiyama Y, Kodama M, Yokota N, Asano K, Kato T, Fukuya K. Post-irradiation annealing effects on microstructure and helium bubbles in neutron irradiated type 304 stainless steel. *J Nucl Mater* 1996; 239: 90–94.
- Jacobs A. Effects of low temperature annealing on the microstructure and grain boundary chemistry of irradiated type 304SS and correlations with IASCC resistance. In: *Proc 7th Int. Symp. on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Houston, TX: NACE, 1995: 1021.
- Jacobs AJ, Wozadlo GP. Irradiation assisted stress corrosion cracking as a factor in nuclear power plant aging. In: *Proc Int Conf on Nuclear Power Plant Aging, Availability Factor and Reliability Analysis*. Materials Park, OH: ASM International, 1985: 173.
- Jacobs AJ, Hale DA, Siegler M. Unpublished data on SCC of irradiated SS in 288°C water and inert gas. San Jose, CA: GE Nuclear Energy, 1986.
- Jacobs AJ, Clausing RE, Miller MK, Shepherd CM. Influence of grain boundary composition on the IASCC susceptibility of type 348 stainless steel. In: *Proc 4th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Houston, TX: NACE, 1990: 14–21.
- Jacobs AJ, Bell GEC, Shepherd CM, Wozadlo GP. High-temperature solution annealing as an IASCC mitigation technique. In: *Proc 5th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. La Grange Park, IL: American Nuclear Society, 1992: 917.
- Jacobs AJ, Wozadlo GP, Nakata K, Kasahara S, Okada T, Kawano TS, Suzuki S. The correlation of grain boundary composition in irradiated stainless steel with IASCC resistance. In: *Proc 6th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Warrendale, PA: The Minerals, Metals, and Materials Society (TMS), 1993: 597.
- Jacobs AJ, Wozadlo GP, Gordon GM. Low-temperature annealing-a process to mitigate irradiation-assisted stress-corrosion cracking. *Corrosion* 1995; 51: 731–737.
- Jenssen A, Jakobsson R. Tensile tests of irradiated control rod material from Barseback 1. Studsvik Report N(K)-01/031. 2001.

- Jenssen HK, Espeland M. PIE of four miniature compact tension (CT) specimens from IFA-639. Institute for Energy Technologie Report IFE/KR/F-2002/142. 2002.
- Jenssen A, Stjarnsater J. Crack growth rates of irradiated commercial stainless steels in BWR and PWR environments. Report 1019028. Palo Alto, CA: EPRI, 2009.
- Jenssen A, Ljungberg L, Walmsley J, Fisher S. The importance of molybdenum on irradiation assisted stress corrosion cracking in austenitic stainless steels. In: Corrosion/96. Houston, TX: NACE, 1996: Paper No. 101.
- Jenssen A, Ljungberg LG, Walmsley J, Fisher S. Importance of molybdenum on irradiation-assisted stress corrosion cracking in austenitic stainless steels. Corrosion 1998; 54: 48–60.
- Jenssen A, Gott K, Efsing P, Andersson P. Crack growth behavior of irradiated type304L stainless steel in simulated BWR environment. In: Proc 11th Int Conf. on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. Stevenson, WA: ANS, 2003: 1015–1026.
- Jenssen A, Sundberg J, Konig M. CIR II program: crack growth testing of fast reactor irradiated type 304L stainless steel in BWR environments. Report 1014976. Palo Alto, CA: EPRI, 2007.
- Jenssen A, Stjarnsater J, Torimaru T, Kitsunai Y. BWRVIP-221: crack growth in high fluence BWR materials-Phase 1: crack growth rate testing of types 304L and 316L at doses ranging from 3.5 to 13 dpa. Report 1019079. Palo Alto, CA: EPRI, 2009.
- Jiao Z, Was GS. The role of irradiated microstructure in the localized deformation of austenitic stainless steels. J Nucl Mater 2010; 407: 34–43.
- Jiao Z, Was GS. Impact of localized deformation on IASCC in austenitic stainless steels. J Nucl Mater 2011a; 408: 246–256.
- Jiao Z, Was GS. Novel features of radiation-induced segregation and radiation-induced precipitation in austenitic stainless steels. Acta Mater 2011b; 59: 1220–1238.
- Jones RH editor. Proc Chemistry and Electrochemistry of Corrosion and SCC: A Symposium Honoring the Contributions of R.W. Staehle. TMS, Feb. 2001.
- Kaji Y, Ugachi H, Tsukada T, Matsui Y, Ohmi M, Nagata N, Dozaki K, Takiguchi H. Comparison of SCC growth rate between in-core and ex-core tests in BWR simulated high temperature water. In: Proc 13th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. Whistler, BC, Canada: CNS, 2007.
- Kaji Y, Miwa Y, Shibata A, Nakano J, Tsukada T, Takakura K, Nakata K. Influence of microstructure on IASCC growth behavior of neutron irradiated type 304 austenitic stainless steels in simulated BWR condition. In: Proc 14th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems-Water Reactors. Virginia Beach, VA: ANS, 2009: 1181–1191.
- Karlsen TM. Final Report on IFA-586/IFA-605: effects of water chemistry on the in-pile crack growth behaviour of double cantilever beam specimens. OECD Halden Reactor Project, HWR-473. 1996.
- Karlsen W. TEM characterization of irradiated CW316 and SA304 stainless steels. VTT Report VTT-R-00801-09. 2009.
- Karlsen TM, Horvath A. Final report on the in-pile crack growth behaviour of irradiated compact tension specimens in IFA-639. OECD Halden Reactor Project, HWR-770. 2004.
- Karlsen W, Aaltonen P. FEG-STEM Study of 12 dpa Barseback 304L and 20 dpa and 30 dpa Chooz A 304SS. VTT Report VTT-R-01215-07. 2007.
- Karlsen TM, Bennett P, Hogberg NW. In-core crack growth rate studies on irradiated austenitic stainless steels in BWR and PWR conditions in the Halden reactor. In: Proc 12th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. Salt Lake City, UT: TMS, 2005a: 337–348.
- Karlsen TM, Espeland M, Horvath A. Summary report on the PWR crack growth rate investigation, IFA-657. OECD Halden Reactor Project, HWR-773. 2005b.
- Karlsen TM, Espeland M, Helin M. Final Results from the BWR Crack Growth Rate Investigation IFA-709. OECD Halden Reactor Project, HWR-977. 2010.
- Kato T, Takahashi H, Izumiya M. Grain-boundary segregation under electron-irradiation in austenitic stainless-steels modified with oversized elements. J Nucl Mater 1992; 189: 167–174.
- Katsura S, Ishiyama Y, Yokota N, Kato T, Nakata K, Fukuya K, Sakamoto H, Asano K. Post-irradiation annealing effects of austenitic stainless steel in IASCC. In: Corrosion/98. Houston, TX: NACE, 1998: Paper No. 132.
- Kenik EA. Radiation-induced segregation in irradiated type-304 stainless-steels. J Nucl Mater 1992; 187: 239–246.
- Kodama M, Fukuya K, Kayano H. Influence of impurities and alloying elements on IASCC in neutron irradiated austenitic stainless steels. In: Proc 16th Int Symp on Radiation Effects on Materials (ASTM STP 1175). West Conshohocken, PA: ASTM International, 1994: 889.
- Kumagai K, Morra MM, Andresen PL, Catlin GM, Nelson JL, Suzuki S, Takagi Y, Horn RM. Effects of K and impurity concentration on crack growth kinetics at alloy 182/A533B weld overlay boundaries in BWRs. In: Proc 14th Int. Conf. on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. ANS, 2009.
- Lam NQ, Kumar A, Wiedersich H. Kinetics of radiation induced segregation in ternary alloys. In: Proc Effects of Radiation on Materials, 11th Conf. ASTM STP 782. West Conshohocken, PA: ASTM International, 1982: 985.
- Lehockey EM, Palumbo G, Lin P. Metall Trans A 1998; 29: 369.
- Li GF, Kaneshima Y, Shoji T. Effects of impurities on environmentally assisted crack growth of solution-annealed austenitic steels in primary water at 325 degrees C. Corrosion 2000; 56: 460–469.
- Lin CC. An overview of radiation chemistry in reactor coolants. In: Proc 2nd Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. LaGrange Park, IL: American Nuclear Society (ANS), 1986: 160–172.
- Manahan MP, Kohli R, Santucci J, Sipush P. A Phenomenological investigation of in-reactor cracking of type-304 stainless-steel control rod cladding. Nucl Eng Des 1989; 113: 297–321.
- Massoud JP, Dubuisson P, Scott P, Chamardine VK. CIR II program: description of the Boris 6 and 7 experiments in the BOR-60 fast breeder reactor. Report 1011787. Palo Alto, CA: EPRI, 2005.
- Maziasz PJ. Overview of microstructural evolution in neutron-irradiated austenitic stainless-steels. J Nucl Mater 1993; 205: 118–145.
- Maziasz PJ, Mchargue CJ. Microstructural evolution in annealed austenitic steels during neutron-irradiation. Int Mater Rev 1987; 32: 190–219.
- Miwa Y, Tsukada T, Jitsukawa S, Kita S, Hamada S, Matsui Y, Shindo M. Effect of minor elements on irradiation assisted stress corrosion cracking of model austenitic stainless steels. J Nucl Mater 1996: 1393–1396.
- Multer I. European operating experience. In: Proc Joint Topical Meeting on Commercial Nuclear Fuel Technology Today. Toronto, Canada: CNA/ANS, 1975.

- Nakahigashi S, Kodama M, Fukuya K, Nishimura S, Yamamoto S, Saito K, Saito T. Effects of neutron irradiation on corrosion and segregation behavior in austenitic stainless steels. *J Nucl Mater* 1992; 179–181: 1061–1064.
- Nakamura T, Koshiishi M, Torimaru T, Kitsunai Y, Takakura K, Nakata K, Ando M, Ishiyama Y, Jenssen A. Correlation between IASCC growth behavior and plastic zone size of crack tip in 3.5 dpa neutron irradiated type 304L SS CT specimen. In: *Proc 13th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Whistler, BC, Canada: CNS, 2007: P0030.
- Nakano J, Karlsen TM, Espeland M. Summary report on the BWR crack growth rate experiment, IFA-658. OECD Halden Reactor Project, HWR-862. 2007.
- Nakano J, Karlsen TM, Espeland M. Summary of results from the PWR crack growth rate investigation IFA-670. OECD Halden Reactor Project, HWR-843. 2008.
- Nelson JL, Andresen PL. Review of current research and understanding of irradiation assisted stress corrosion cracking. In: *Proc 5th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. LaGrange Park, IL: American Nuclear Society, 1992: 10.
- Norris DIR, Baker C, Taylor C, Titchmarsh JM. A study of radiation induced sensitization in 20/25/Nb steel by compositional profile measurements at grain boundaries. In: *Materials for nuclear reactor core applications*. London, UK: BNES, 1987a: Paper No. 52.
- Norris DIR, Baker C, Titchmarsh JM. Compositional profiles at grain boundaries in 20% Cr/25% Ni/Nb stainless steel. In: *Proc. radiation-induced sensitization of stainless steels*. Berkeley Nuclear Labs. London, UK: Central Electricity Generating Board, 1987b: 86–98.
- Norring K, Engstrom J, Norberg P. Intergranular stress corrosion cracking in steam generator tubing, testing of alloy 690 and alloy 600 tubes. In: *Proc 3rd Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Warrendale, PA: New York, NY: The American Institute of Mining, Metallurgical, and Petroleum Engineers (AIME), 1988: 587–593.
- Okamoto PR, Rehn LE. Radiation-induced segregation in binary and ternary alloys. *J Nucl Mater* 1979; 83: 2–23.
- Okamoto PR, Wiedersich H. Segregation of alloying elements to free surfaces during irradiation. *J Nucl Mater* 1974; 53: 336–345.
- Onchi T, Dohi K, Soneda N, Navas M, Castano ML. Neutron fluence dependent intergranular cracking of thermally sensitized type 304 stainless steel at 290°C in inert gas and in water. In: *Proc 11th Int. Conf. on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. LaGrange Park, IL: American Nuclear Society (ANS), 2003: 1, 111.
- Ooki S, Tanaka Y, Takamori K, Suzuki S, Tanaka S, Saito Y, Nakamura T, Kato T, Chatani K, Kodama M. Study on SCC growth behavior of BWR core shroud. In: *Proc 12th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Salt Lake City, UT: TMS, 2005: 365–376.
- Othon M, Morra M. EBSD Characterization of the deformation behavior of alloy 182 weld metal. *Microsc Microanal* 2008; 14(Suppl. S2): 934–935.
- Palumbo G, King PJ, Aust KT. *Scripta Mater* 1991; 21: 1775.
- Pan Y, Adams BL, Olson T, Panayotou N. *Acta Mater* 1996; 44: 4685.
- Pashos TJ, Baroch CJ, Arlt WH, Williamson HE, Duncan RN, Hoffmann JP. Failure experience with stainless steel clad fuel rods in Vbwr. *T. Am Nucl Soc* 1964; 7: 416.
- Pasupathi V, Klingensmith RW. Investigation of stress corrosion in clad fuel elements and fuel performance in the connecticut Yankee reactor. NP-2119. Palo Alto, CA: EPRI, 1981.
- Pathania R, Gott K, Scott PM. An overview of the cooperative IASCC research (CIR) program. In: *Proc 13th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Whistler, BC, Canada: CNS, 2007: P0056.
- Perks JM, Marwick AD, English CA. Fundamental aspects of radiation induced segregation in Fe-Cr-Ni alloys. In: *Proc radiation-induced sensitisation of stainless steels*. Berkeley, Gloucestershire, UK: Central Electricity Generating Board, Berkeley Nuclear Labs, 1987: 15.
- Proc 1st–14th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors. NACE/ANS/TMS, 1983–2009.
- Proc 1st–6th Int Symp Fontevraud Contribution of Materials Investigation to the Resolution of Problems Encountered in Pressurized Water Reactors. 1986–2006.
- Proc Water Chemistry of Nuclear Reactor Systems, No.5. London, UK: British Nuclear Energy Society, October 23–27, 1989.
- Ritter S, Seifert HP. The effect of chloride and sulfate transients on the stress corrosion cracking behavior of low alloy RPV steels under simulated BWR environment. In: *Corrosion 2004*. NACE, 2004: Paper No. 04680.
- Schaffer LD. Army Reactor Program Progress Report ORNL 3231. 1962.
- Scott P. A review of irradiation assisted stress-corrosion cracking. *J Nucl Mater* 1994; 211: 101–122.
- Scott PM. Materials reliability program: a review of the cooperative irradiation assisted stress corrosion cracking research program (MRP-98). Palo Alto, CA: EPRI, 2003.
- Scott P. Final review of the cooperative irradiation-assisted stress corrosion cracking research program. Final Report, 1020986. EPRI, 2010.
- Scott PM, Meunier M-C, Deydier D, Silvestre S, Trenty A. An analysis of baffle/former bolt cracking in French PWRs. In: *ASTM STP 1401, Environmentally Assisted Cracking: Predictive Methods for Risk Assessment and Evaluation of Materials, Equipment and Structures*. West Conshohocken, PA: ASTM International, 2000: 210–223.
- Seeger A. On the theory of radiation damage and radiation hardening. In: *Proc 2nd United Nations Int Conf. on Peaceful Uses of Atomic Energy*, Vol. 6. New York, NY: United Nations, 1958: 250.
- Shoji T. Progress in the mechanistic understanding of BWR SCC and its implication to the prediction of SCC growth behavior in plants. In: *Proc 11th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Stevenson, WA: ANS, 2003: 588–599.
- Shoji T, Lu Z, Murakami H. Formulating stress corrosion cracking growth rates by combination of crack tip mechanics and crack tip oxidation kinetics. *Corros Sci* 2010; 52: 769–779.
- Singh BN, Foreman AJE, Trinkaus H. Radiation hardening revisited: role of intracascade clustering. *J Nucl Mater* 1997; 249: 103–115.
- Speidel MO, Magdowski R. Stress corrosion cracking of stabilized austenitic stainless steels in various types of nuclear power plants. In: *Proc 9th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Warrendale, PA: The Metallurgical Society of The American Institute of Mining, Metallurgical, and Petroleum Engineers (AIME), 1999: 325.
- Staehle RW, Forty AJ, VanRooyen D, editors. *Fundamental aspects of stress corrosion cracking*. Ohio State Univ: Houston, TX: NACE, 1967.

- Staehele RW, Hochmann J, McCright RD, Slatern JE, editors. Stress corrosion cracking and hydrogen embrittlement of iron-base alloys. Firminy, France: NACE, Houston, 1977.
- Stoller R. Primary damage formation in irradiated materials. *J Met* 1996; 48: 23.
- Storrer JT, Locke DH. High burnup irradiation experience in vulcan. *Nucl Eng Inter* 1970; 2: 93–99.
- Stroh AN. The existence of microcracks after cold-work. *Philos Mag* 1957; 2: 1–4.
- Sumiya R, Tanaka S, Nakata K, Takakura K, Ando M, Torimaru T, Kitsunai Y. K Validity criterion of neutron irradiated type 316L stainless steel CT specimen for SCC growth test. In: *Proc 13th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Whistler, BC, Canada: CNS, 2007.
- Swann PR. Dislocation substructure vs. transgranular stress corrosion susceptibility of single phase alloys. *Corrosion* 1963; 19: 102t.
- Takakura K, Nakata K, Tanaka S, Nakamura T, Chatani K, Kaji Y. Crack growth behavior of neutron irradiated L-grade austenitic stainless steels in simulated BWR conditions. In: *Proc 14th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Virginia Beach, VA: ANS, 2009: 1192–1203.
- Taylor DF. Response of electrochemical sensors to ionizing radiation in high temperature aqueous environments. In: *CORROSION/90*. Houston, TX: NACE, 1990: Paper No. 501.
- Taylor DF, Caramihas CA. High temperature aqueous crevice corrosion in alloy 600 and 304L stainless steel. In: *Localized crack chemistry and mechanics of environmentally assisted cracking*. New York, NY: The American Institute of Mining, Metallurgical, and Petroleum Engineers (AIME), 1983.
- Thomas LE, Bruemmer SM. Insights into environmental degradation mechanisms from analytical transmission electron microscopy of SCC cracks. In: *Proc 9th Int. Symp. on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Warrendale, PA: The Metallurgical Society of The American Institute of Mining, Metallurgical, and Petroleum Engineers (AIME), 1999: 41.
- Thompson AW, Bernstein IM. The role of metallurgical variables in hydrogen assisted environmental fracture. In: *Advances in corrosion science and technology*, Vol. 7. New York, NY: Plenum Press, 1980: 53.
- Toloczko MB, Andresen PL, Bruemmer SM. SCC crack growth of cold-worked 316L SS in simulated BWR oxidizing and hydrogen water chemistry conditions. In: *Proc 13th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Canadian Nuclear Soc, 2007.
- U.S.NRC. CRACK INDICATIONS IN MARATHON CONTROL ROD BLADES. Event Number: 46348. <http://www.nrc.gov/reading-rm/doc-collections/event-status/event/>. 2010.
- Valo M. Fracture toughness and tensile properties of irradiated austenitic stainless steel components removed from service. VTT Report DNo REA 35/91. 1993.
- Van Dyck S. IASCC mechanisms-controlling material factors: SCC crack-growth tests on selected Boris-6 tailored heats. Palo Brussels, Belgium: SCK-CEN, 2009.
- Walmsley J, Spellward P, Fisher S, Jenssen A. Microchemical characterization of grain boundaries in irradiated steels. In: *Proc 7th Int Symp on Environmental Degradation of Materials in Nuclear Power Systems-Water Reactors*. Houston, TX: NACE, 1997: 985.
- Was GS. Recent developments in understanding irradiation assisted stress corrosion cracking. In: *Proc 11th Int Conf on Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. LaGrange Park, IL: American Nuclear Society (ANS), 2003: 965–985.
- Was GS. Fundamentals of radiation materials science: metals and alloys. Berlin: Springer, 2007.
- Was GS, Allen T. Radiation-induced segregation in multicomponent alloys-effect of particle type. *Mater Charact* 1994; 32: 239–255.
- Was GS, Andresen PL. Irradiation-assisted stress-corrosion cracking in austenitic alloys. *J Min Met Mat S* 1992a; 44: 8–13.
- Was GS, Andresen PL. Irradiation assisted stress corrosion cracking in austenitic alloys. *J Met* 1992b; 44: 8–13.
- Was GS, Andresen PL. Stress corrosion cracking behavior of alloys in aggressive nuclear reactor core environments. *Corrosion* 2007; 63: 19–45.
- Was GS, Allen TR, Busby JT, Gan J, Damcott D, Carter D, Atzmon M. Microchemistry of proton irradiated austenitic alloys under conditions relevant to LWR core components. In: *Proc Mater Res Soc*, Vol. 540. Warrendale, PA: Materials Research Society, 1999a: 421–432.
- Was GS, Allen TR, Busby JT, Gan J, Damcott D, Carter D, Atzmon M, Kenik EA. Microchemistry and microstructure of proton-irradiated austenitic alloys: toward an understanding of irradiation effects in LWR core components. *J Nucl Mater* 1999b; 270: 96–114.
- Was GS, Busby JT, Allen T, Kenik EA, Jenssen A, Bruemmer SM, Gan J, Edwards AD, Scott PM, Andresen PL. Emulation of neutron irradiation effects with protons: validation of principle. *J Nucl Mater* 2002; 300: 198–216.
- Watanabe H, Muroga T, Yoshida N. The temperature dependent role of phosphorus and titanium in microstructural evolution of Fe-Cr-Ni alloys irradiated in FFTF. *J Nucl Mater* 1996; 228: 261–274.
- Wiedersich H, Okamoto PR, Lam NQ. A theory of radiation-induced segregation in concentrated alloys. *J Nucl Mater* 1979; 83: 98–108.
- Yamashita H. SCC growth behavior of BWR core shroud materials. *Int J Pres Ves Pip* 2008; 85: 582–592.
- Yonezawa T, Iwamura T, Fujimoto K, Ajiki K. Optimized chemical composition and heat treatment conditions of 316 CW and high chromium austenitic stainless steels for PWR baffle bolts. In: *Proc 9th Int Symp On Environmental Degradation of Materials in Nuclear Power Systems – Water Reactors*. Warrendale, PA: The Minerals, Metals, and Materials Society (TMS), 1999: 1015.
- Yonezawa T, Fujimoto K, Iwamura T, Nishida S. Improvement of IASCC resistance for austenitic stainless steels in PWR environment. In: *Environmentally assisted cracking: predictive methods for risk assessment and evaluation of materials, equipment, and structures (ASTM 1401)*. West Conshohocken, PA: ASTM International, 2000a: 224.
- Yonezawa T, Fujimoto K, Kanasaki H, Iwamura T, Nakada S, Ajiki K, Urata S. *Corros Eng* 2000b; 49: 655.
- Young LM, Andresen PL, Angelini TM. Crack tip strain rate: estimates based on continuum theory and experimental measurement. In: *Corrosion* 2001. NACE, 2001: Paper No. 01131.
- Zinkle SJ, Maziasz PJ, Stoller RE. Dose dependence of the microstructural evolution in neutron-irradiated austenitic stainless-steel. *J Nucl Mater* 1993; 206: 266–286.